

Serving modes

MLOPS DEPLOYMENT AND LIFE CYCLING



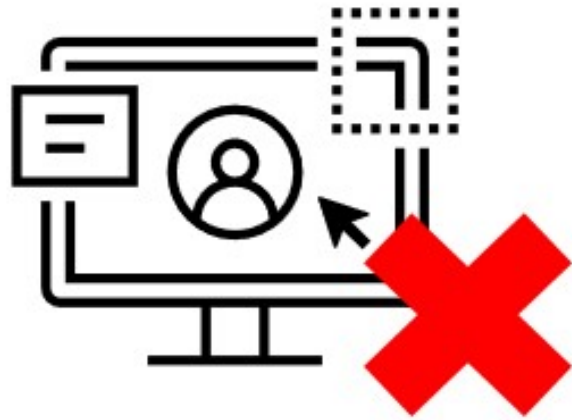
Nemanja Radojkovic

Senior Machine Learning Engineer

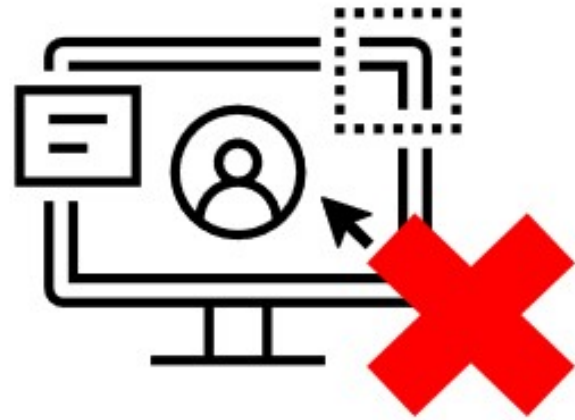
Model as a service



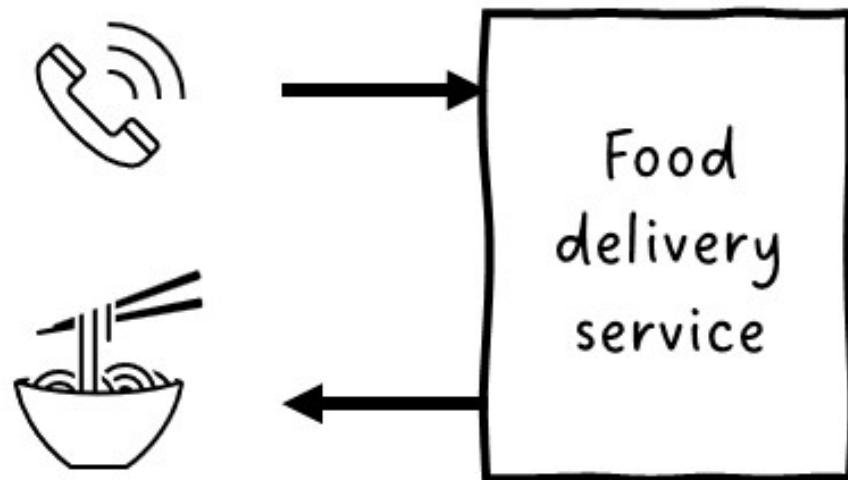
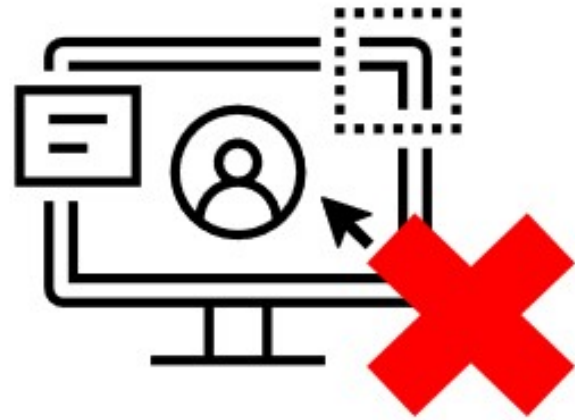
Model as a
service



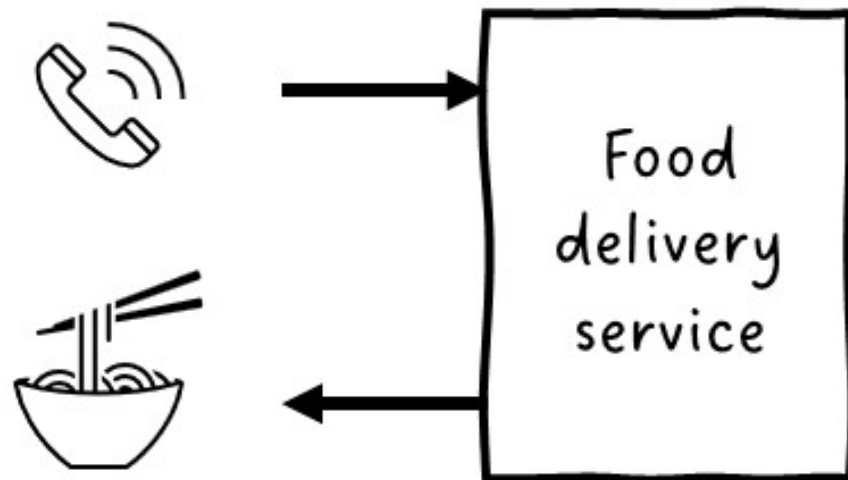
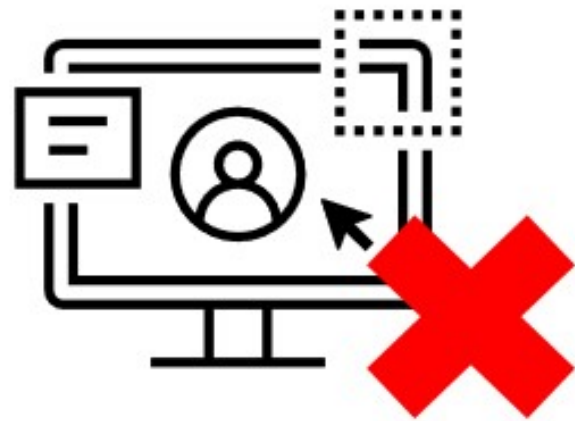
Model as a
service



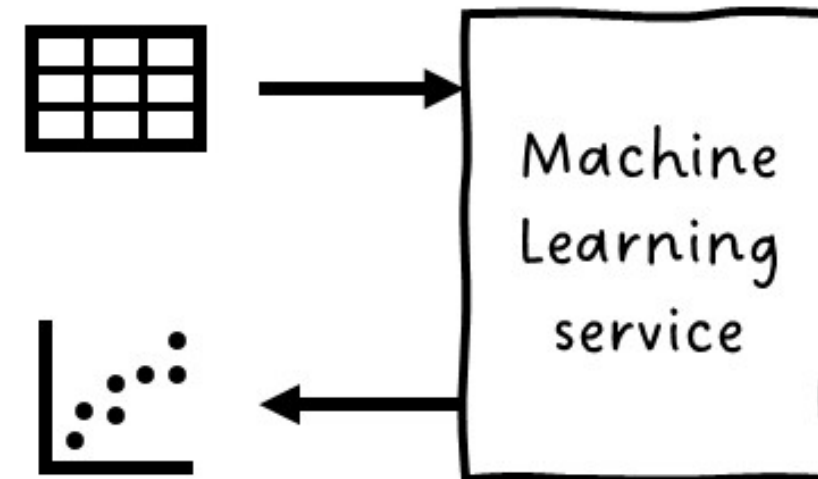
Model as a
service



Model as a service



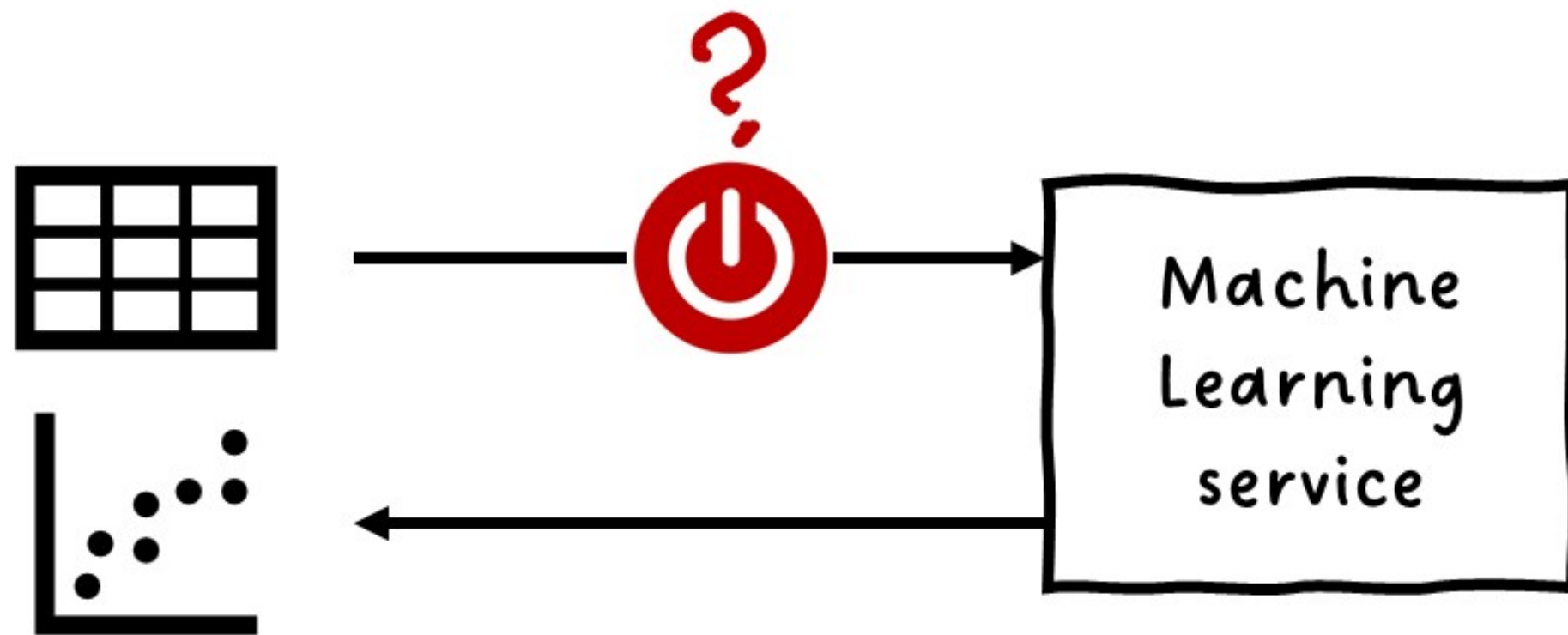
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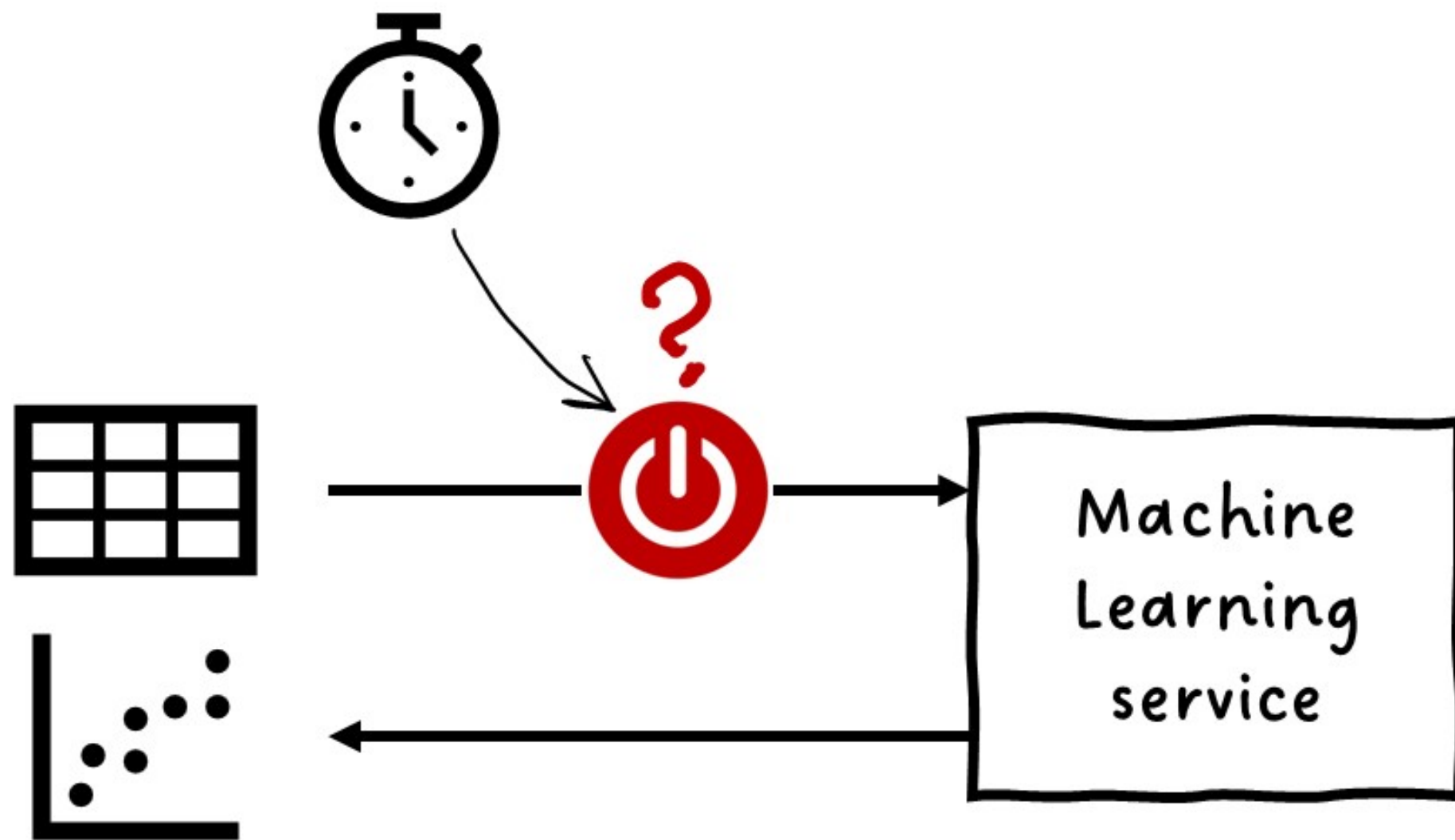


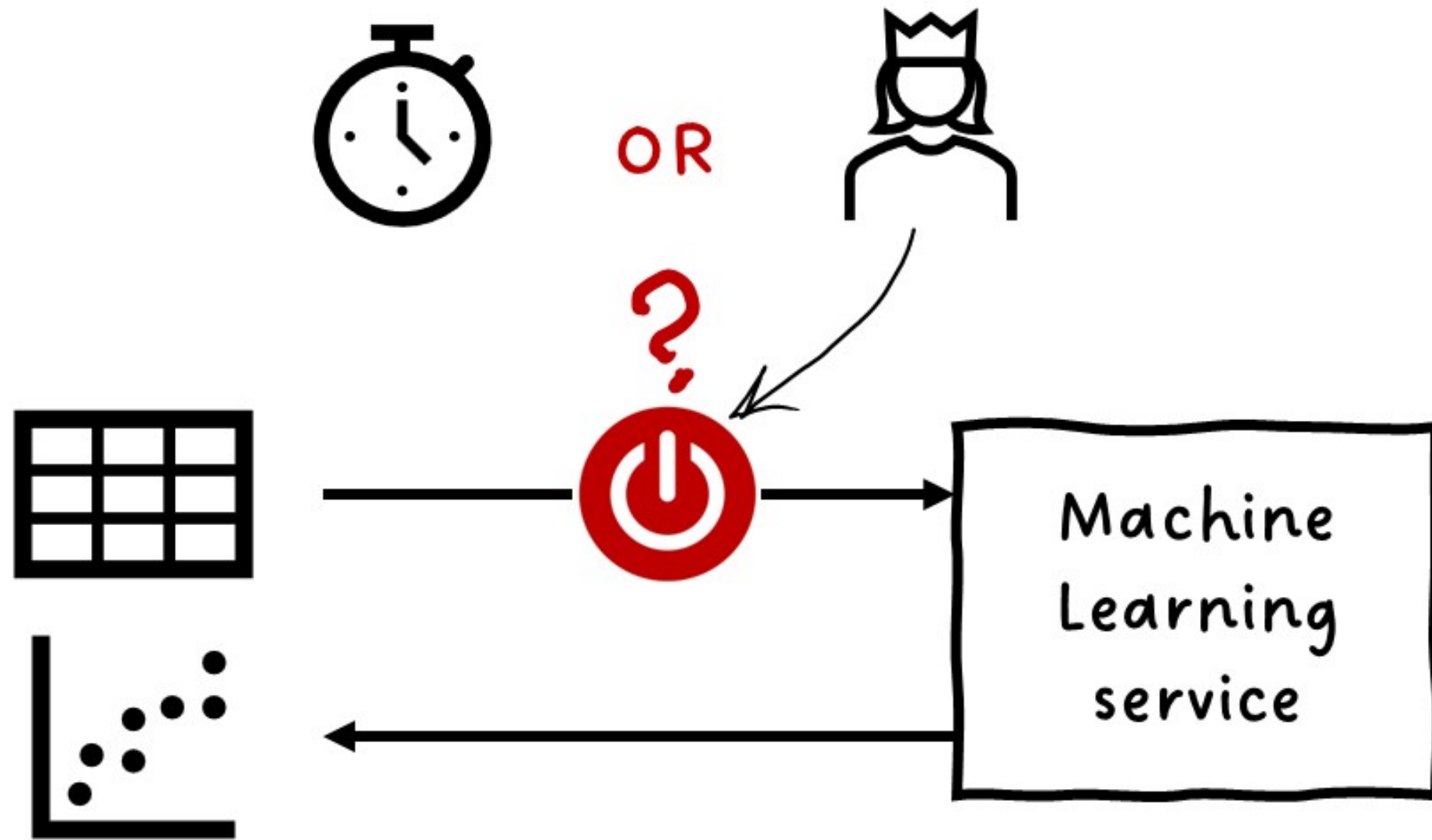
Serving and serving mode

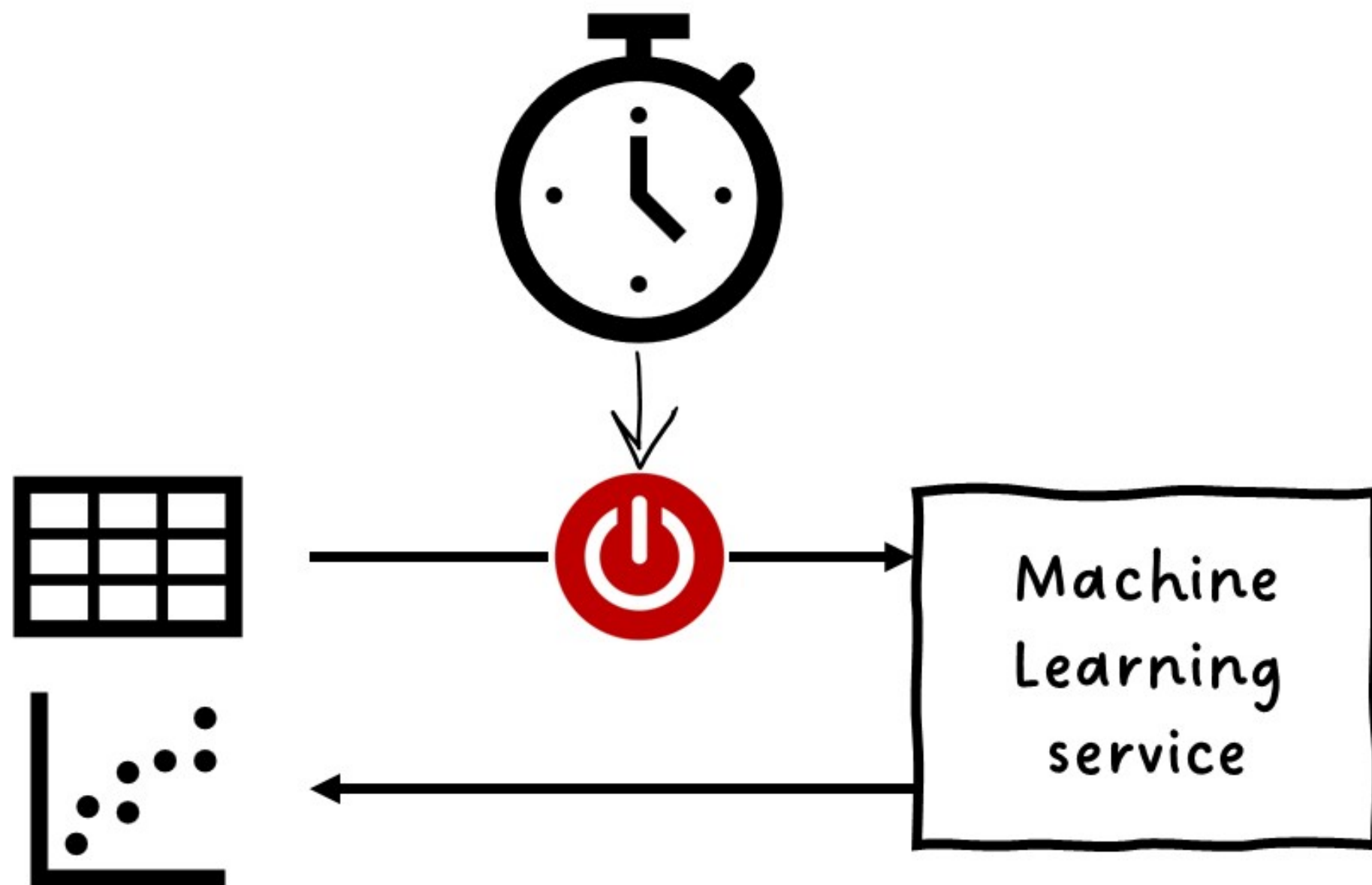
- Providing prediction service == Model serving
- Implementation of a specific type of serving == Serving mode

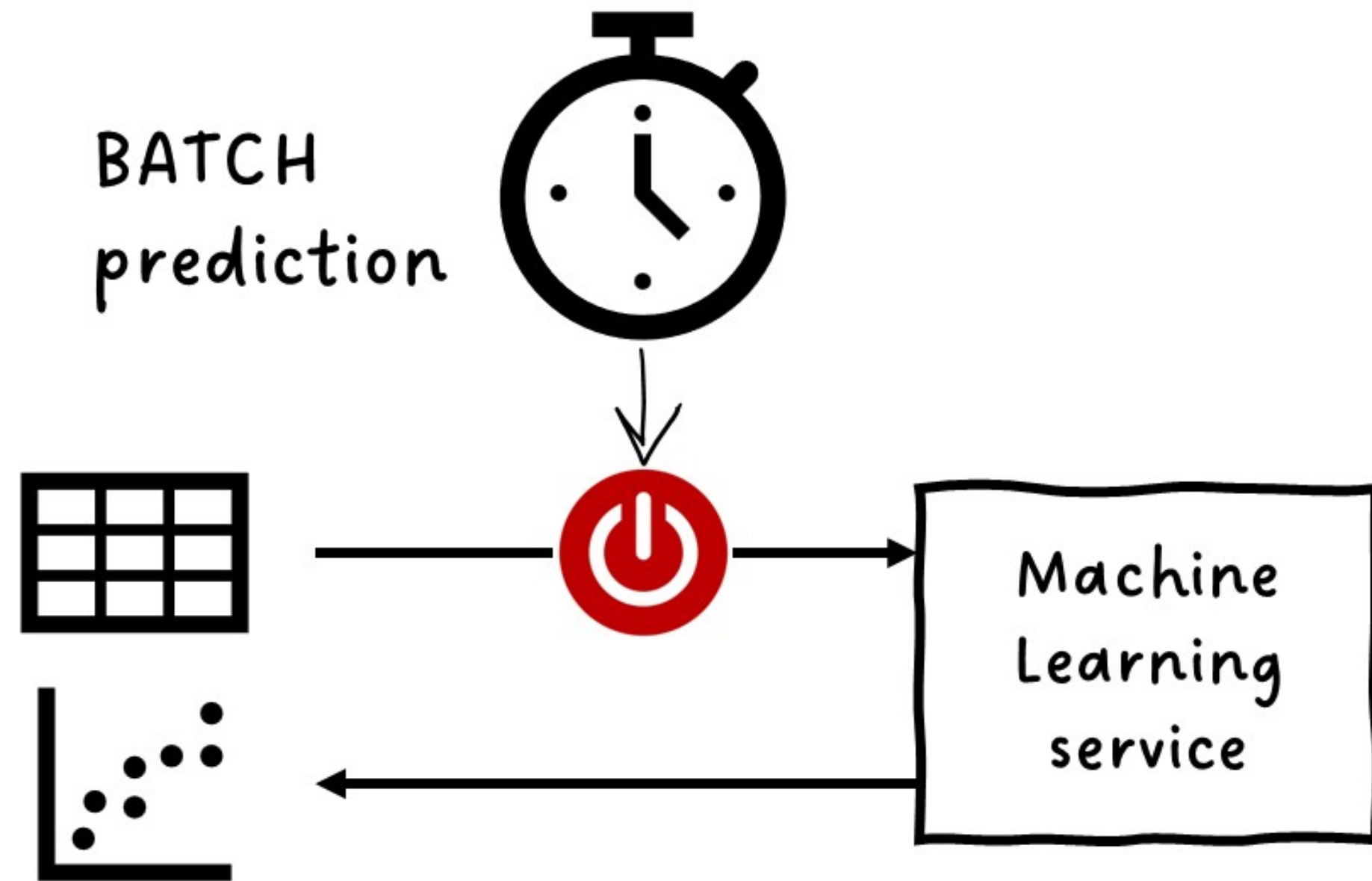
Choose carefully!

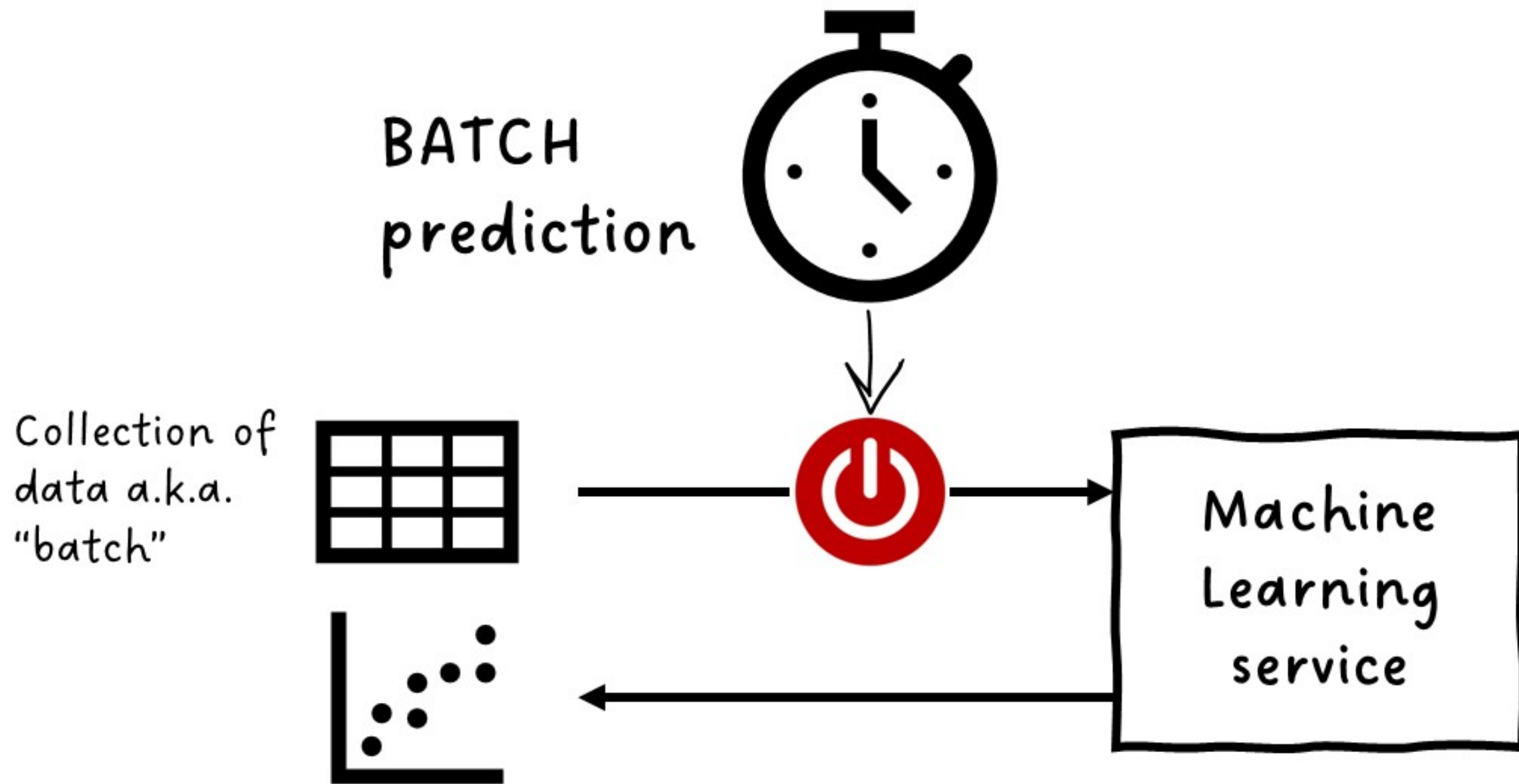


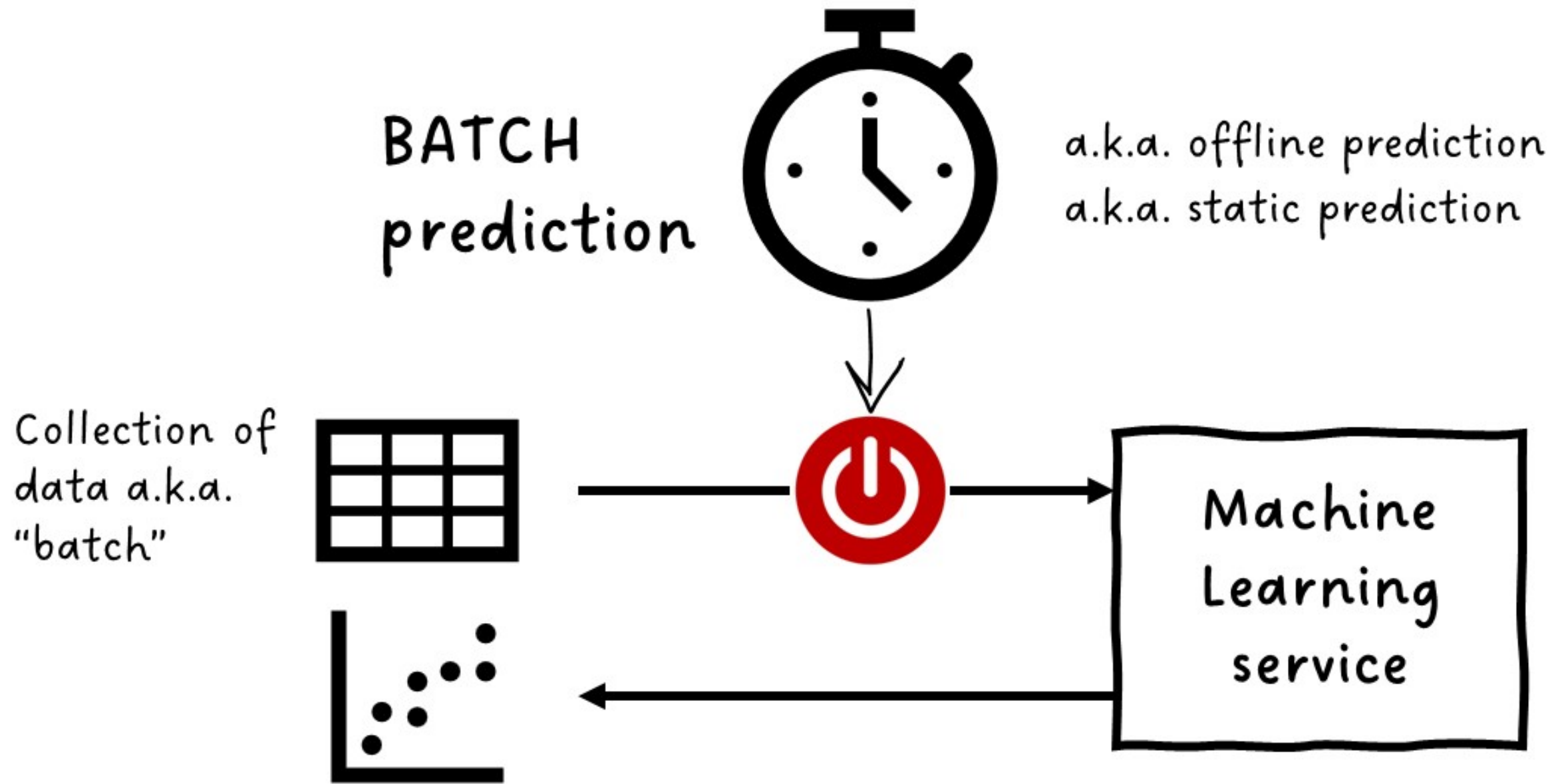






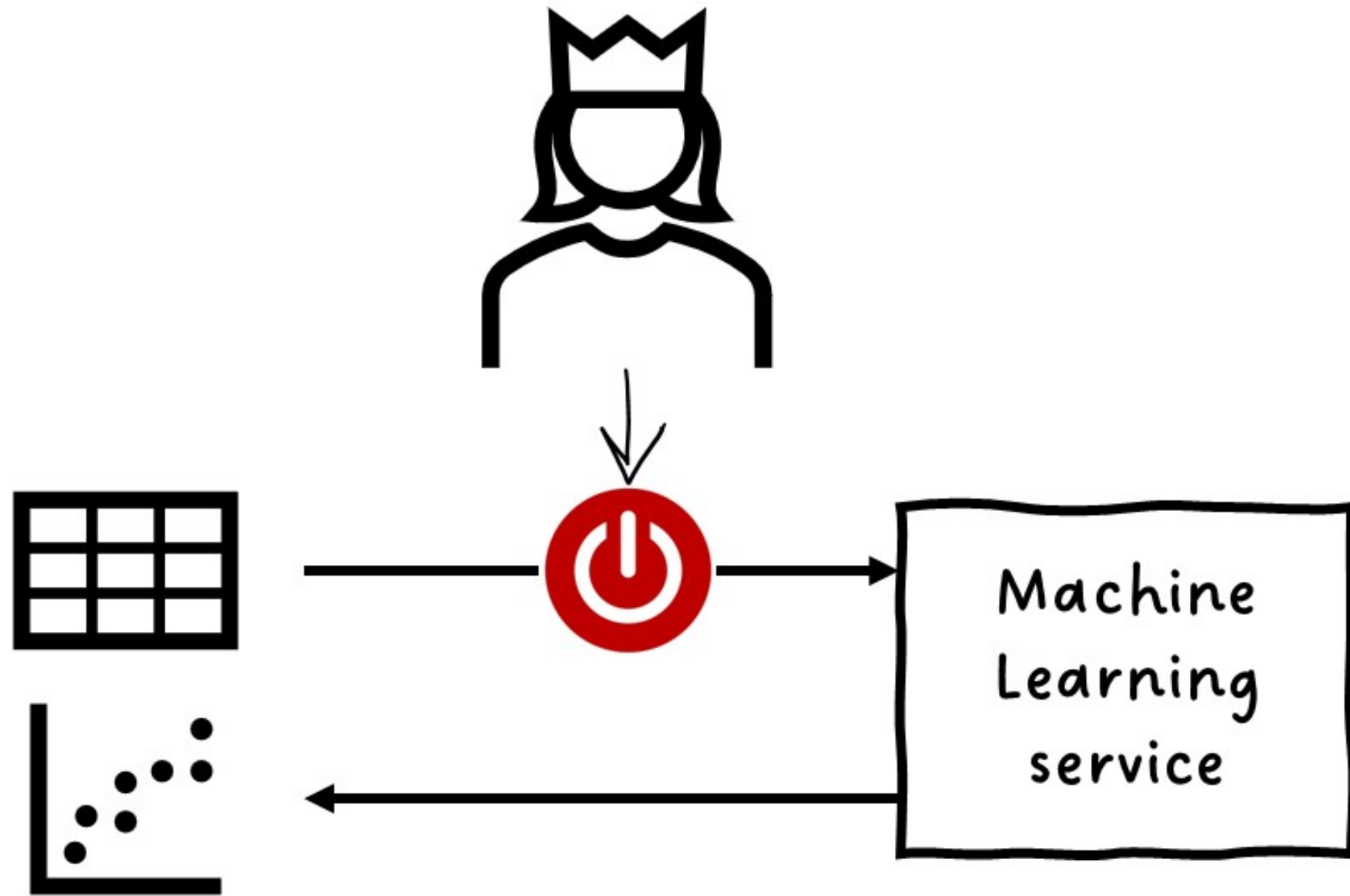




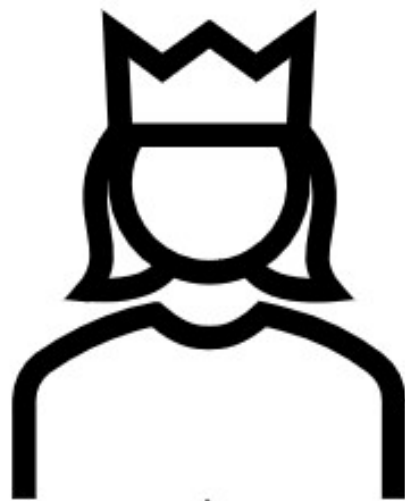


Batch prediction: Keep it simple

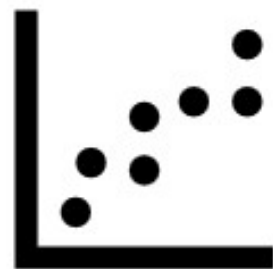
- Batch prediction is the simplest
- If use case allows it, go for it
- Good fit: monthly generation of sales forecasts



On-demand
prediction

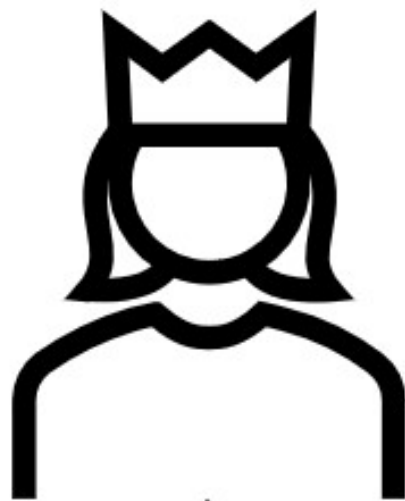


a.k.a. online prediction
a.k.a. dynamic prediction

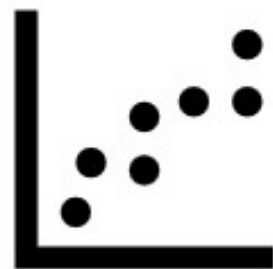


Machine
Learning
service

On-demand
prediction

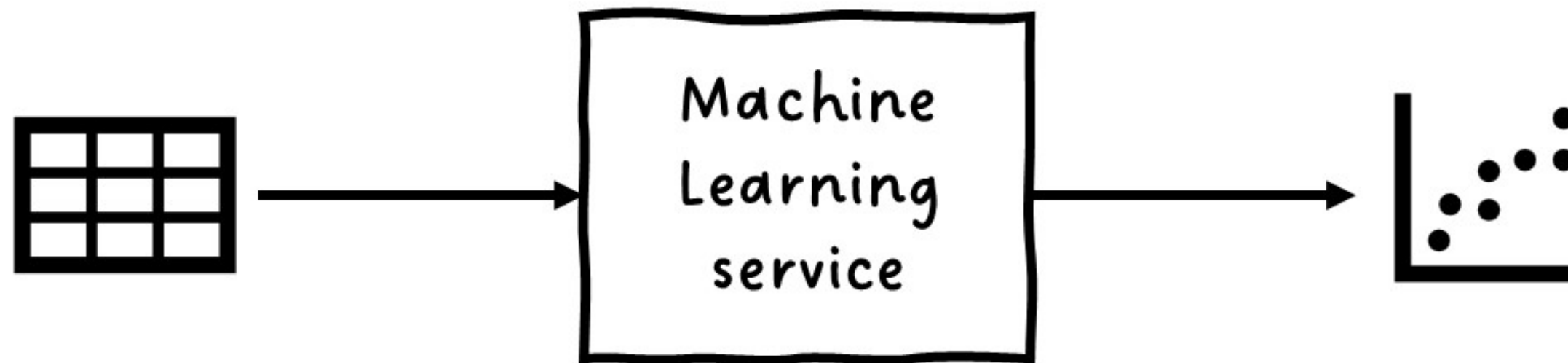


a.k.a. online prediction
a.k.a. dynamic prediction



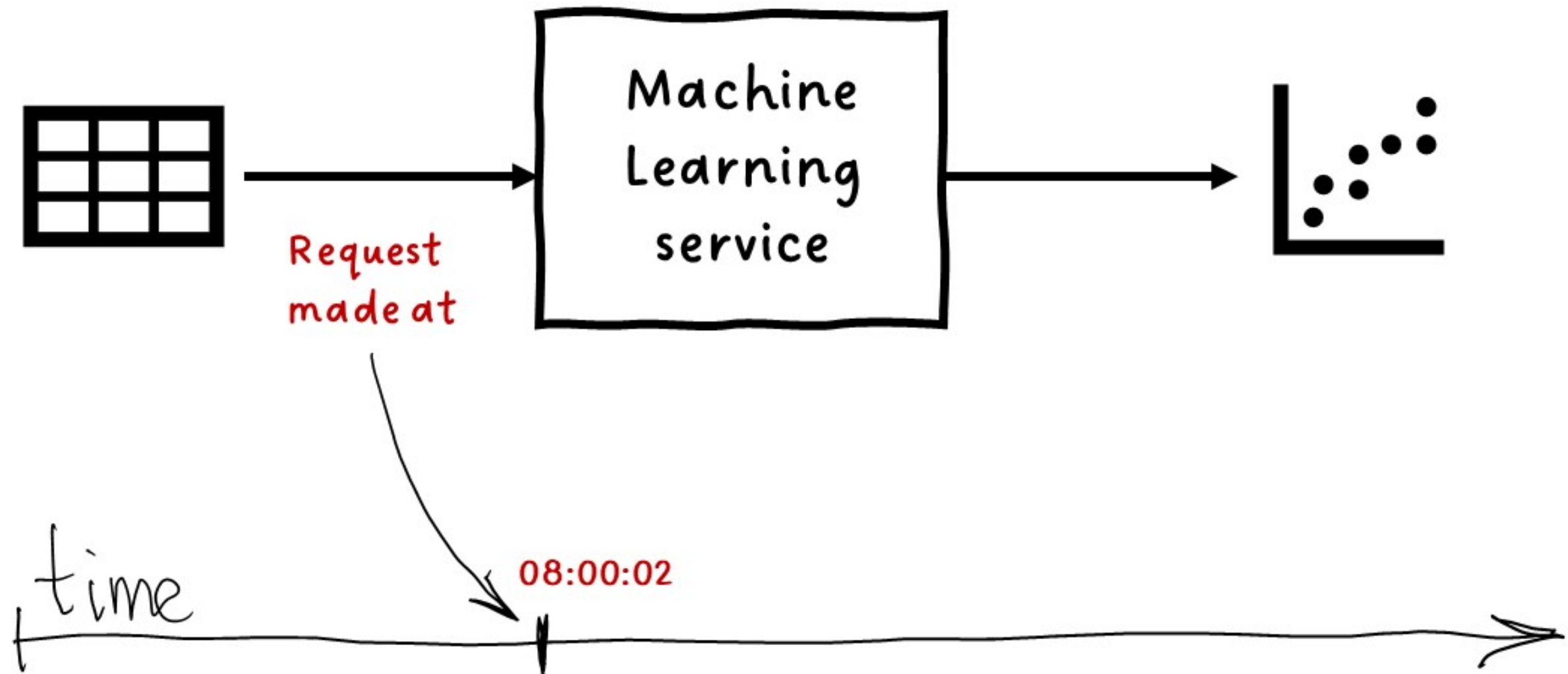
Machine
Learning
service

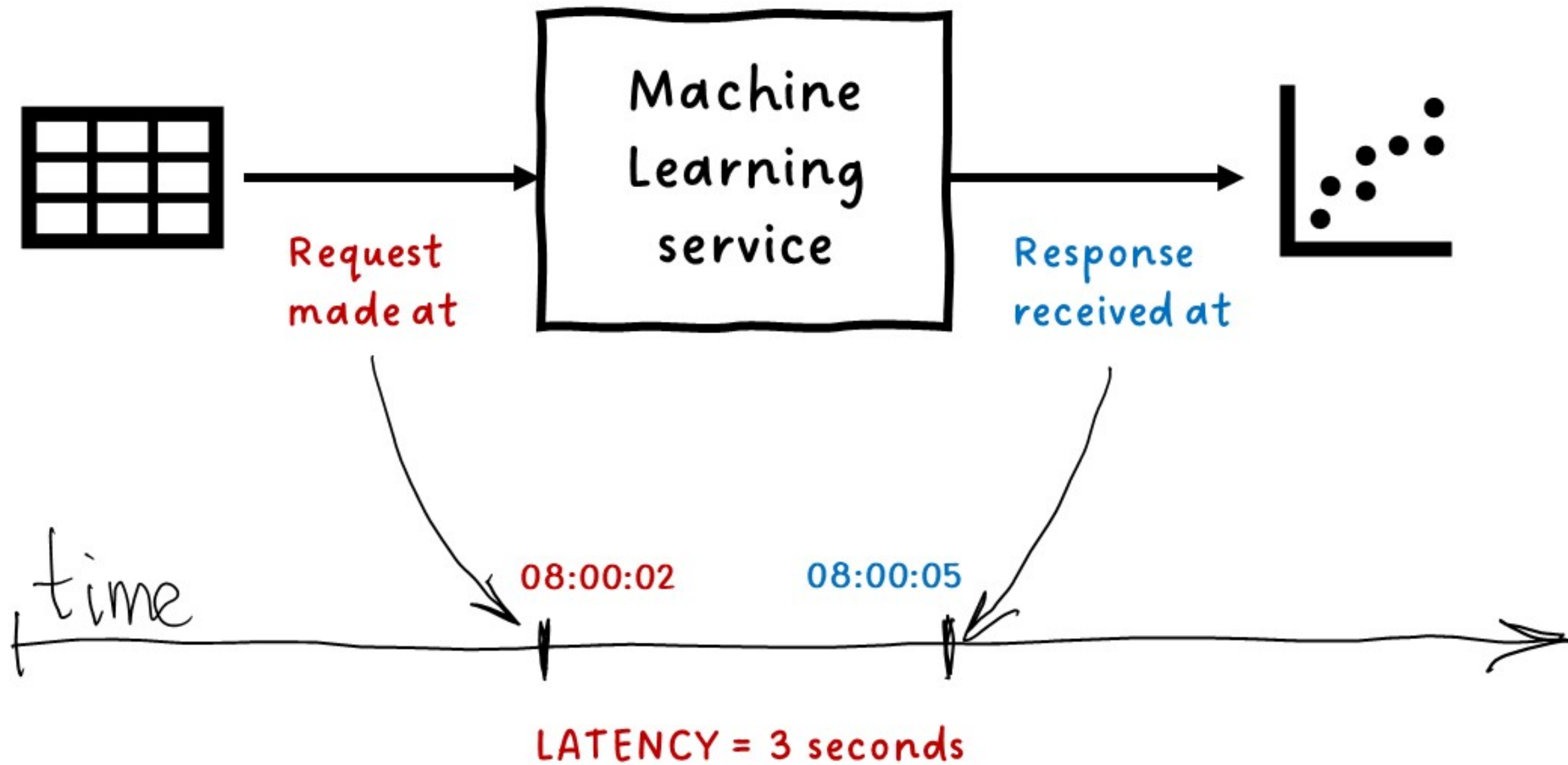
Response time
becomes
important!



time

A long horizontal arrow pointing to the right, representing the progression of time. The word "time" is written in a cursive font above the arrow's starting point.





Acceptable latency

What is acceptable?

- < 1 hour?
- < 1 minute?
- < 1 second?
- < 1 millisecond?

Near-real time prediction a.k.a. Stream processing

Acceptable latency $\approx$ X minutes

Also known as *stream processing* (requests and responses form "data streams")

Real-time prediction

Acceptable latency < 1 sec

Example:

- Credit card fraud detection
- Late prediction as good as useless

When latency is a priority

- Weaker, but faster model more valuable than a stronger, but slower one
- Models deployed to end user devices to reduce latency => "edge deployment"
 - ML-infused smartphone apps:
 - navigation apps
 - unlocking via facial recognition
 - image filters

Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING

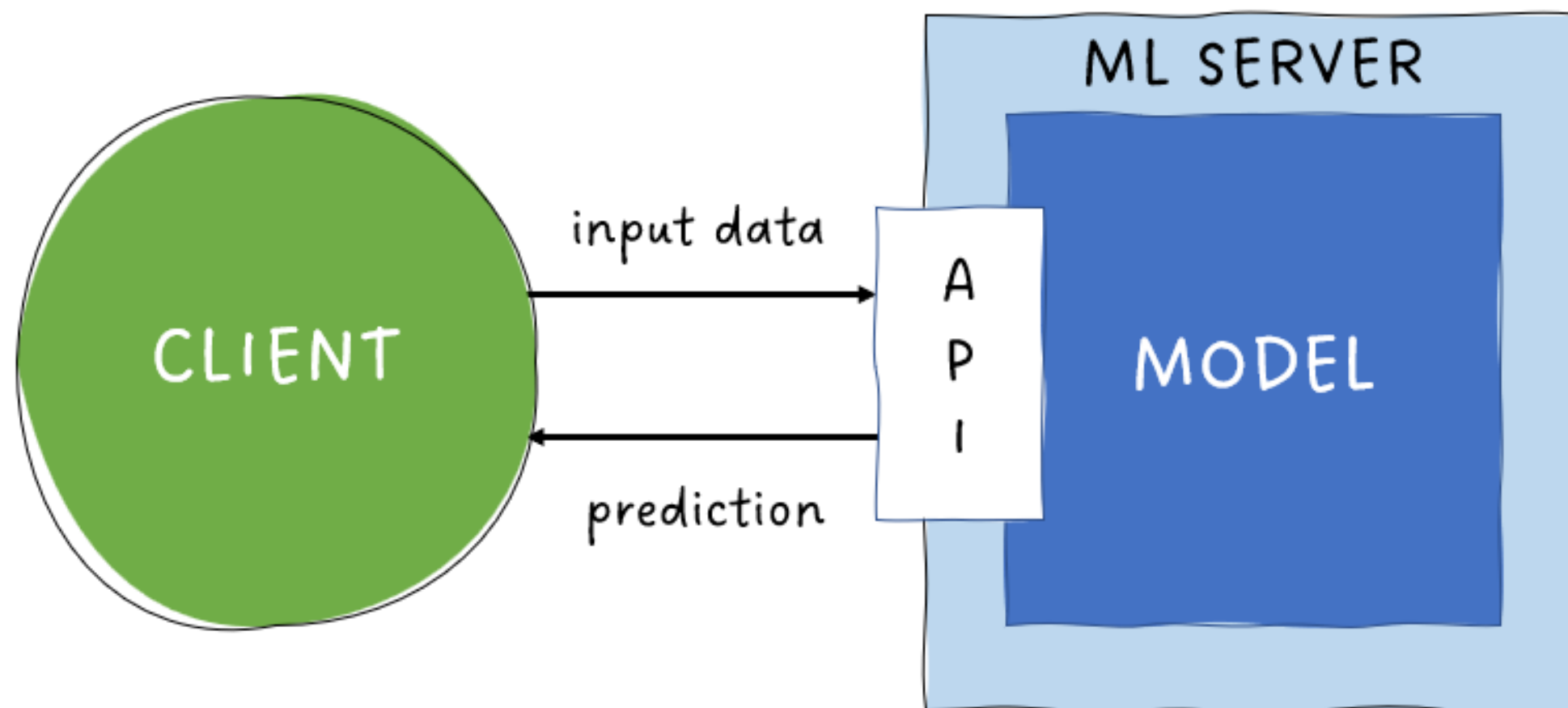
Building the API

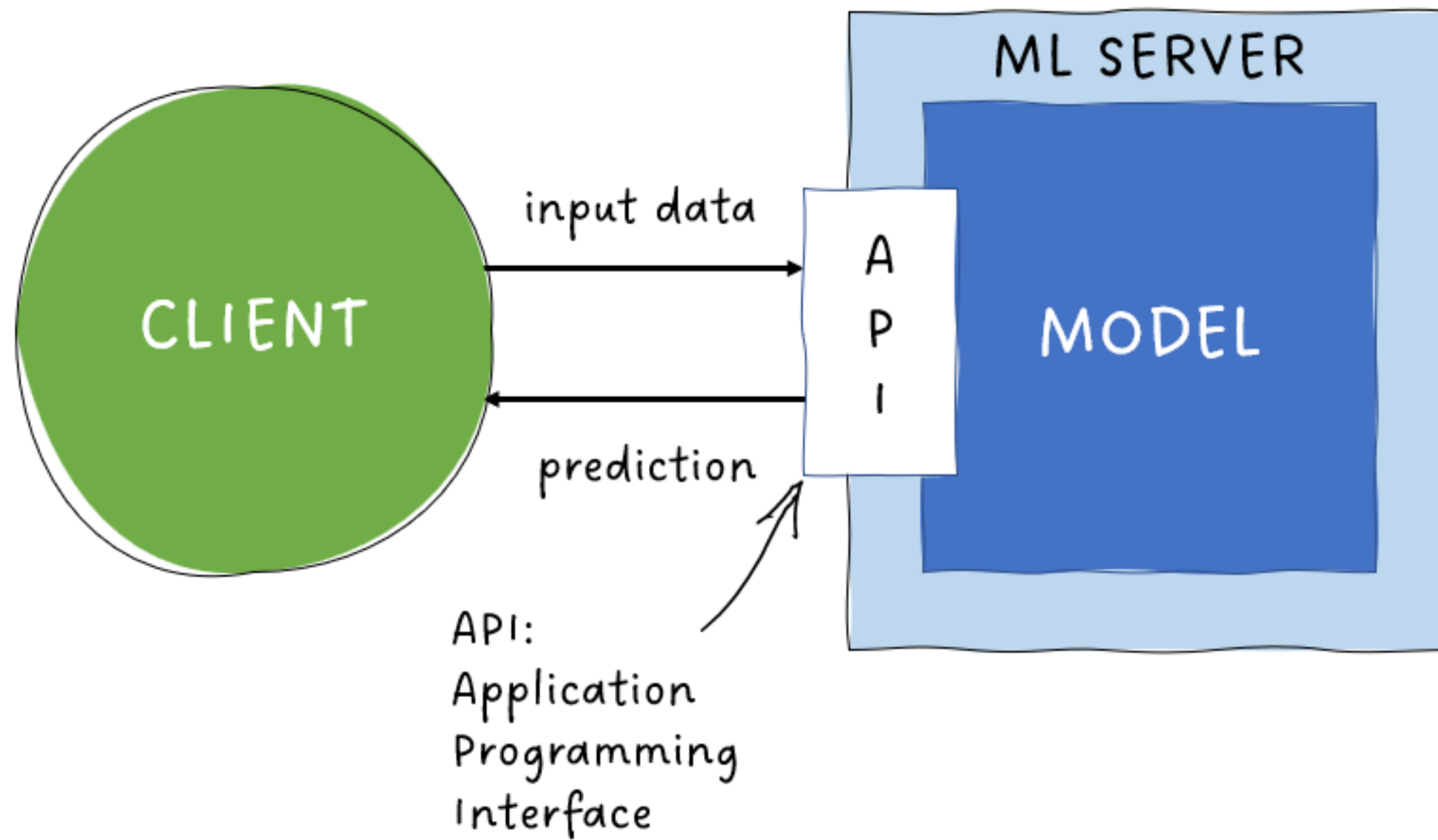
MLOPS DEPLOYMENT AND LIFE CYCLING

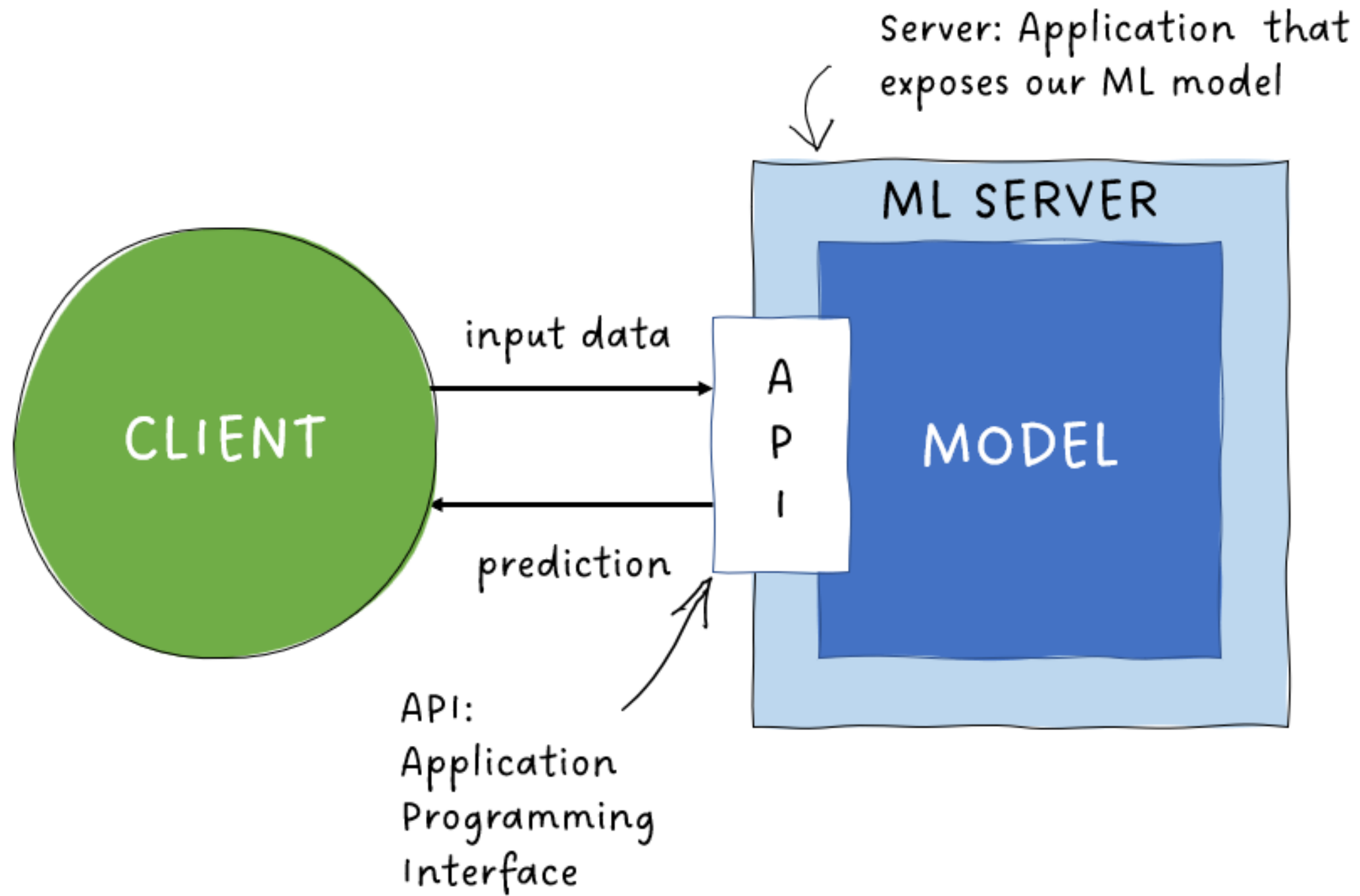


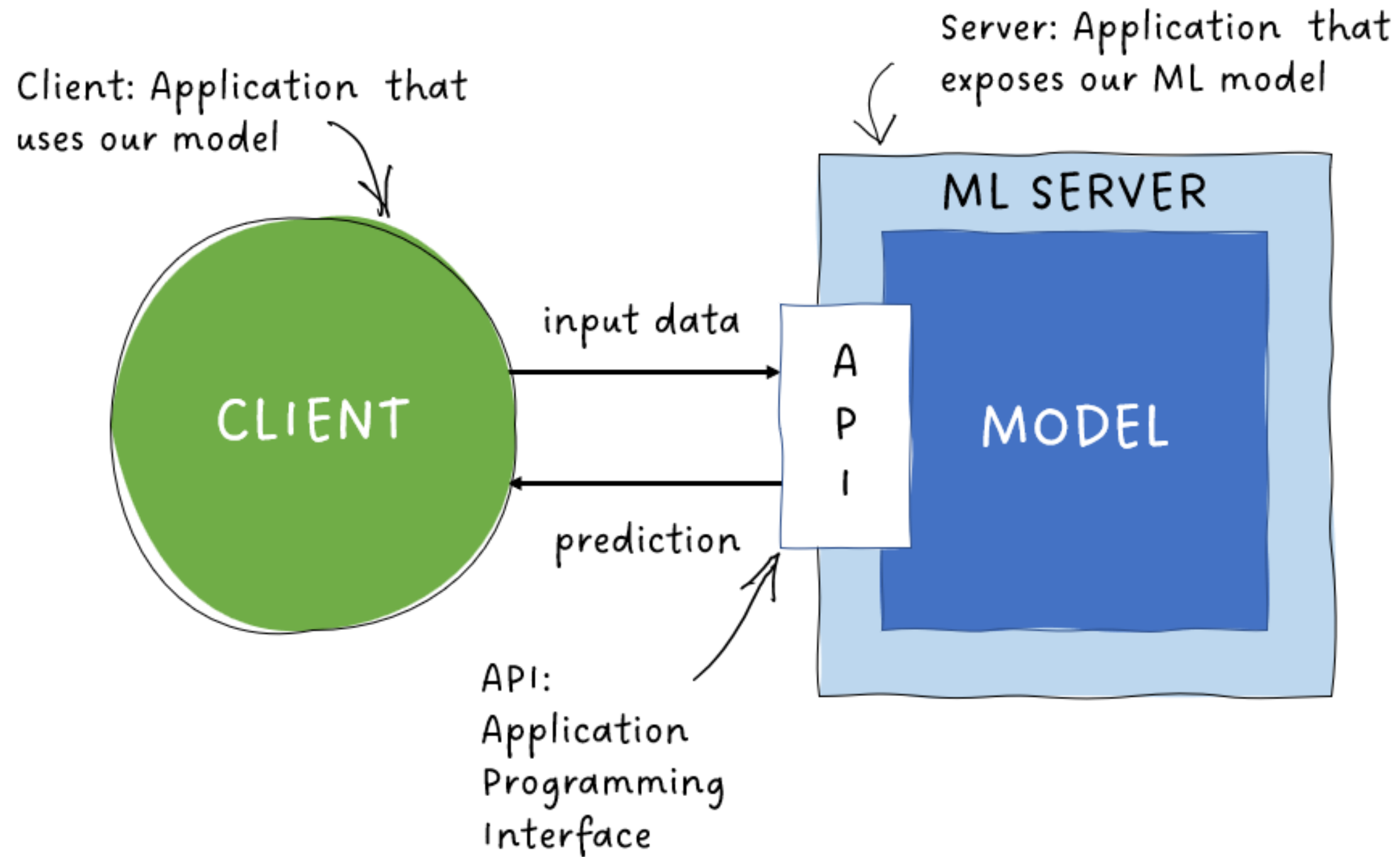
Nemanja Radojkovic

Senior Machine Learning Engineer









Service to the client

Running procedures on our server

Example: Sentiment analysis API for Amazon
Comprehend



Access remote databases

Example: Twitter API, New York Times API



Twitter API

API architectures

1. **REST**: REpresentational State Transfer
2. **RPC**: Remote Procedure Call
3. **SOAP**: Simple Object Access Protocol

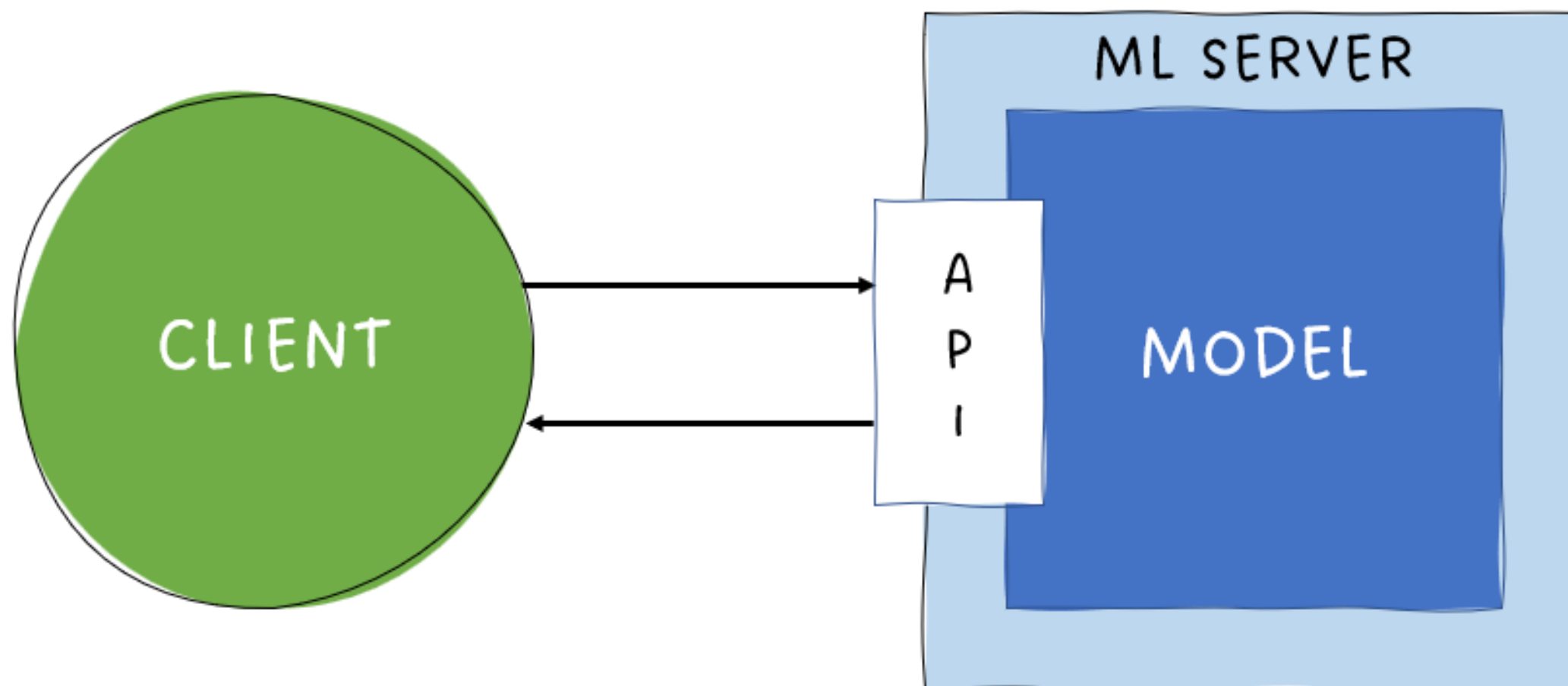
In this course: Architecture-agnostic view

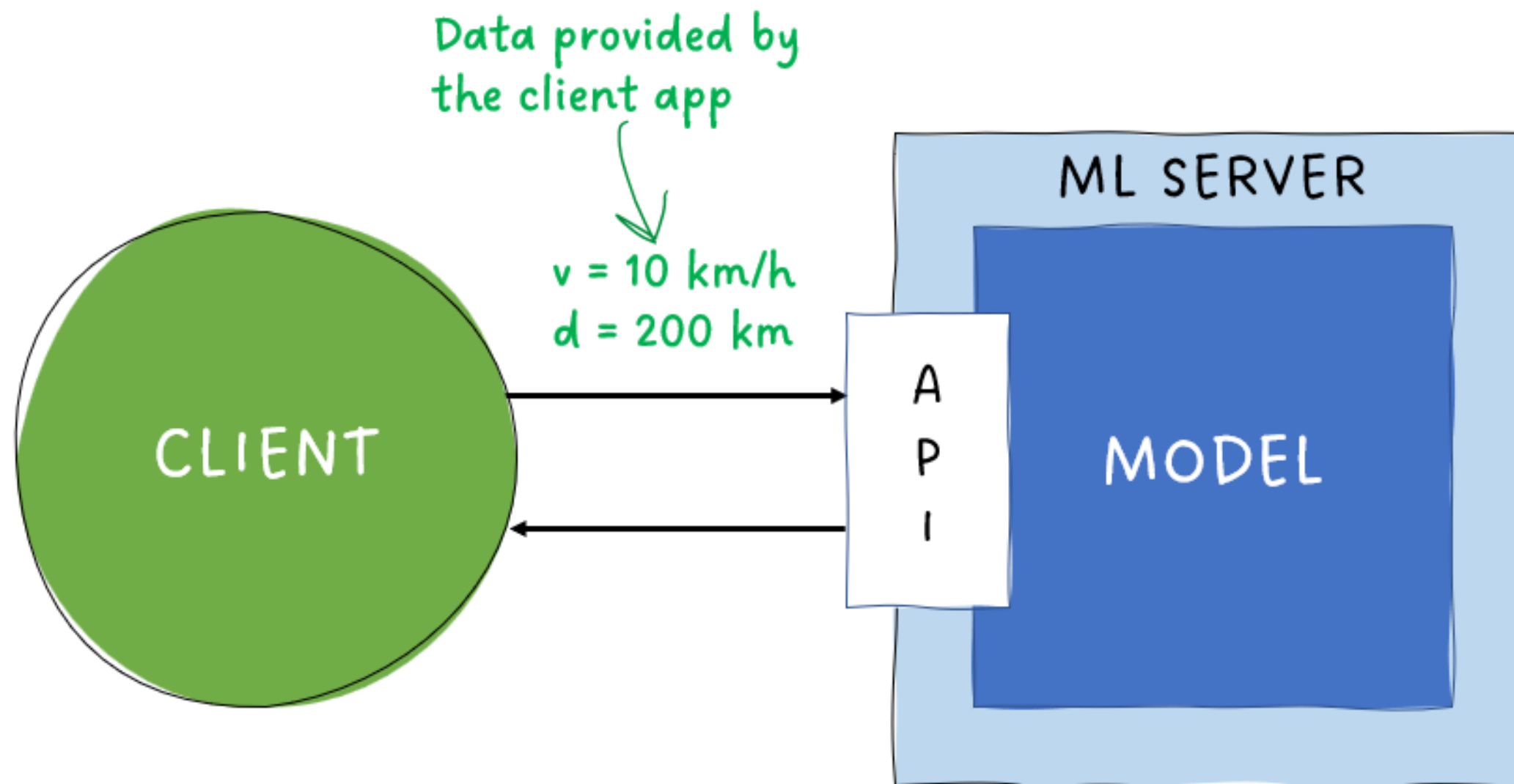
Example: Navigation app

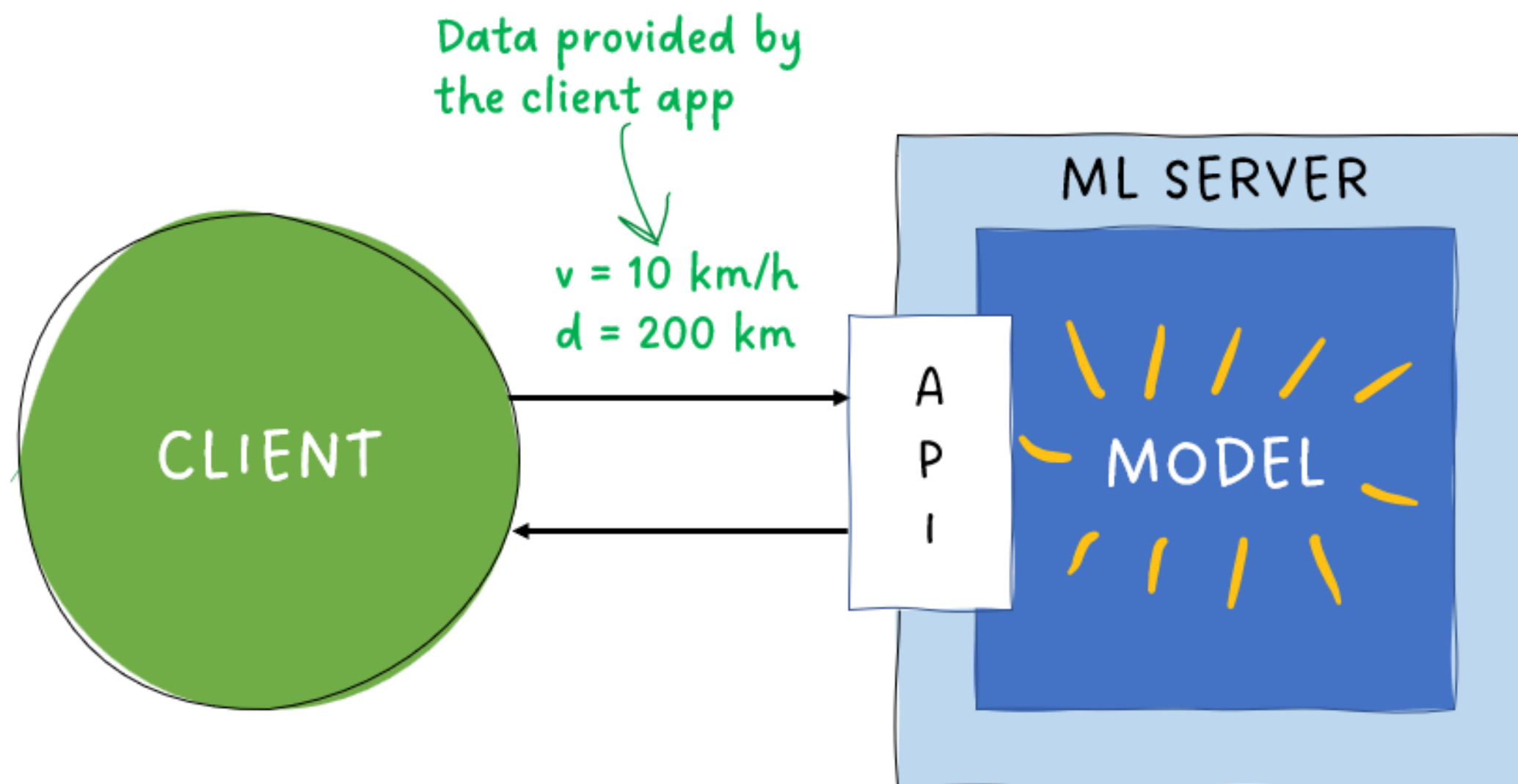


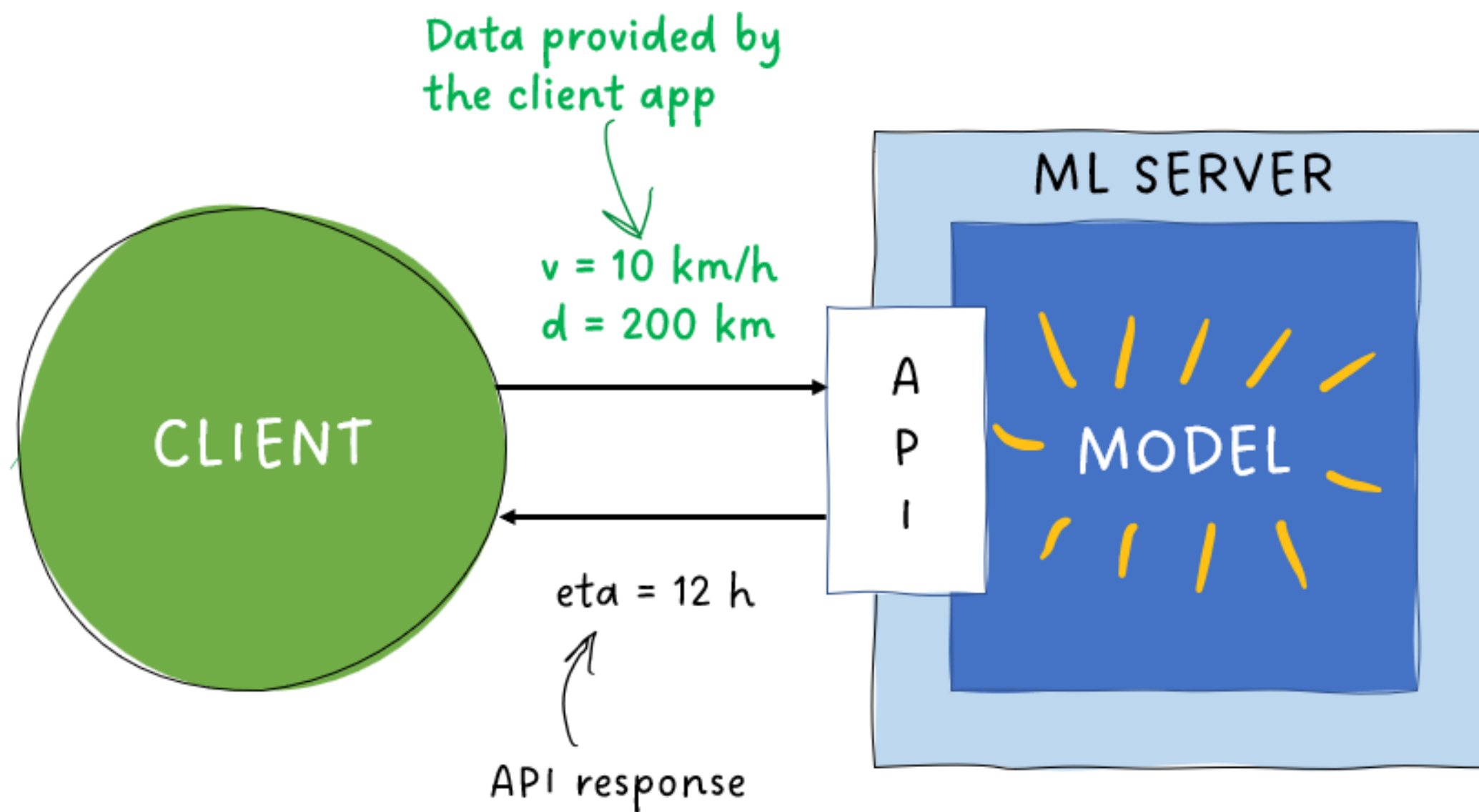
API purpose: Enable client apps to calculate the estimated time of arrival

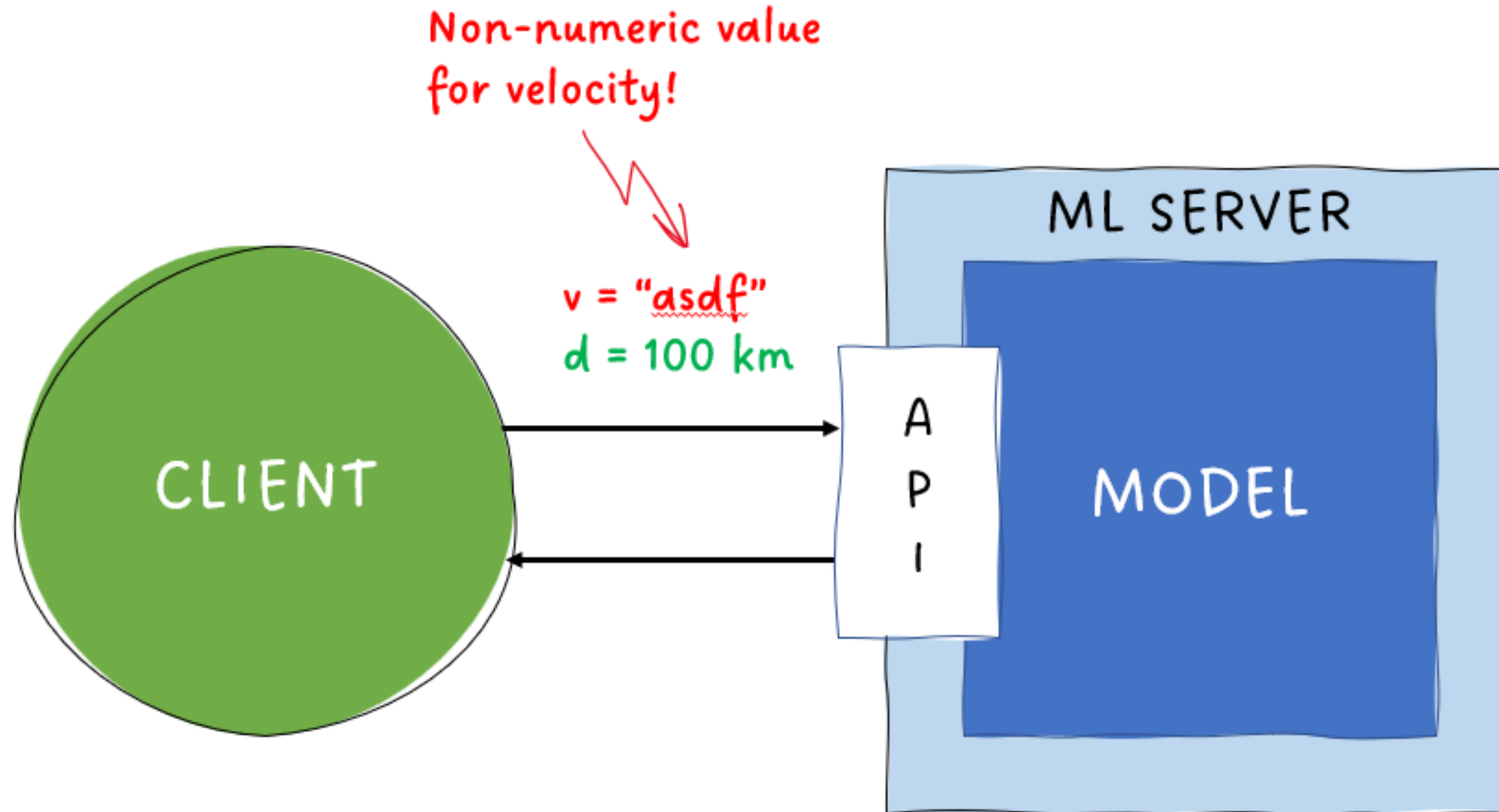
¹ Image: Pixabay, <https://www.stockvault.net/user/profile/161904>

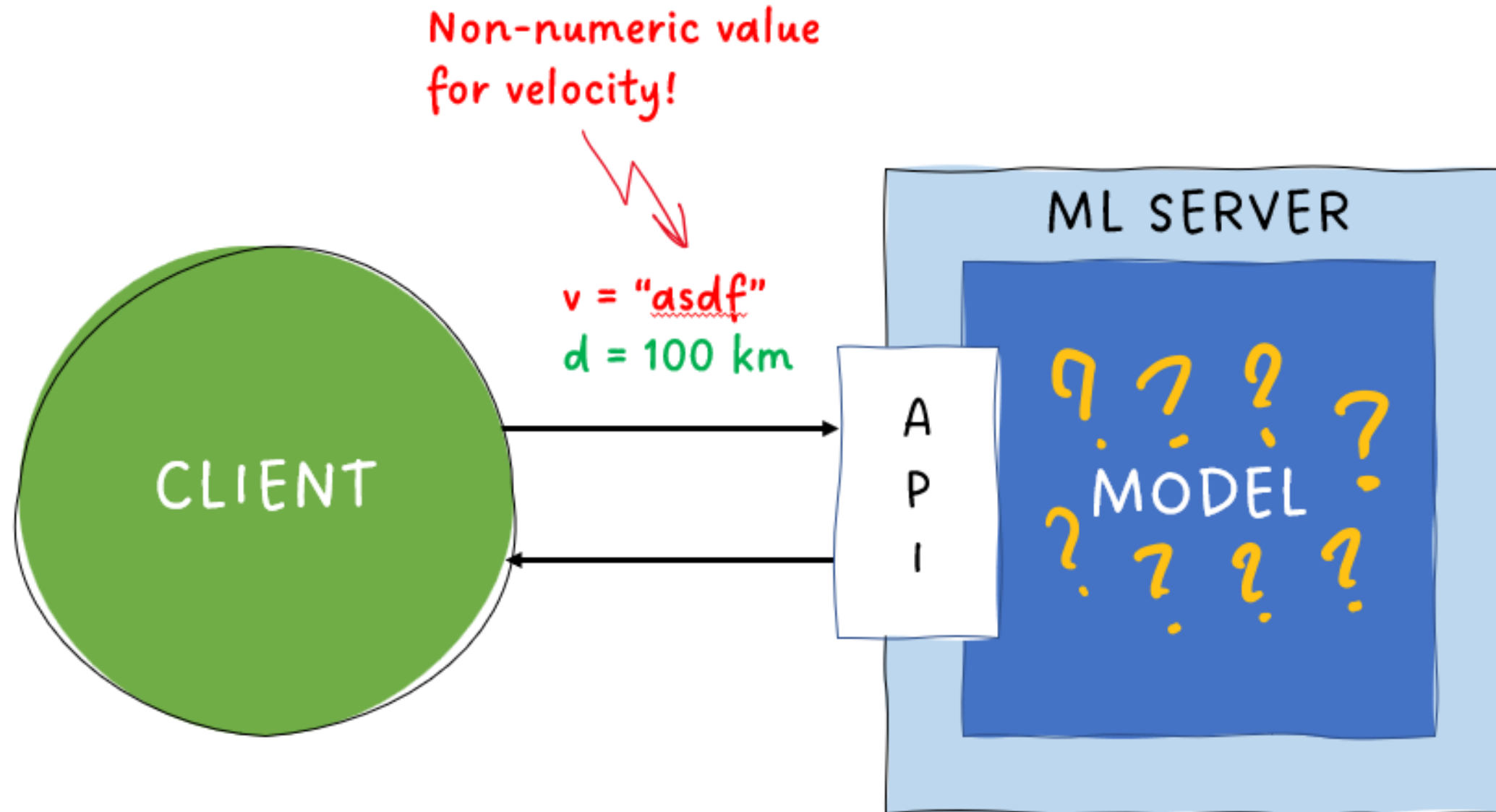


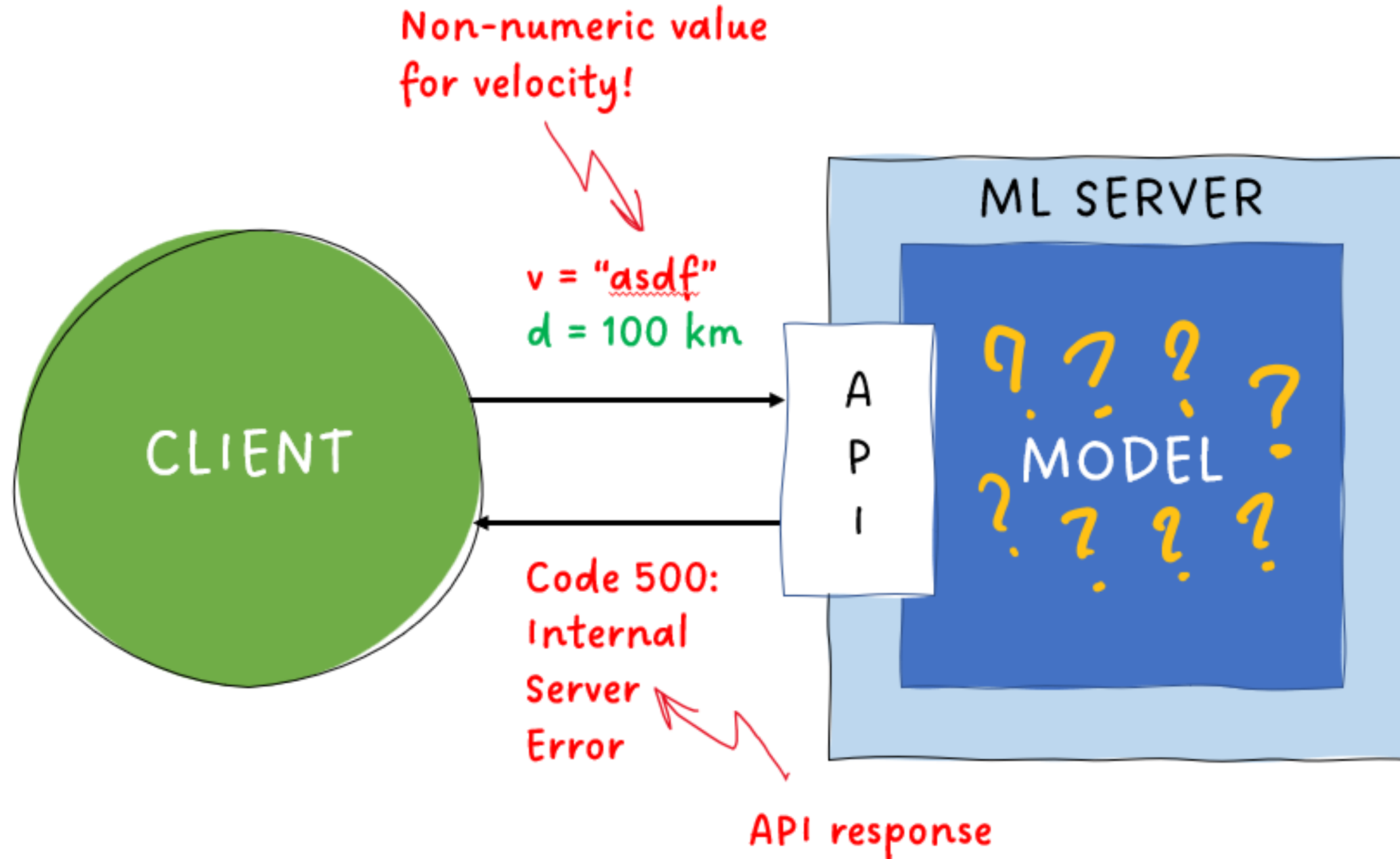


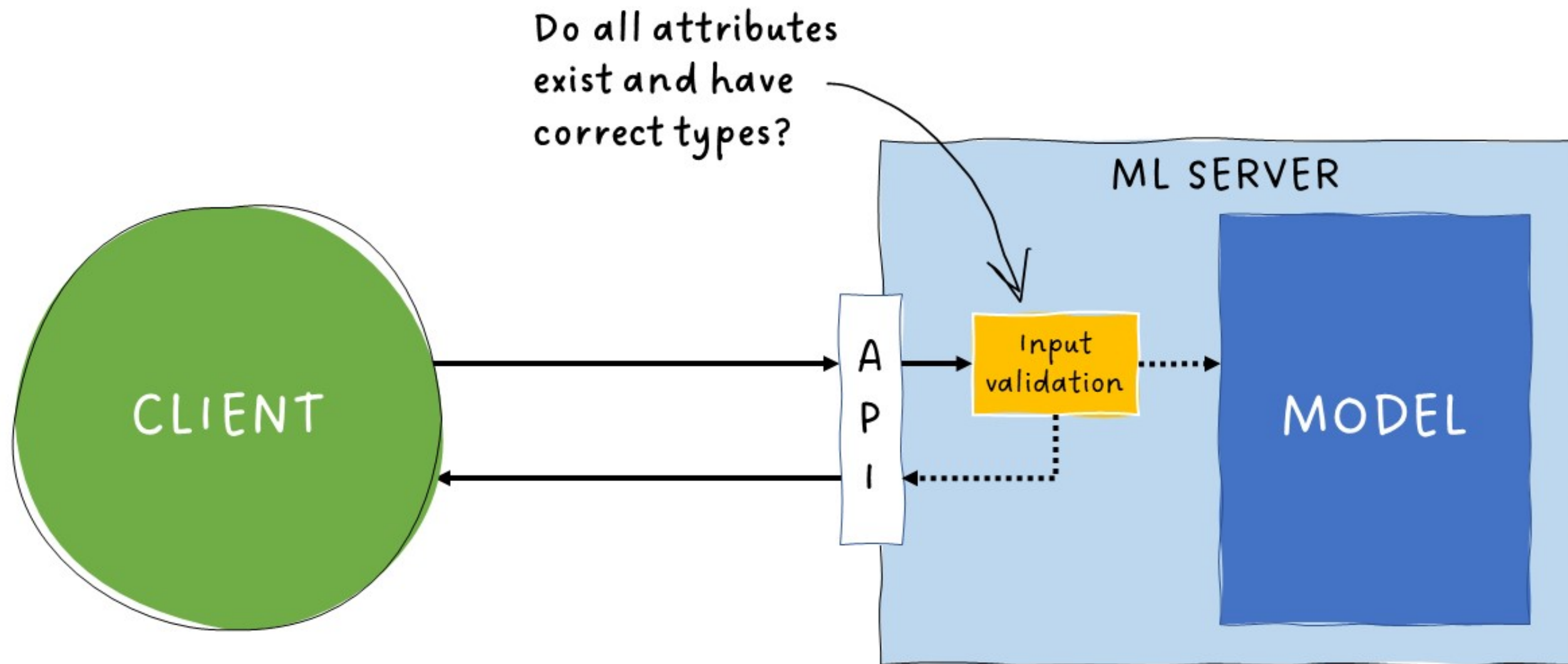


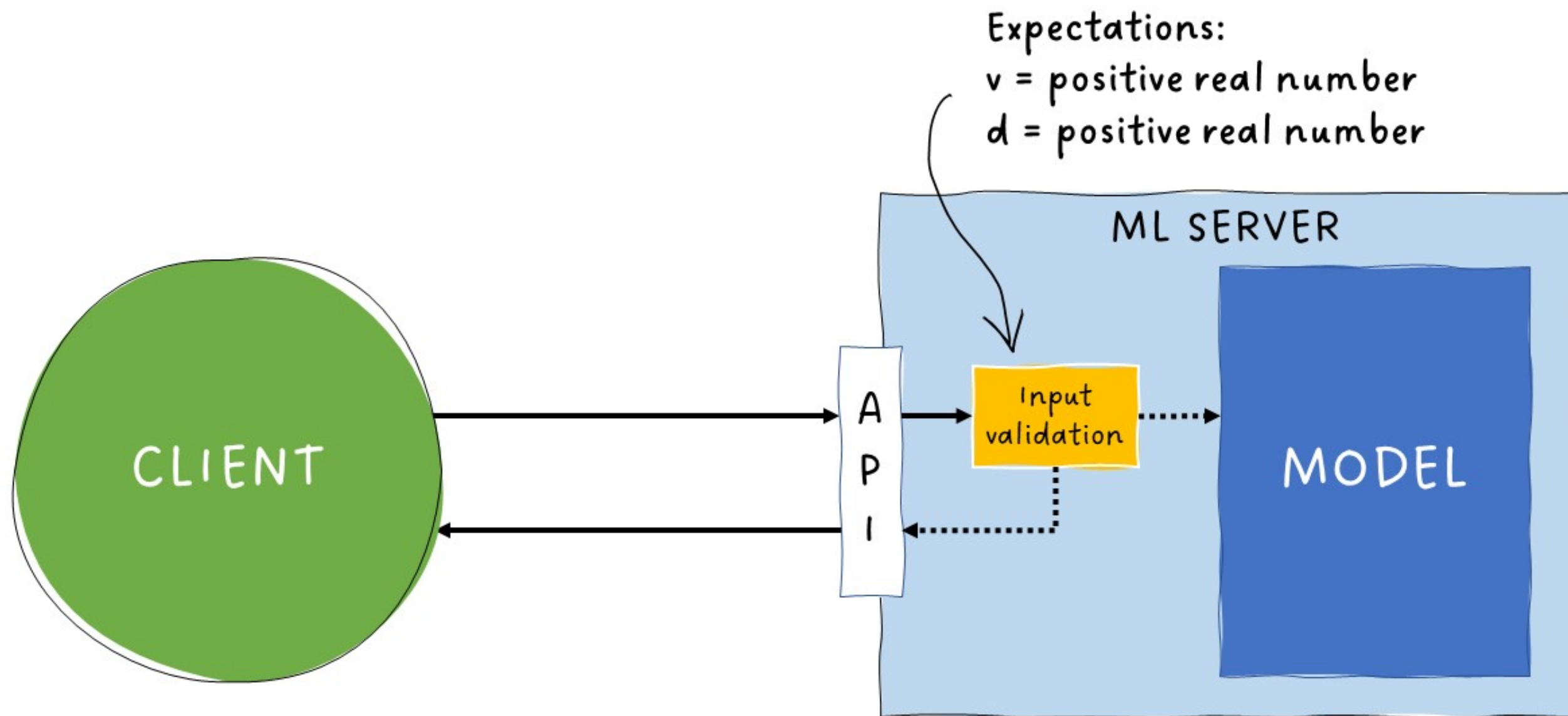


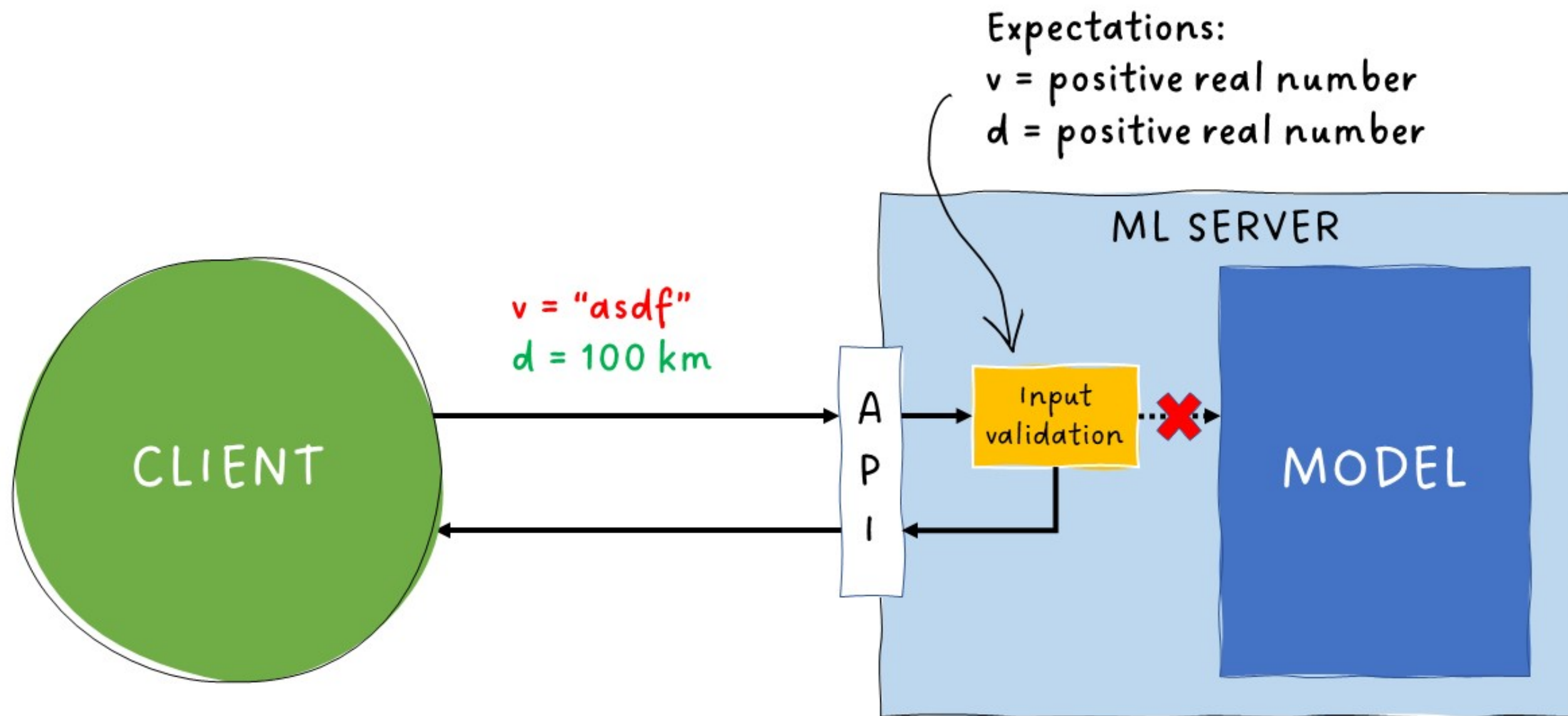


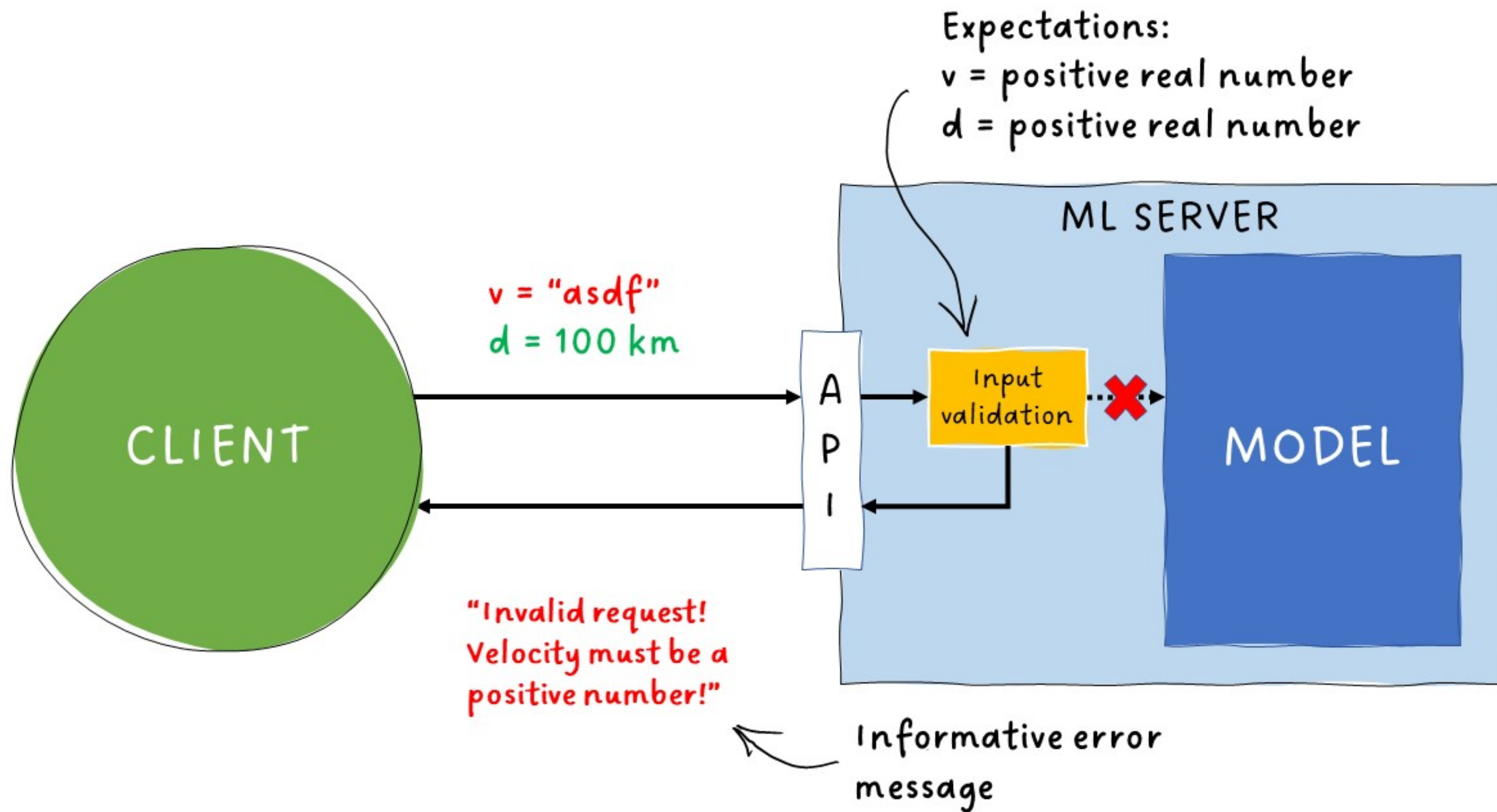


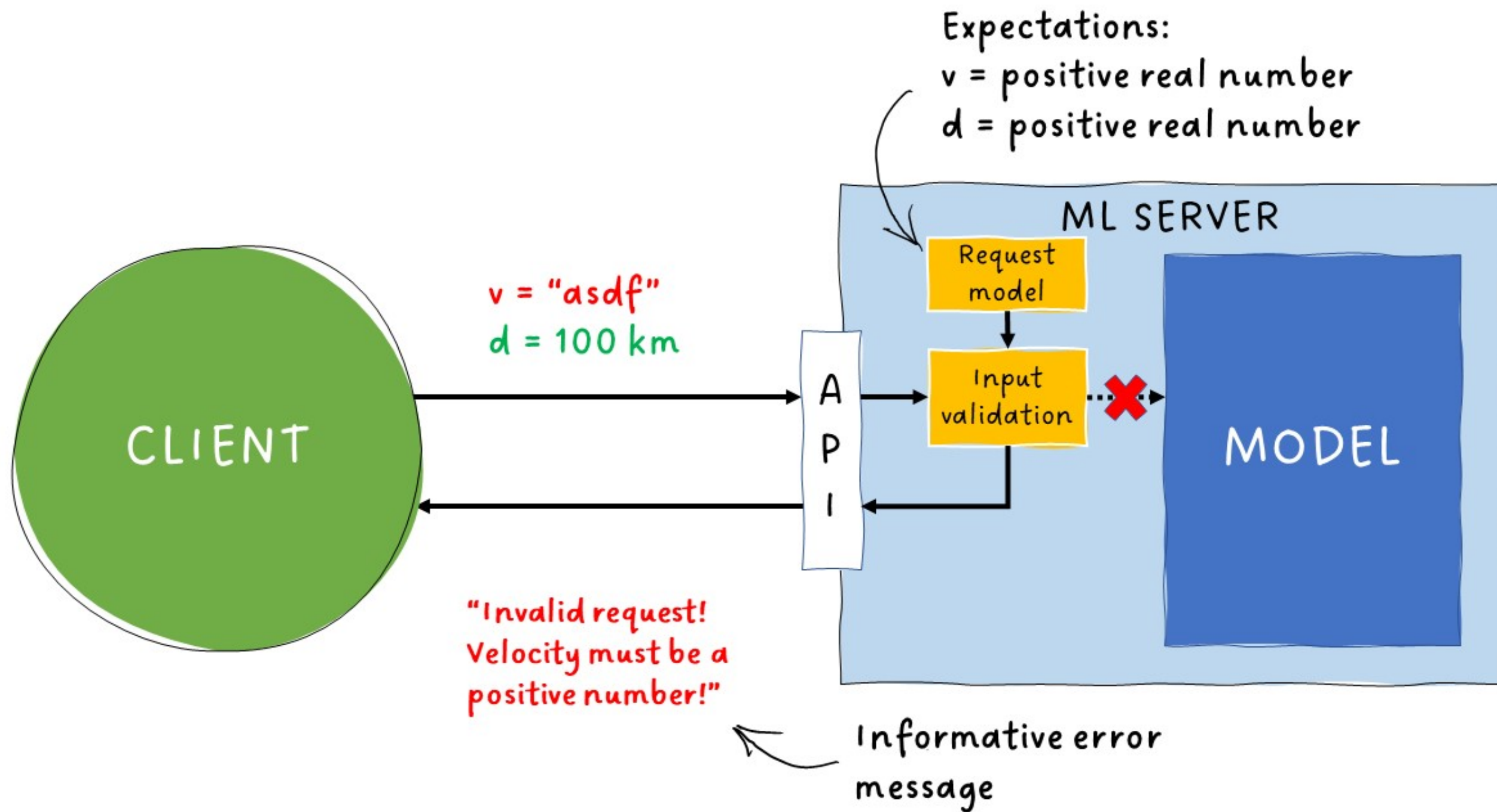


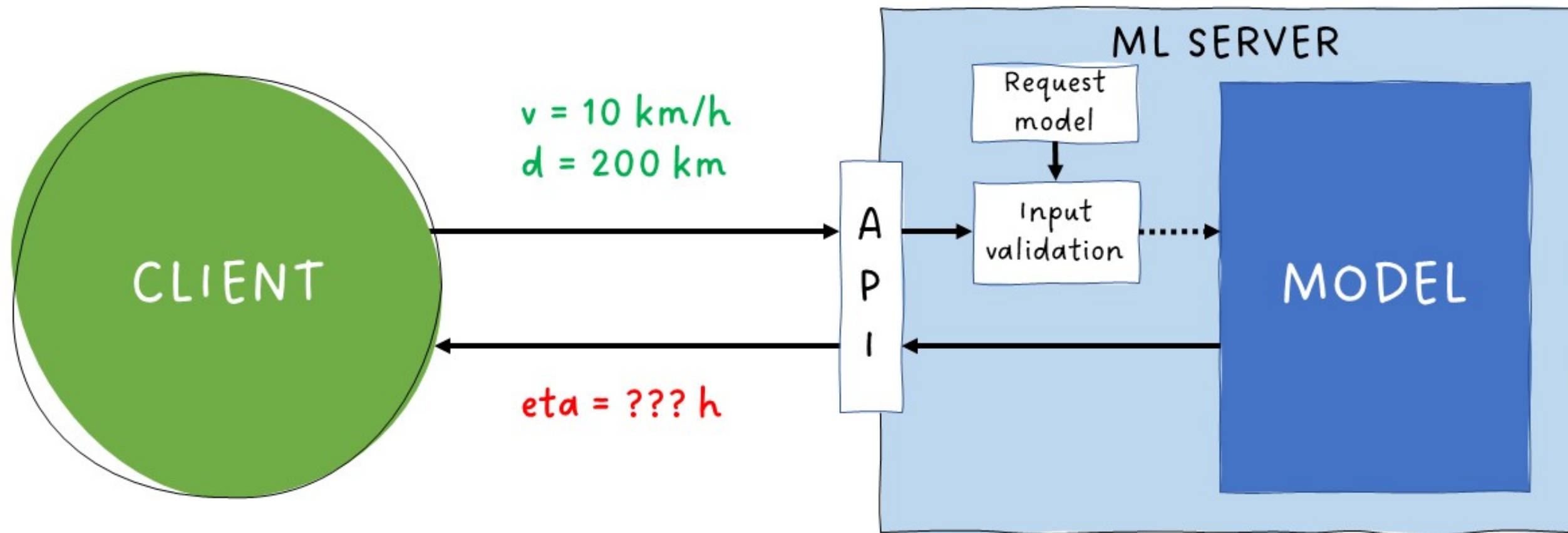


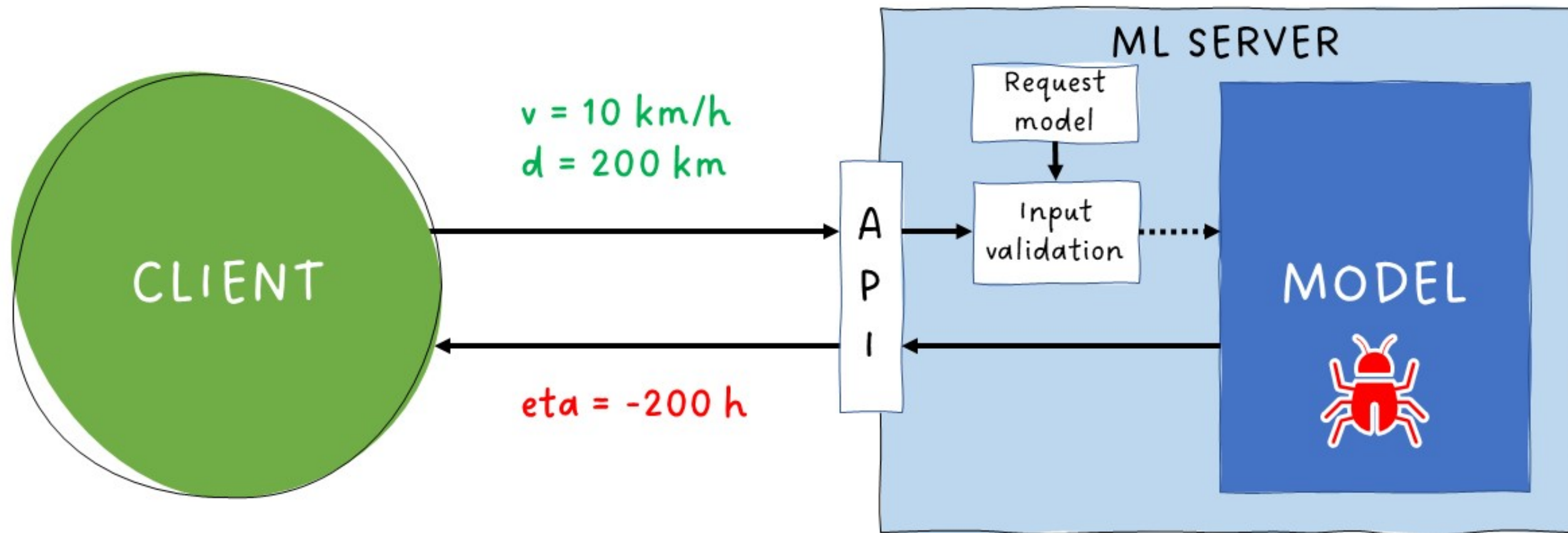


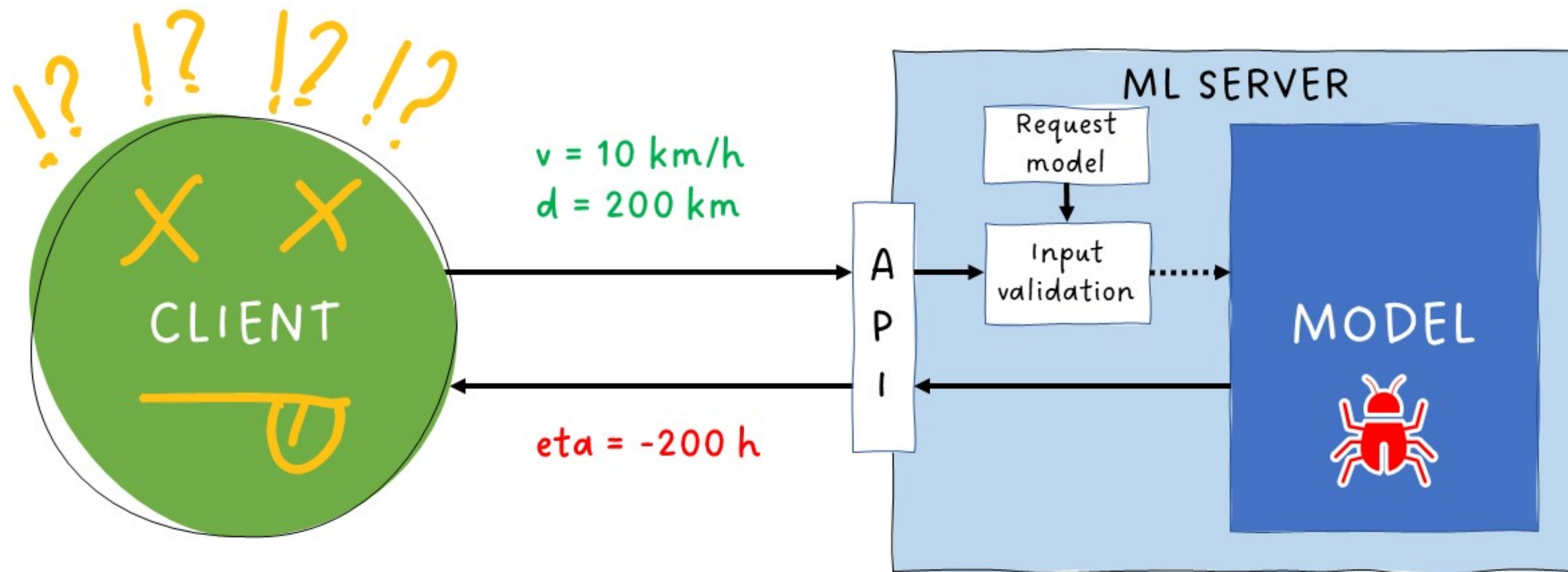


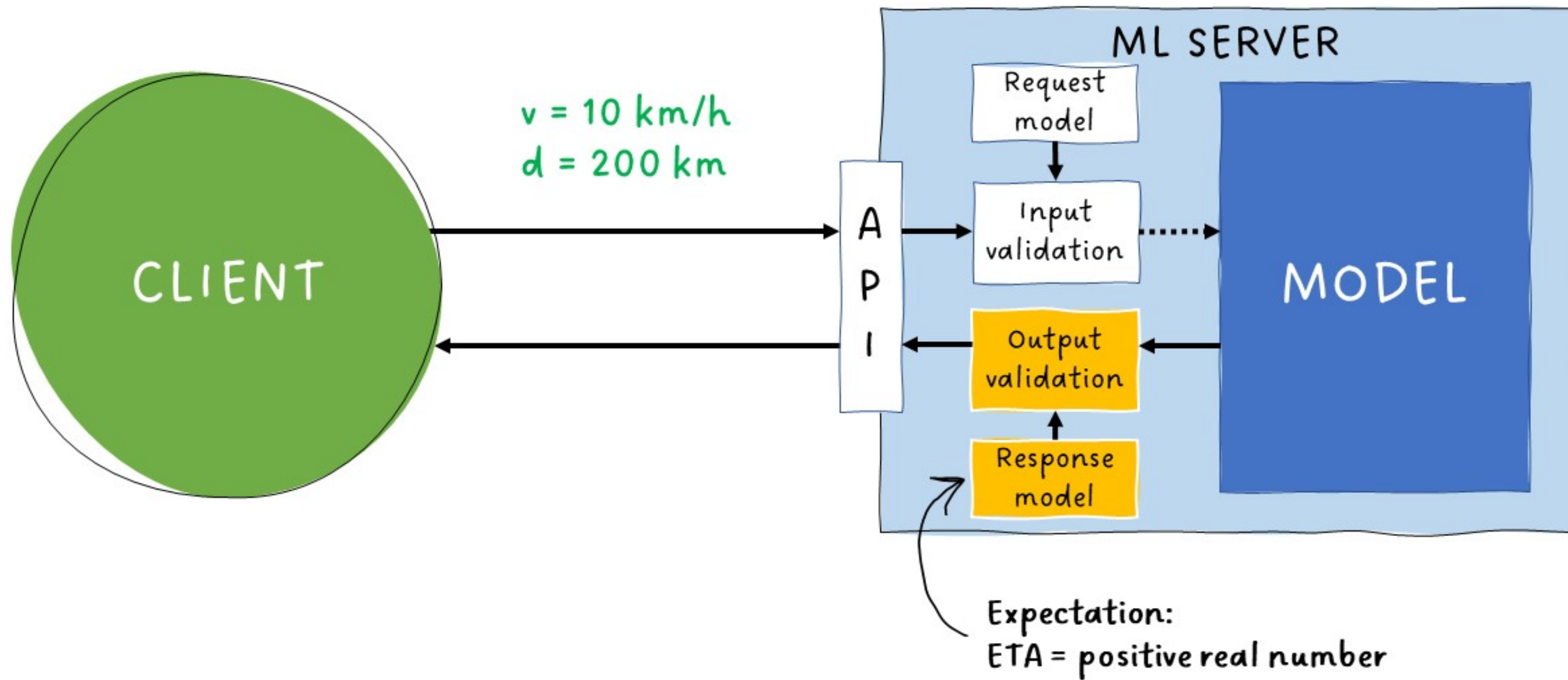


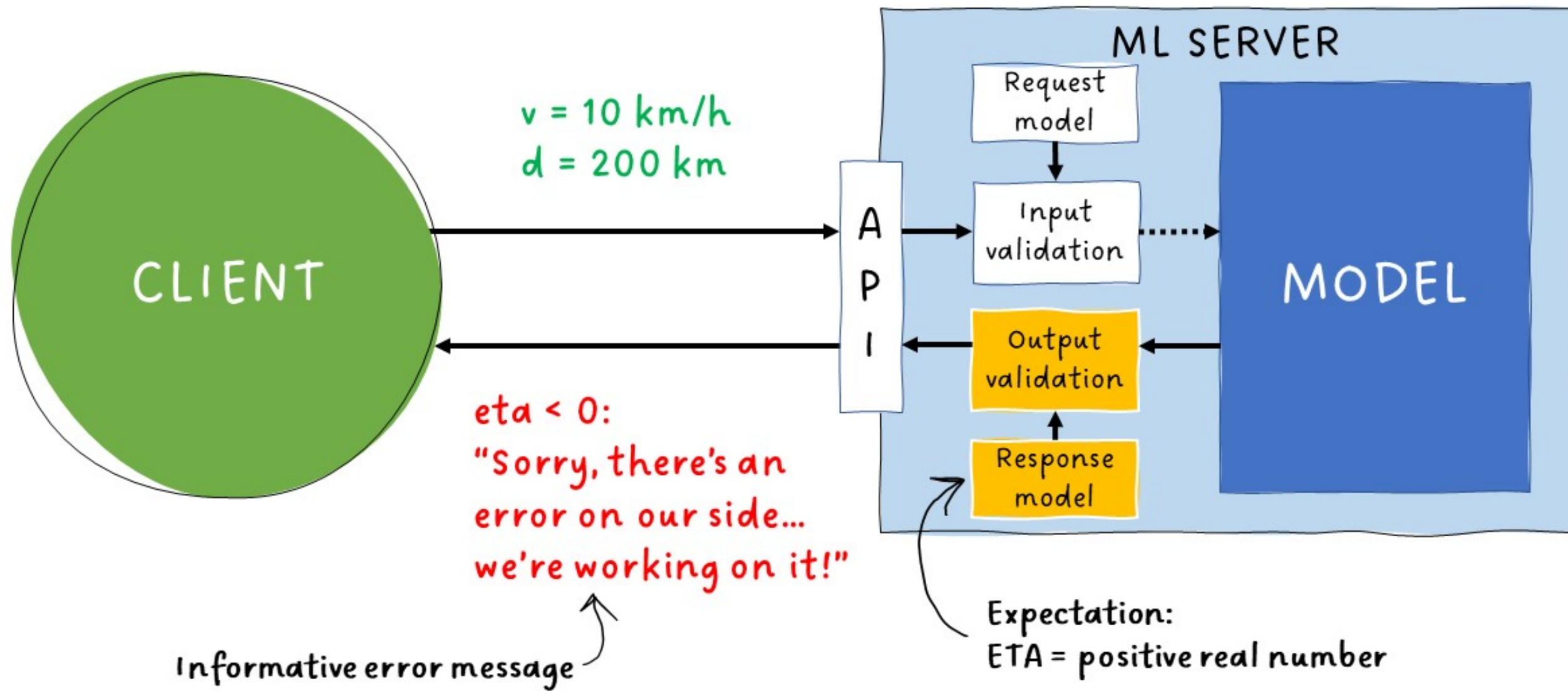


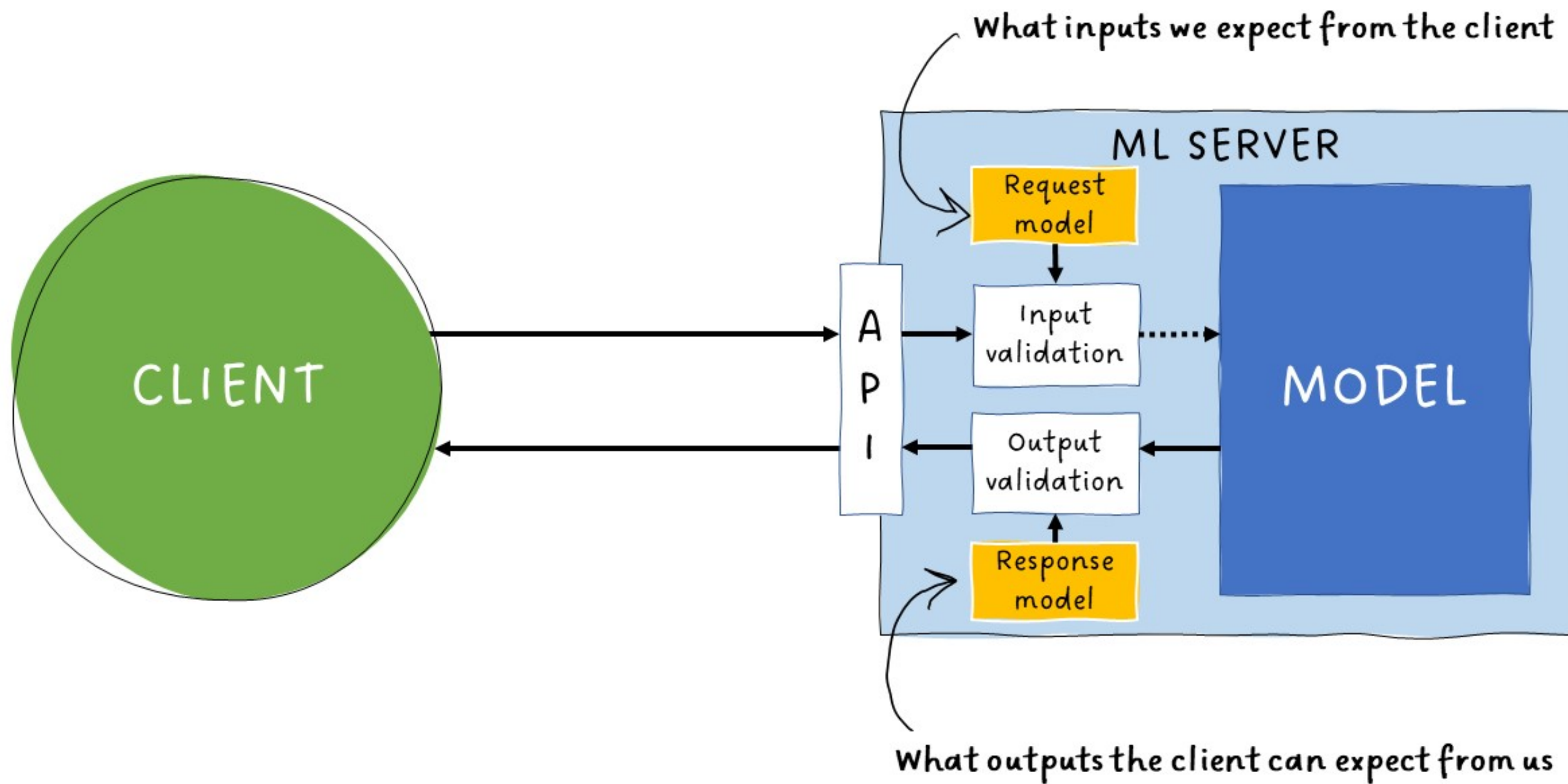


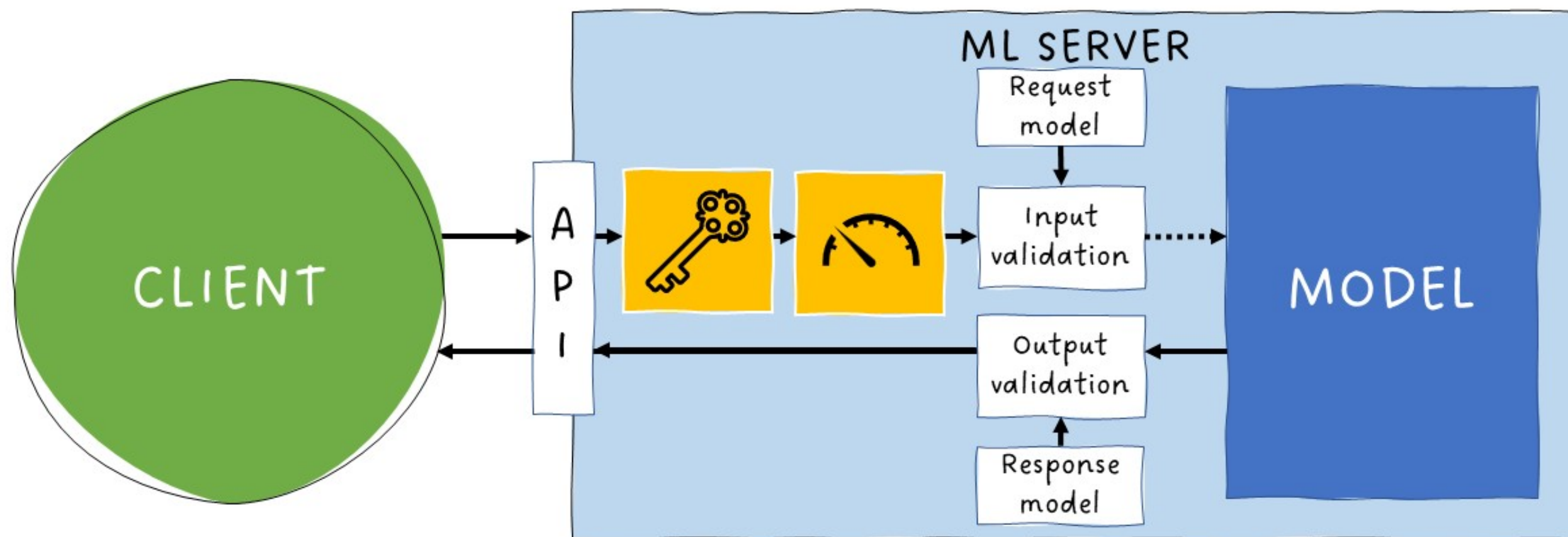


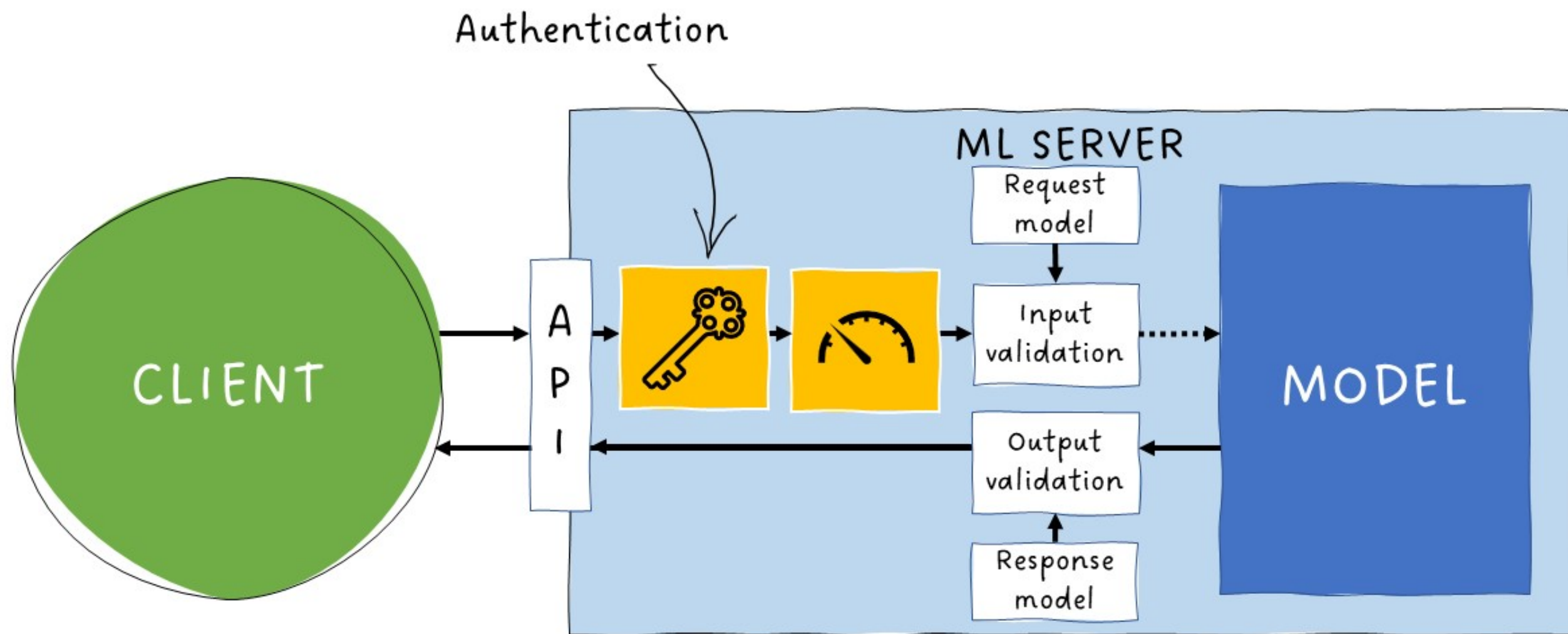


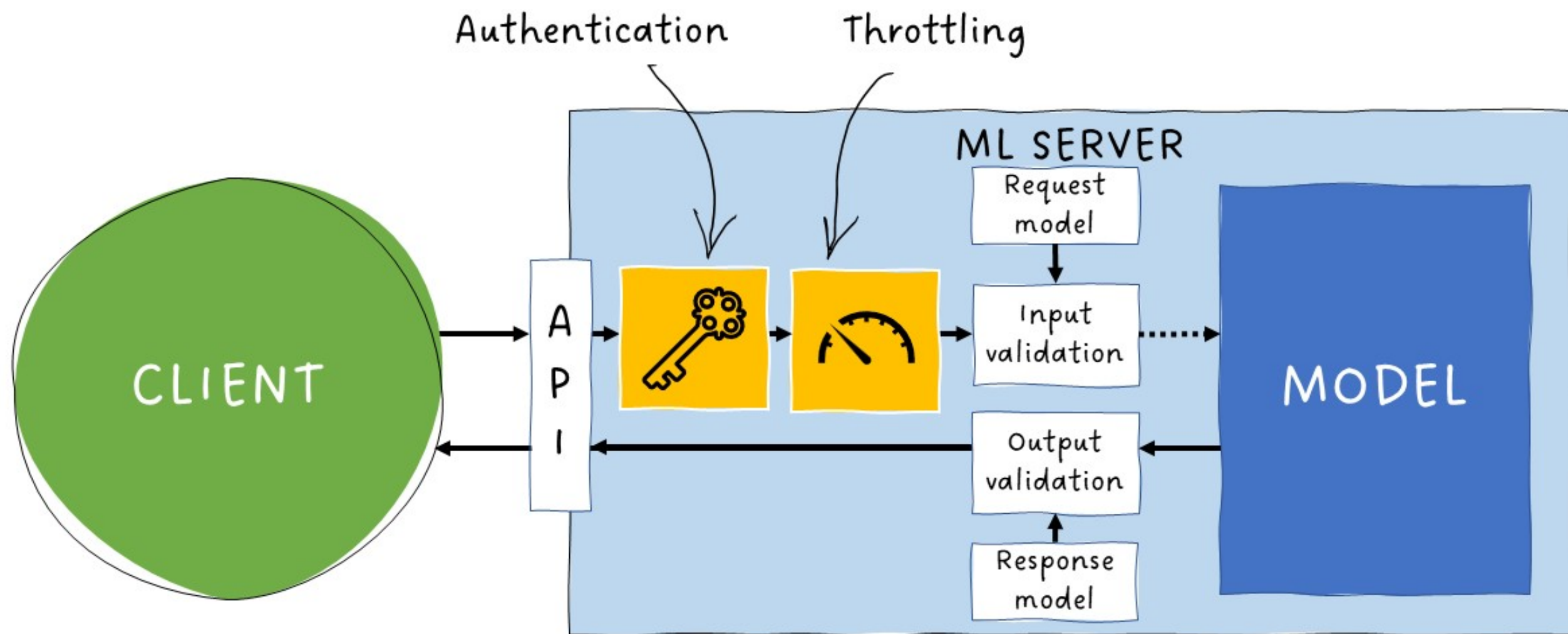














- open-source API framework for Python
- all essential features out of the box
- build and launch REST APIs in no time

¹ <https://fastapi.tiangolo.com>

Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING

Deployment progression and testing

MLOPS DEPLOYMENT AND LIFE CYCLING



Nemanja Radojkovic
Senior Machine Learning Engineer

Not so fast!



- Don't rush to deploy without thorough testing!

Agenda

TEST TYPES

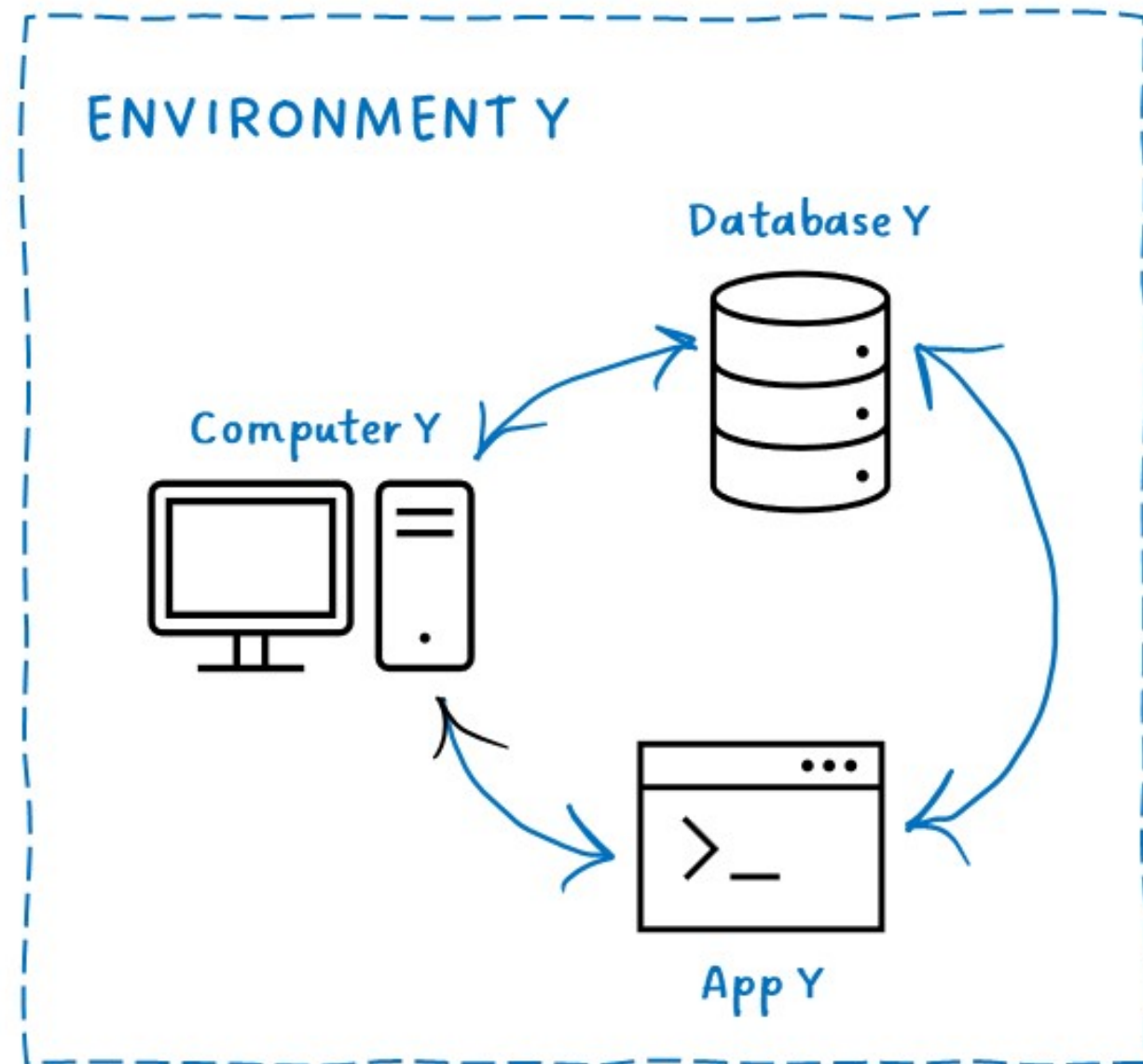
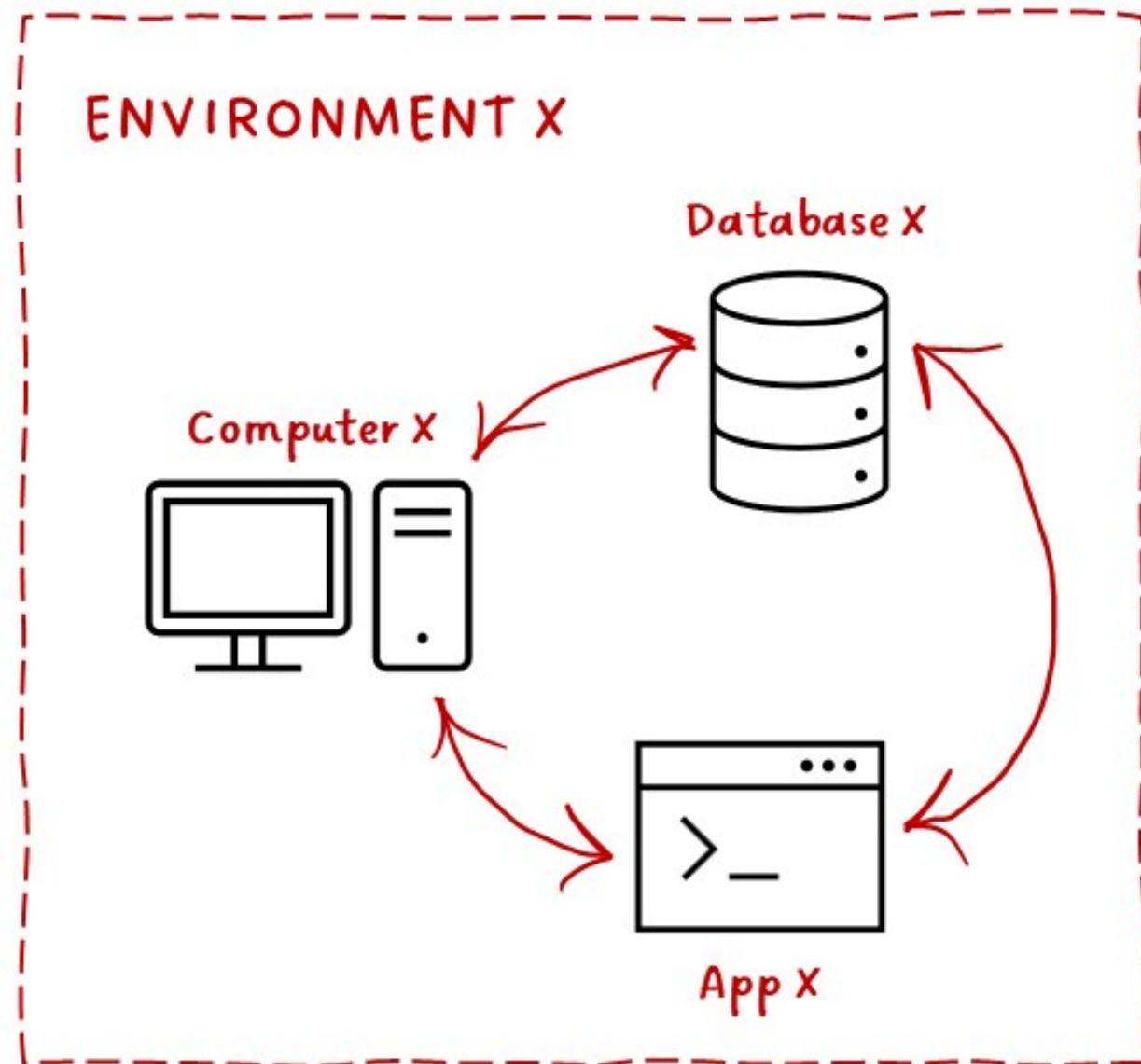
- Unit tests
- Integration tests
- Smoke tests
- Load tests
- Acceptance tests

DEV, DEPLOY, TEST ENVIRONMENTS

- Development
- Test
- Staging
- Production

OUT OF SCOPE: Predictive model performance

IN SCOPE: Testing the overall ML app



Unit testing

Checking if independent code units behave as expected

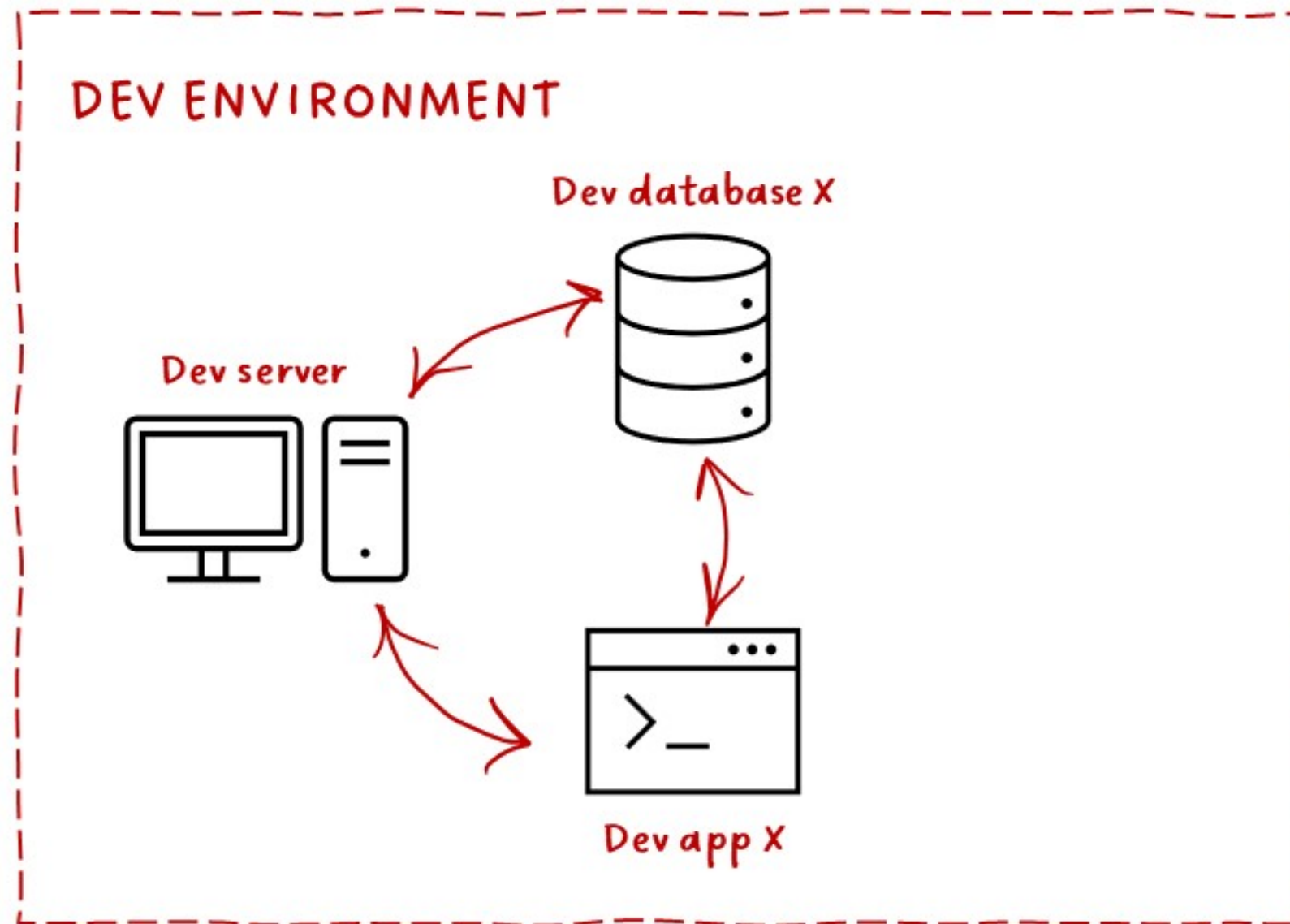
EXAMPLE

Function to test:

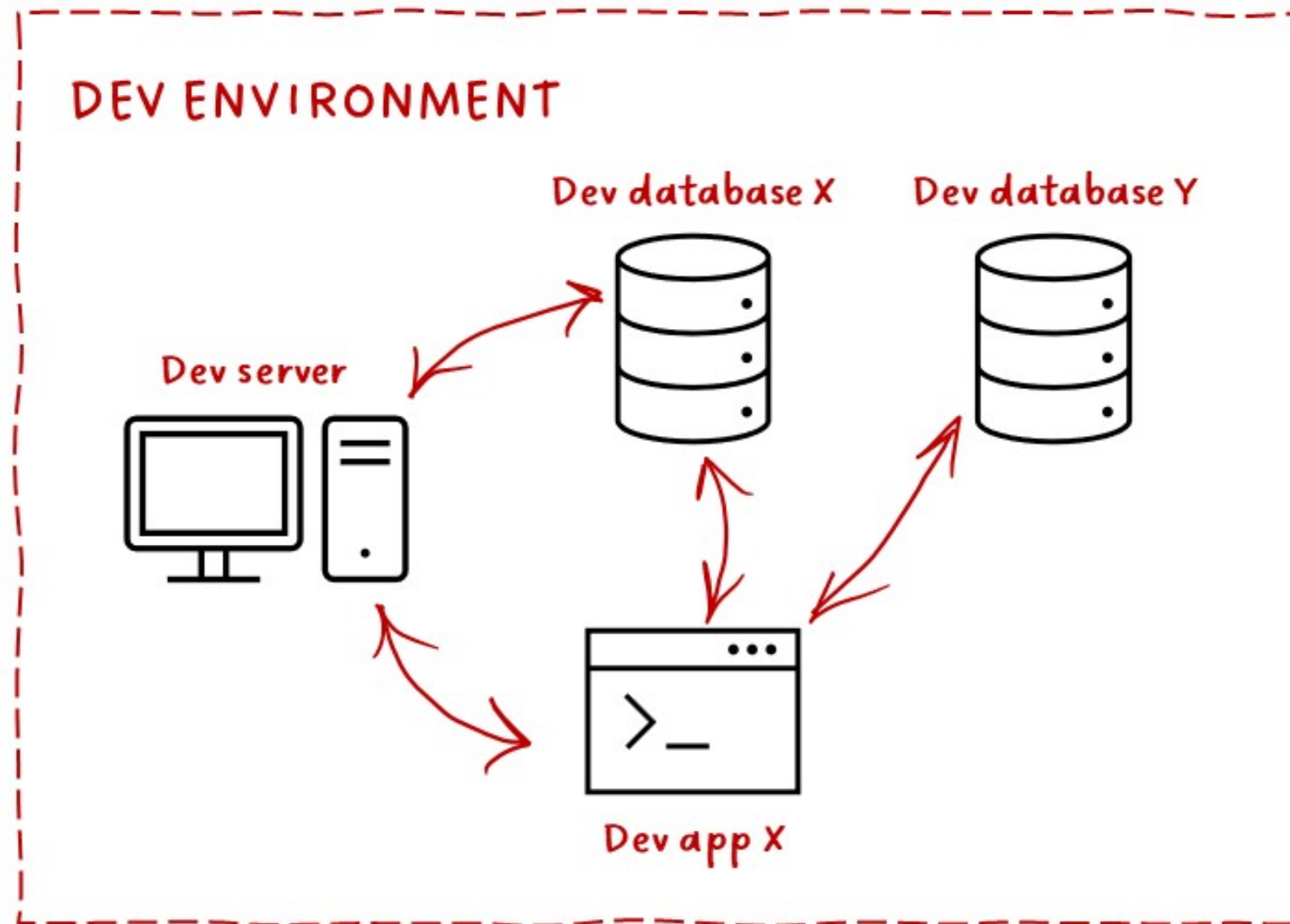
```
def add_two_numbers(x, y):  
    return x + y
```

Unit test:

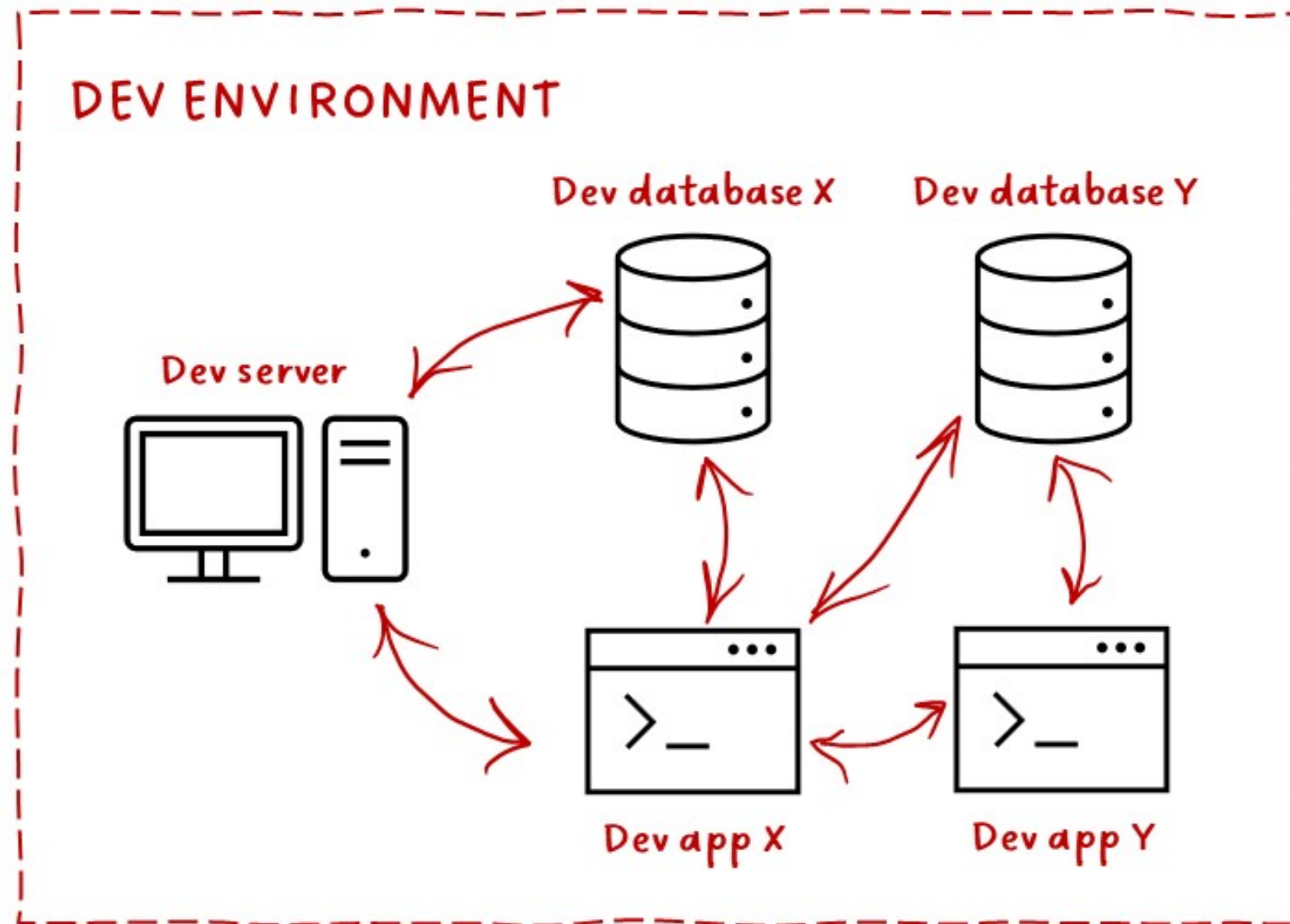
```
add_two_numbers(1, 1) == 2  
add_two_numbers(2, 3) == 5
```



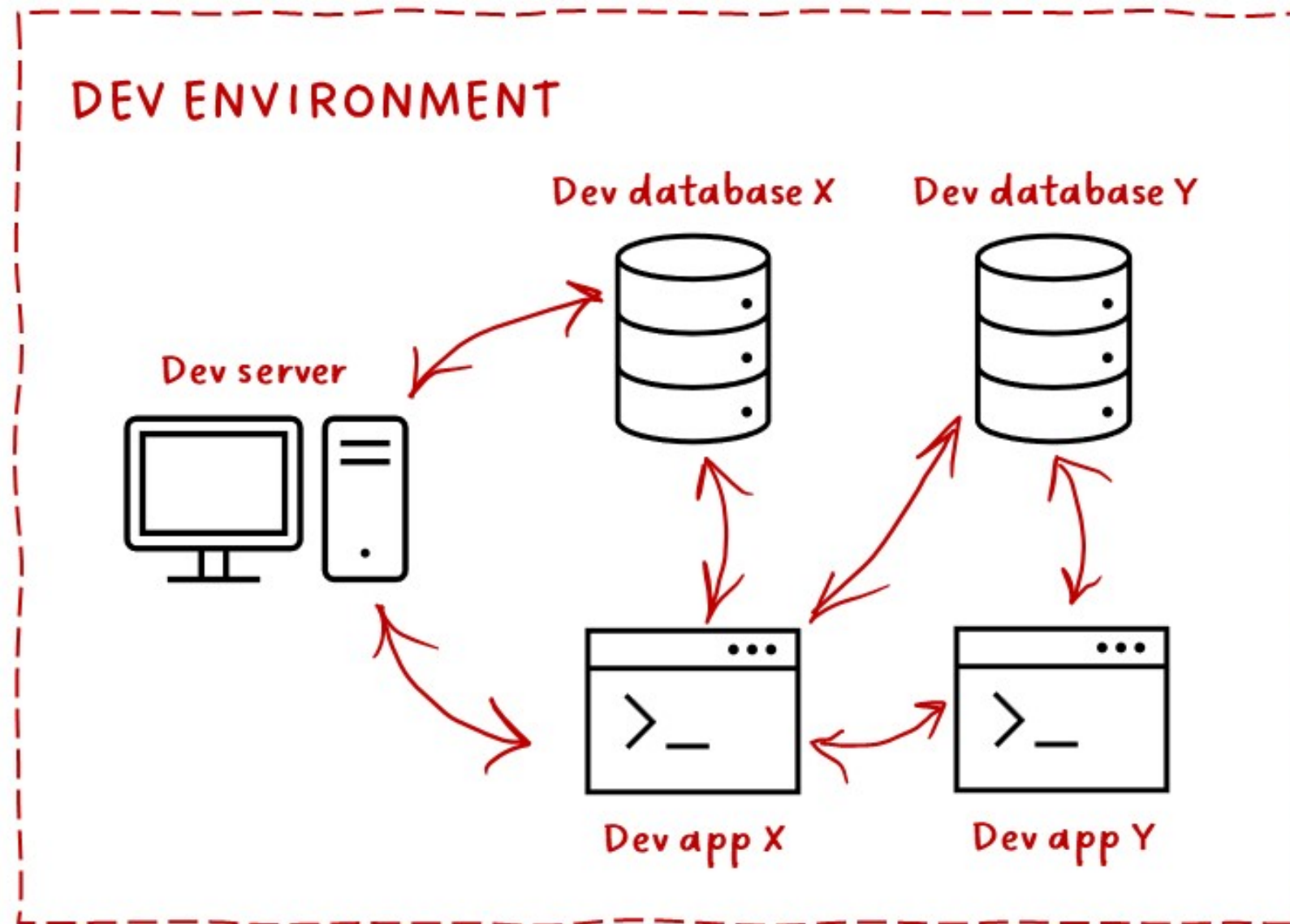
- Experimentation
- Constantly changing



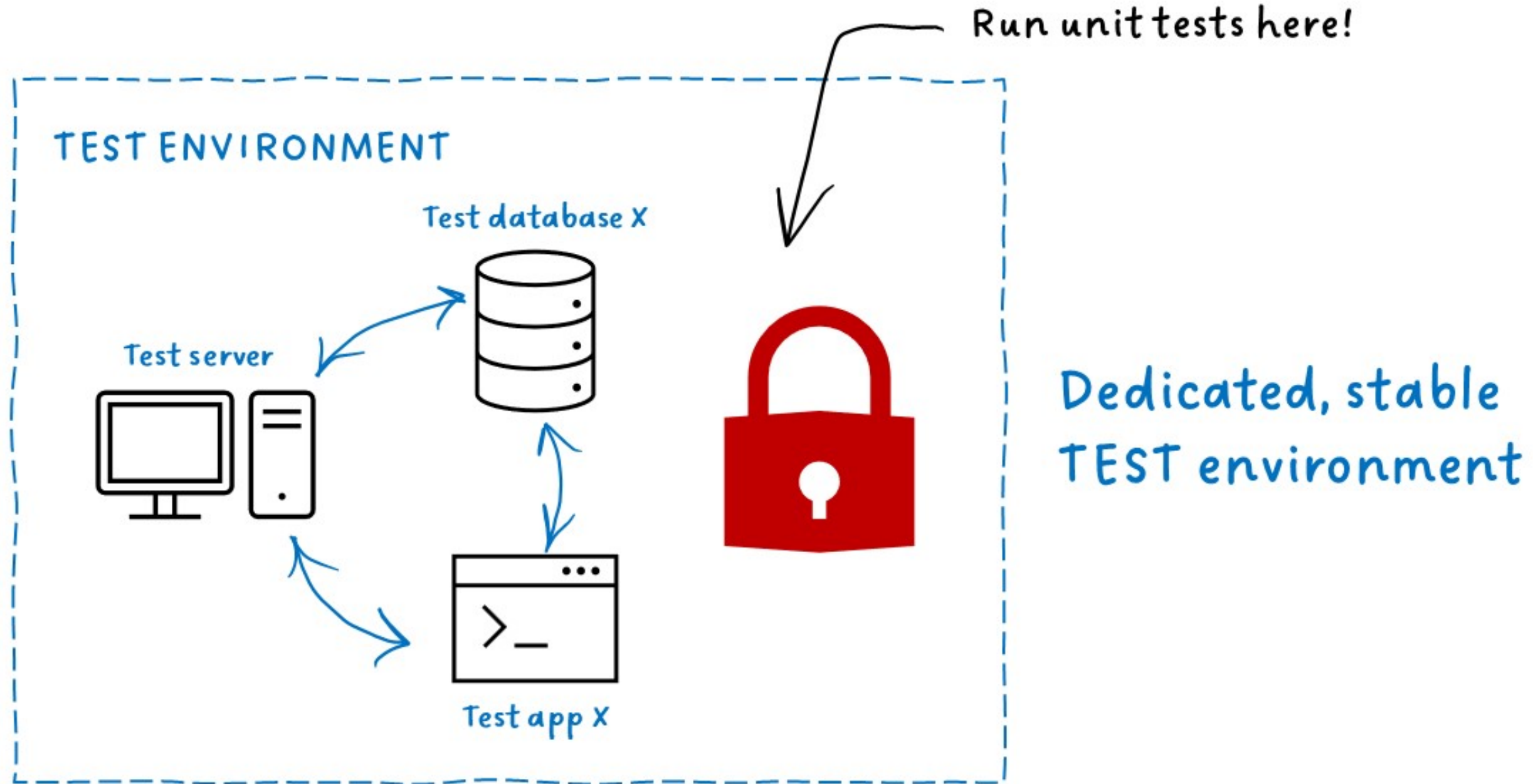
- Experimentation
- Constantly changing

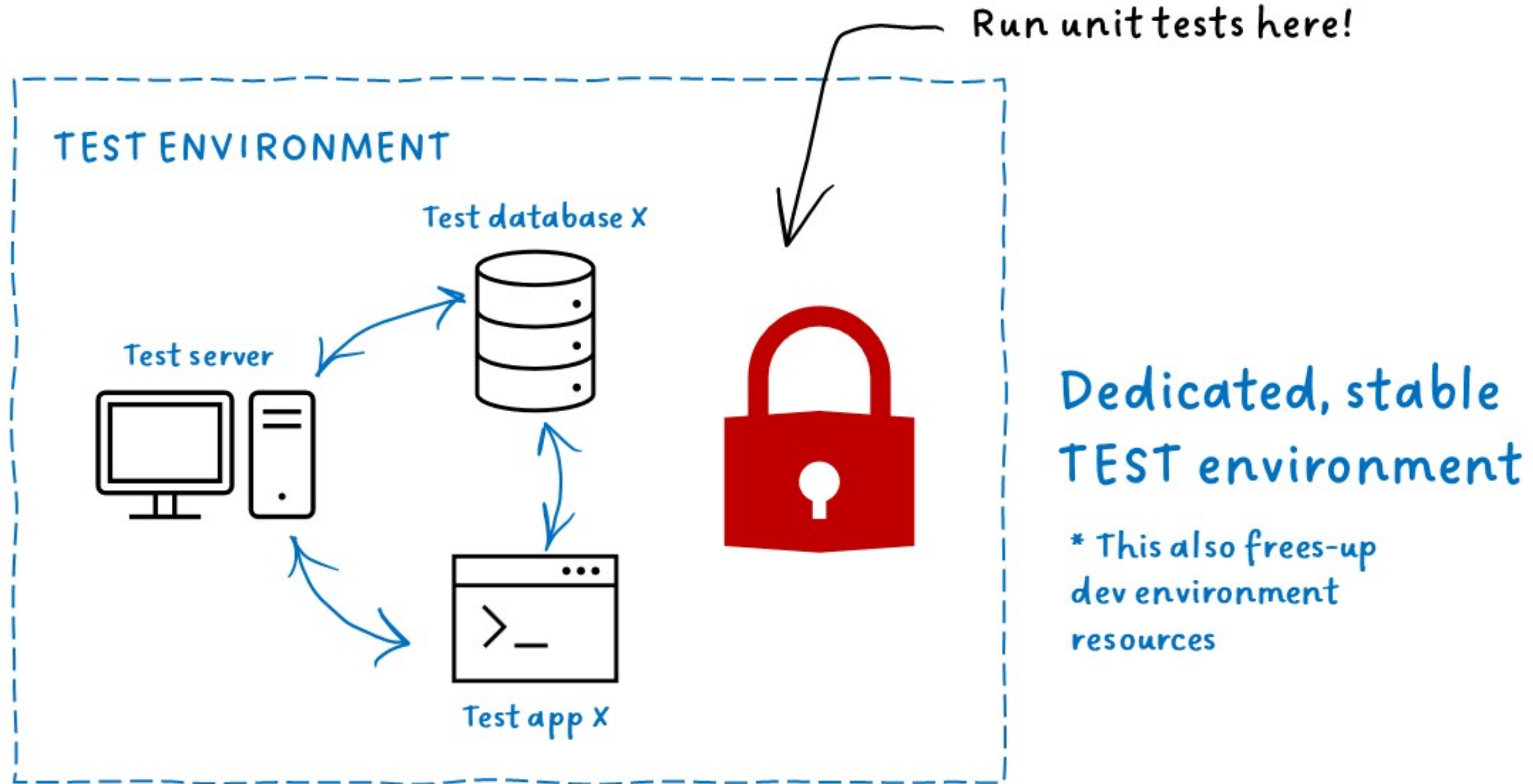


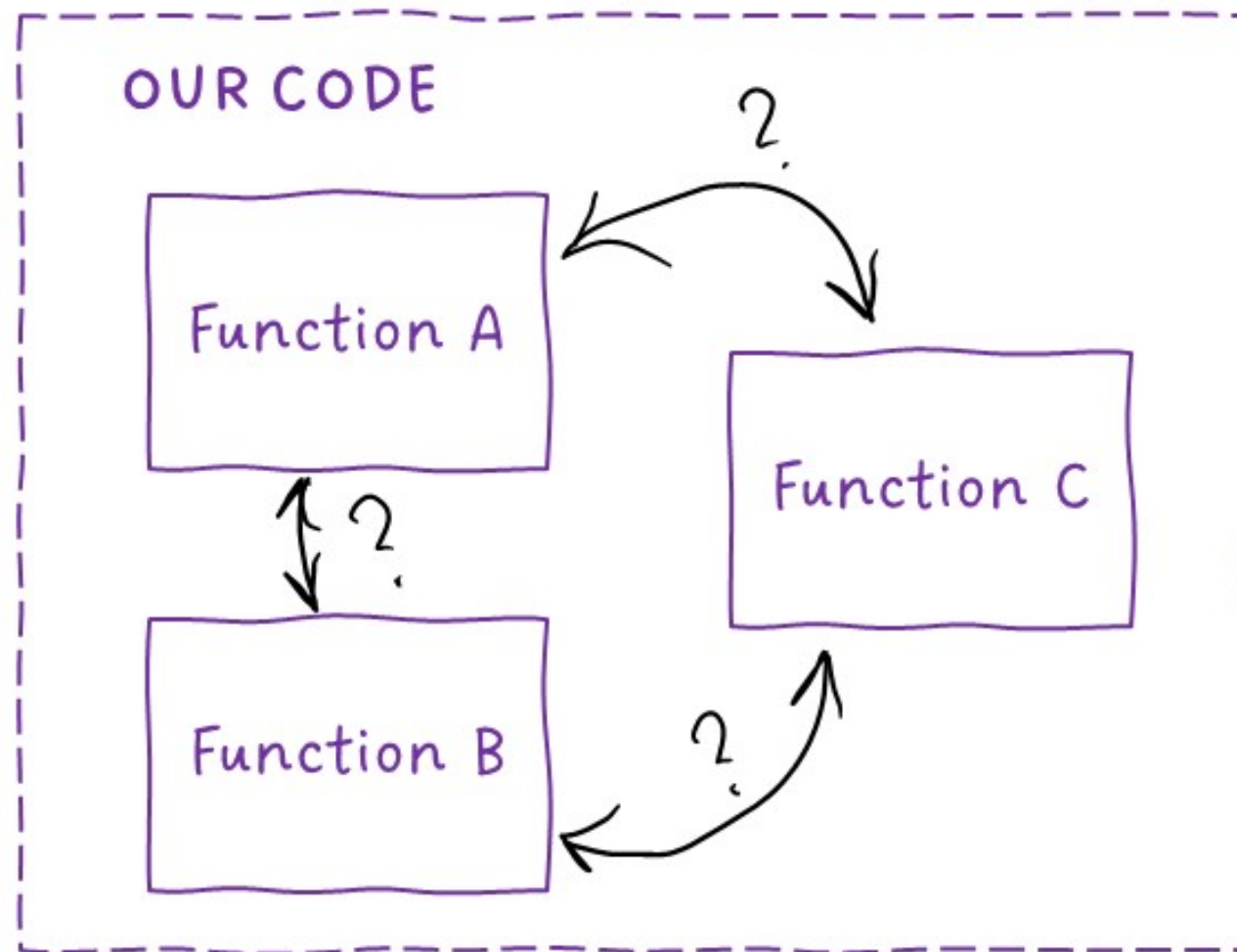
- Experimentation
- Constantly changing

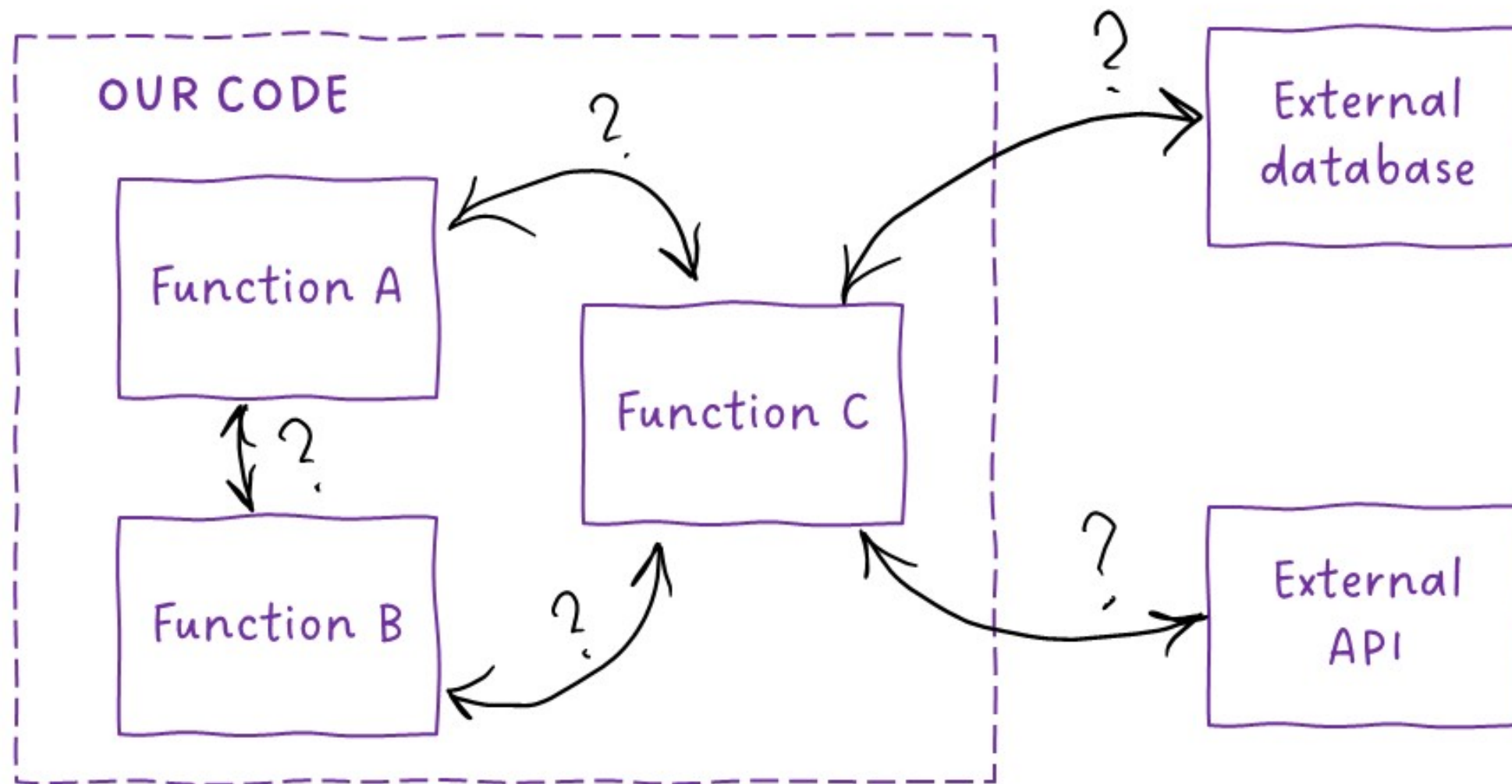


- Experimentation
- Constantly changing
- Tests require stability

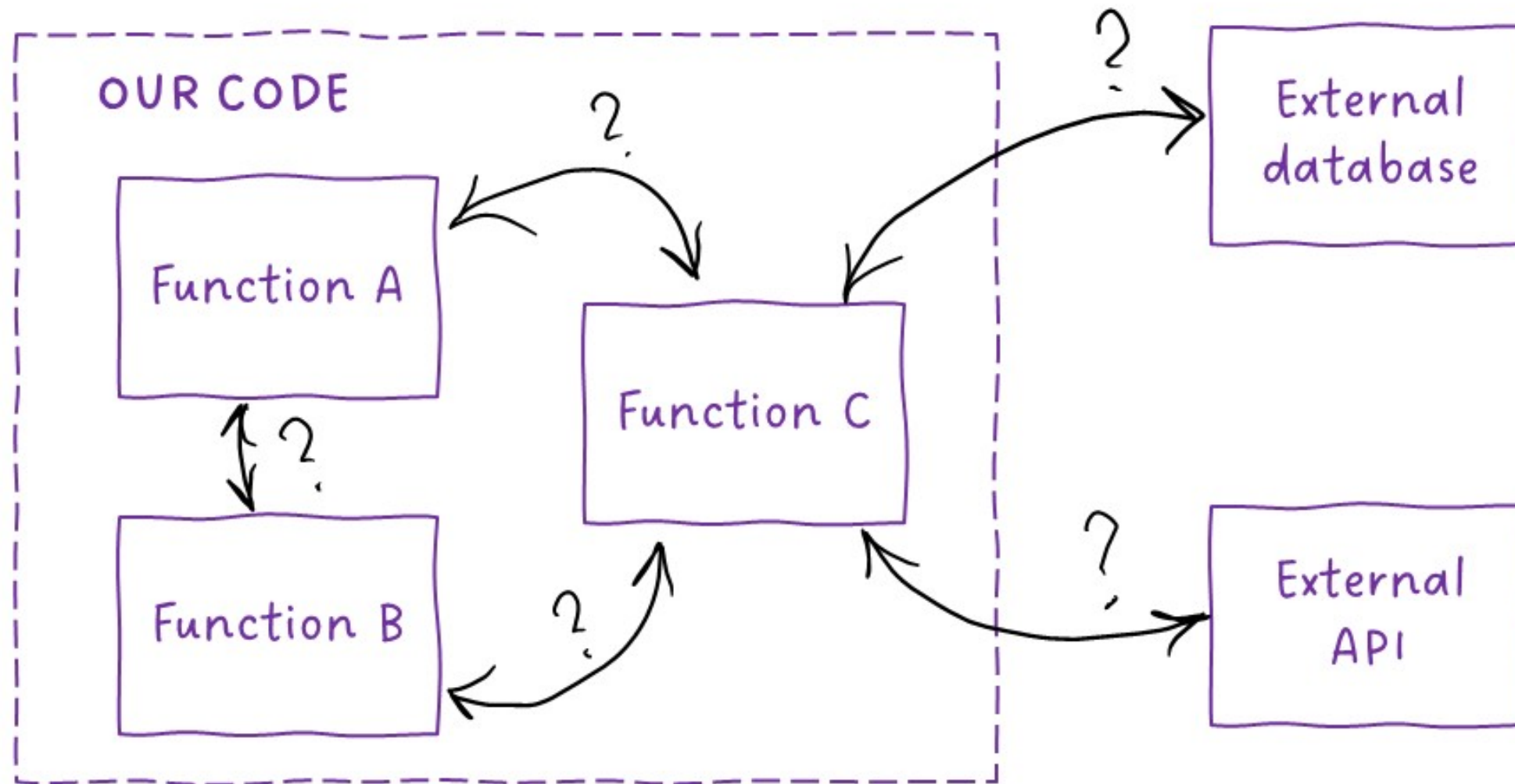




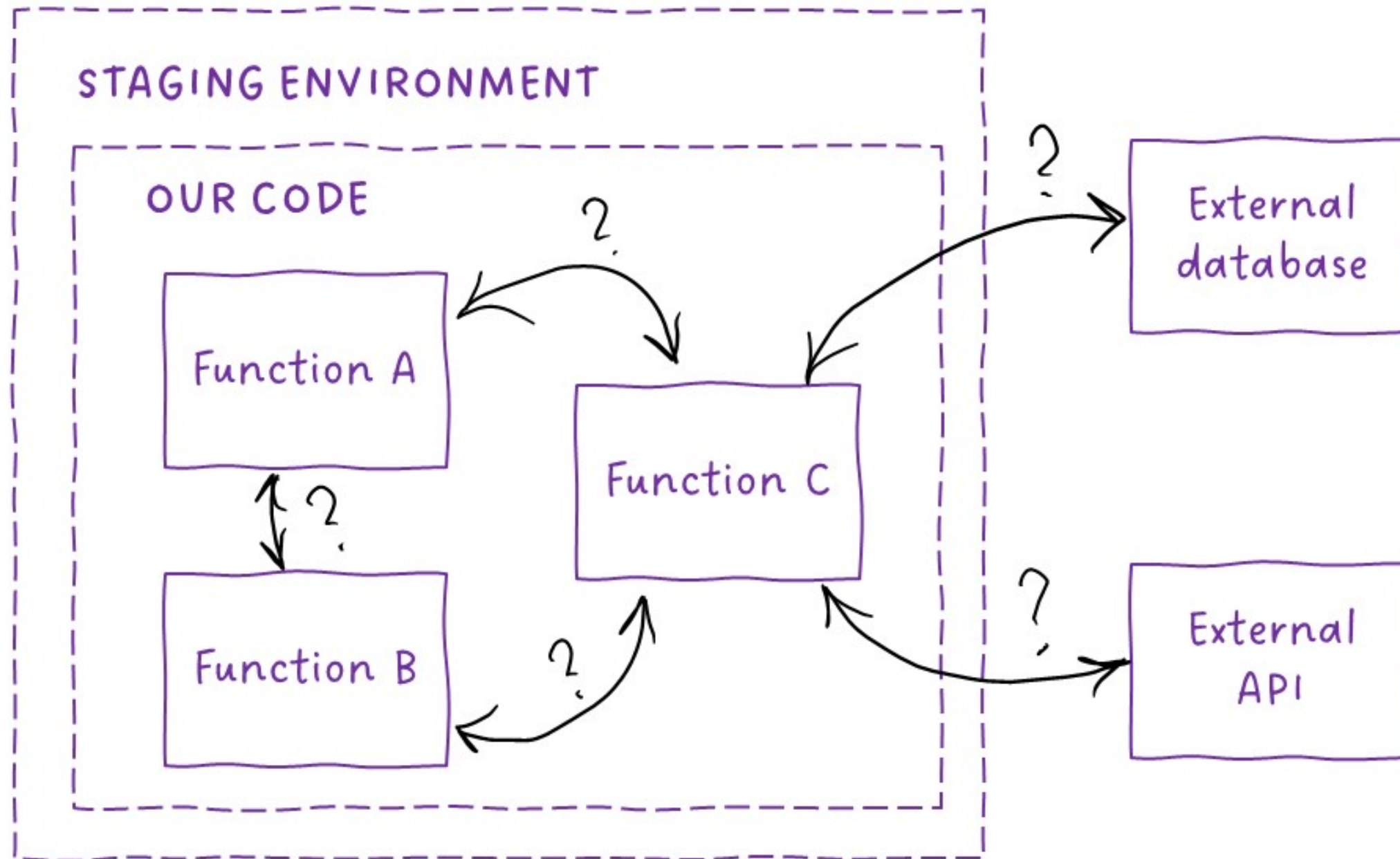


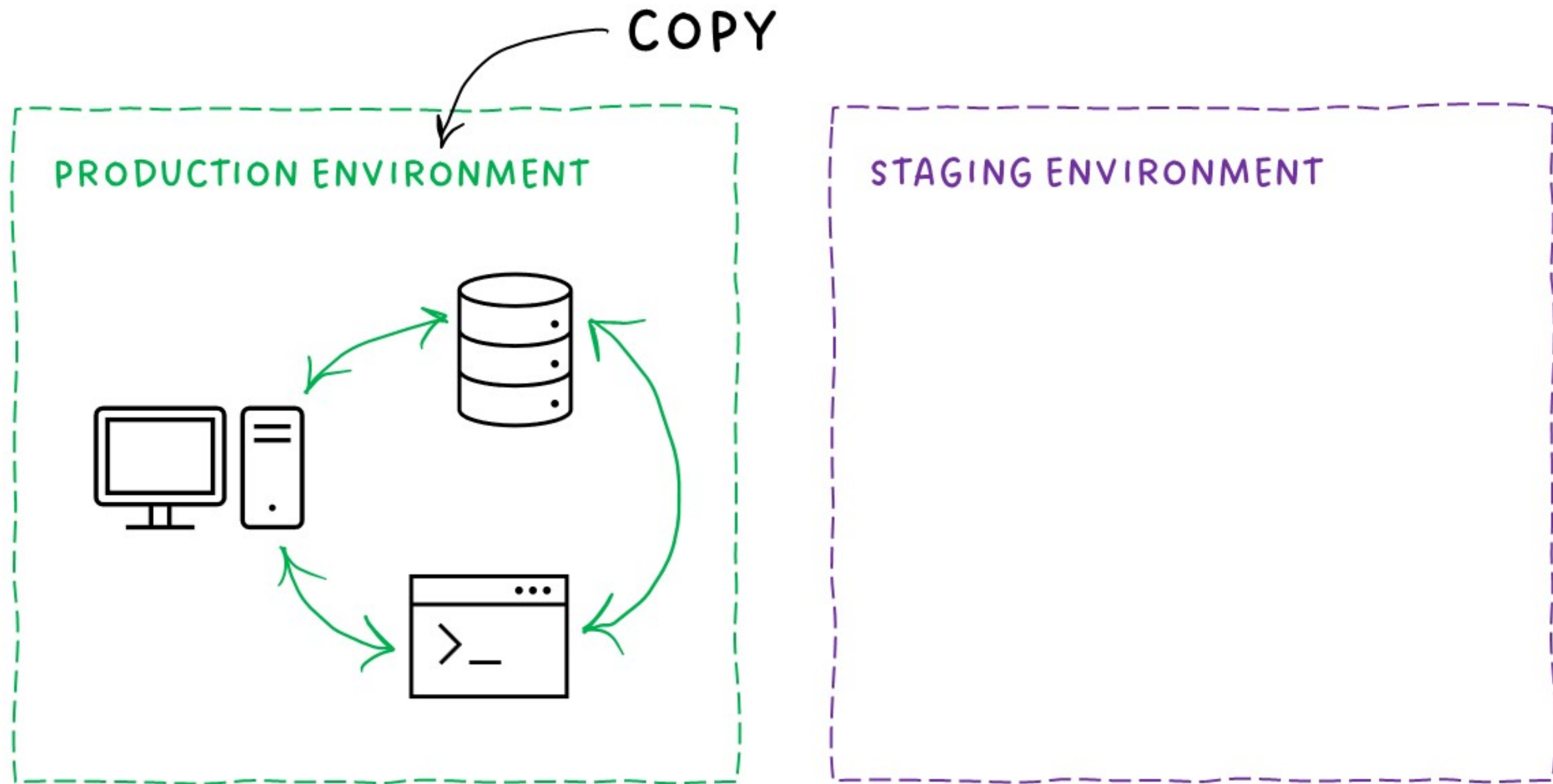


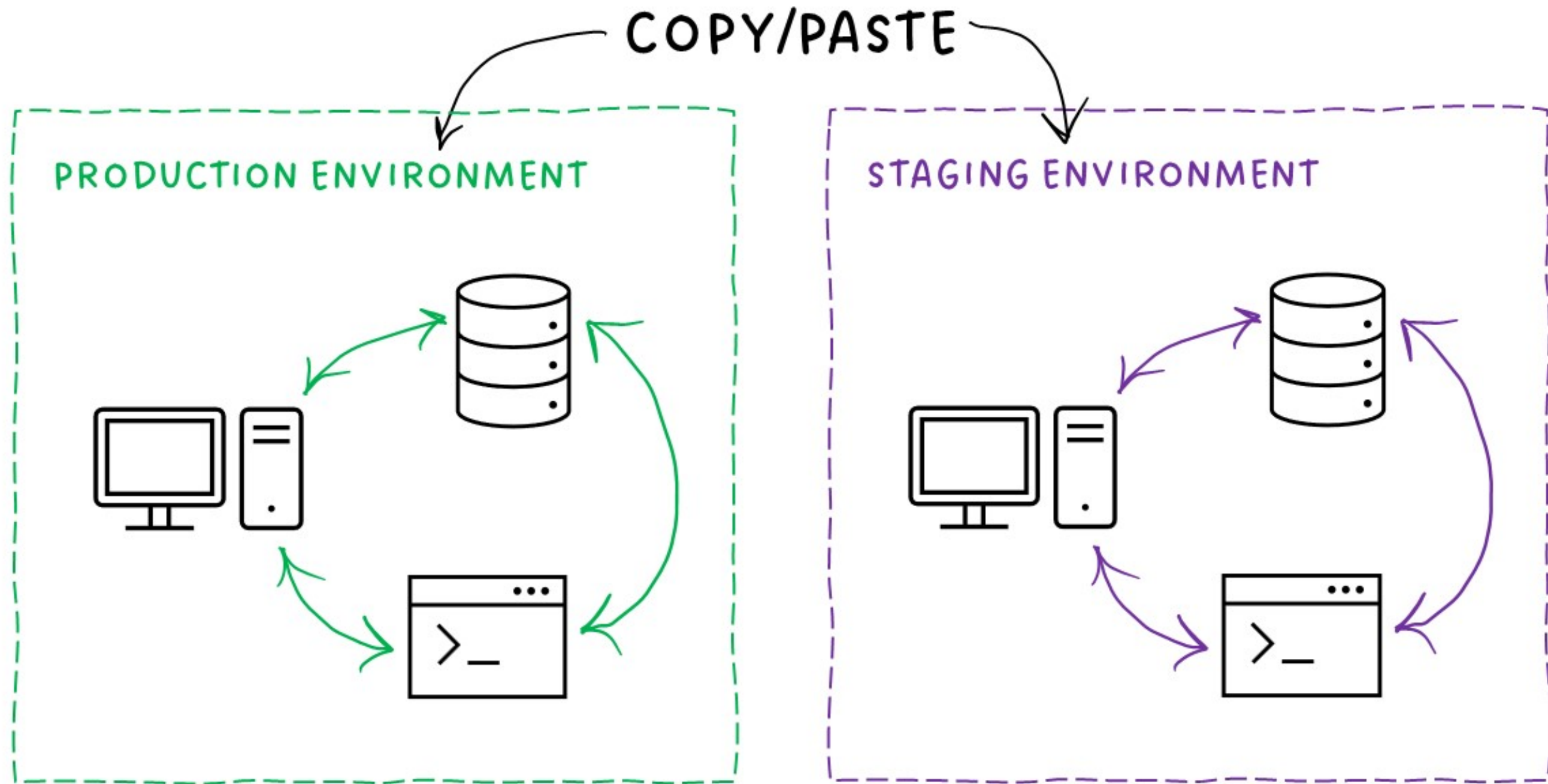
INTEGRATION TESTING

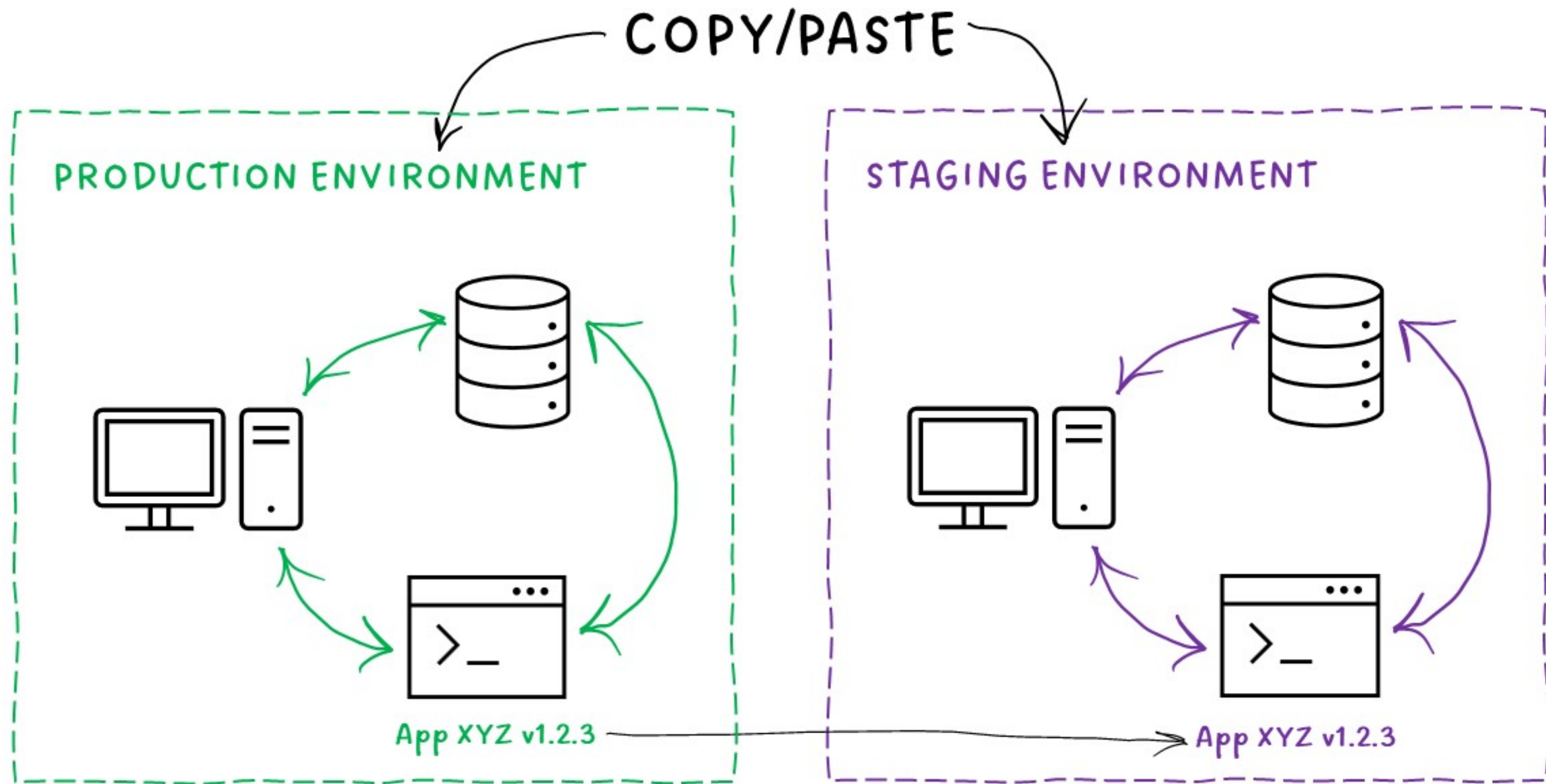


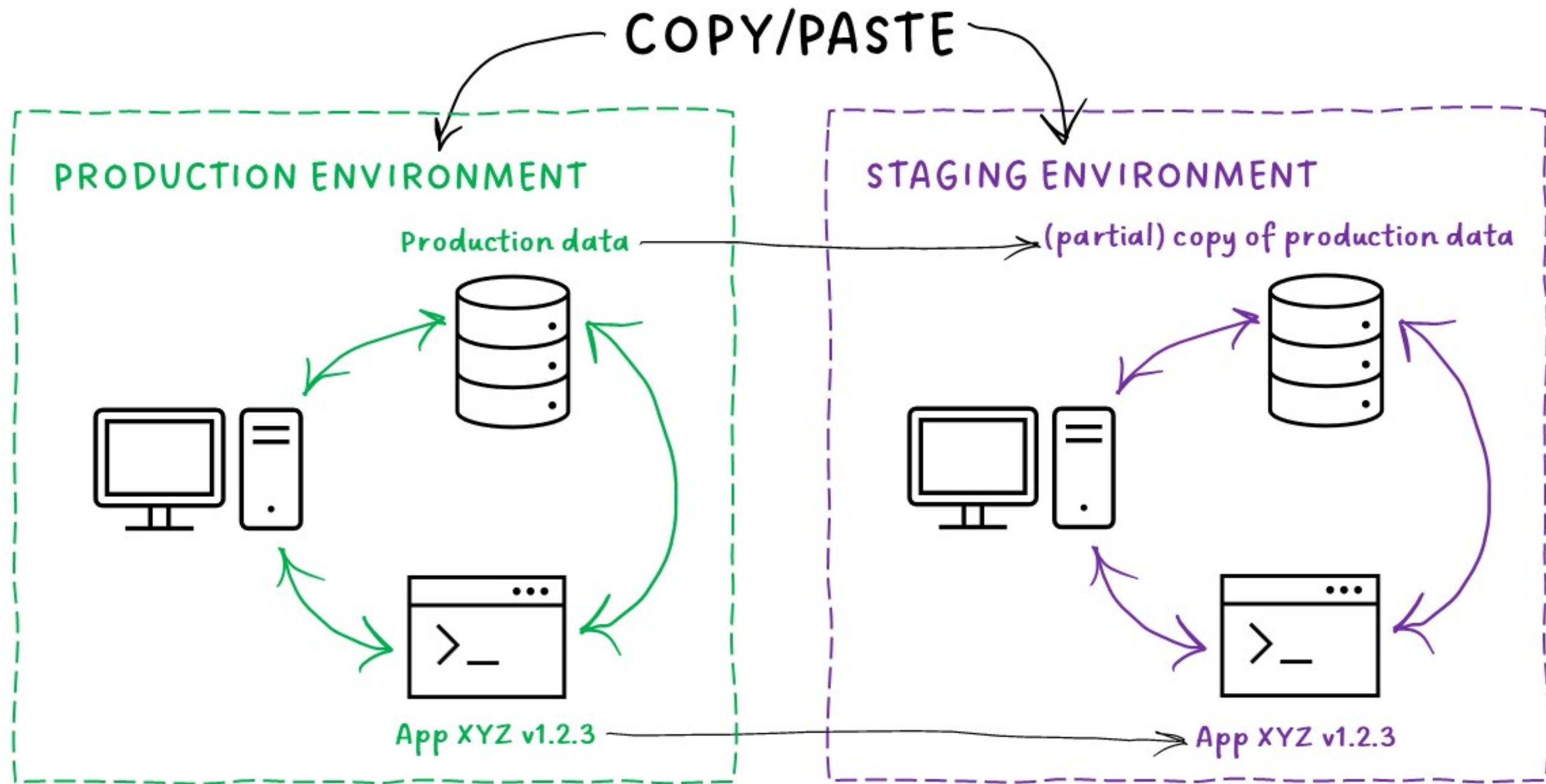
INTEGRATION TESTING



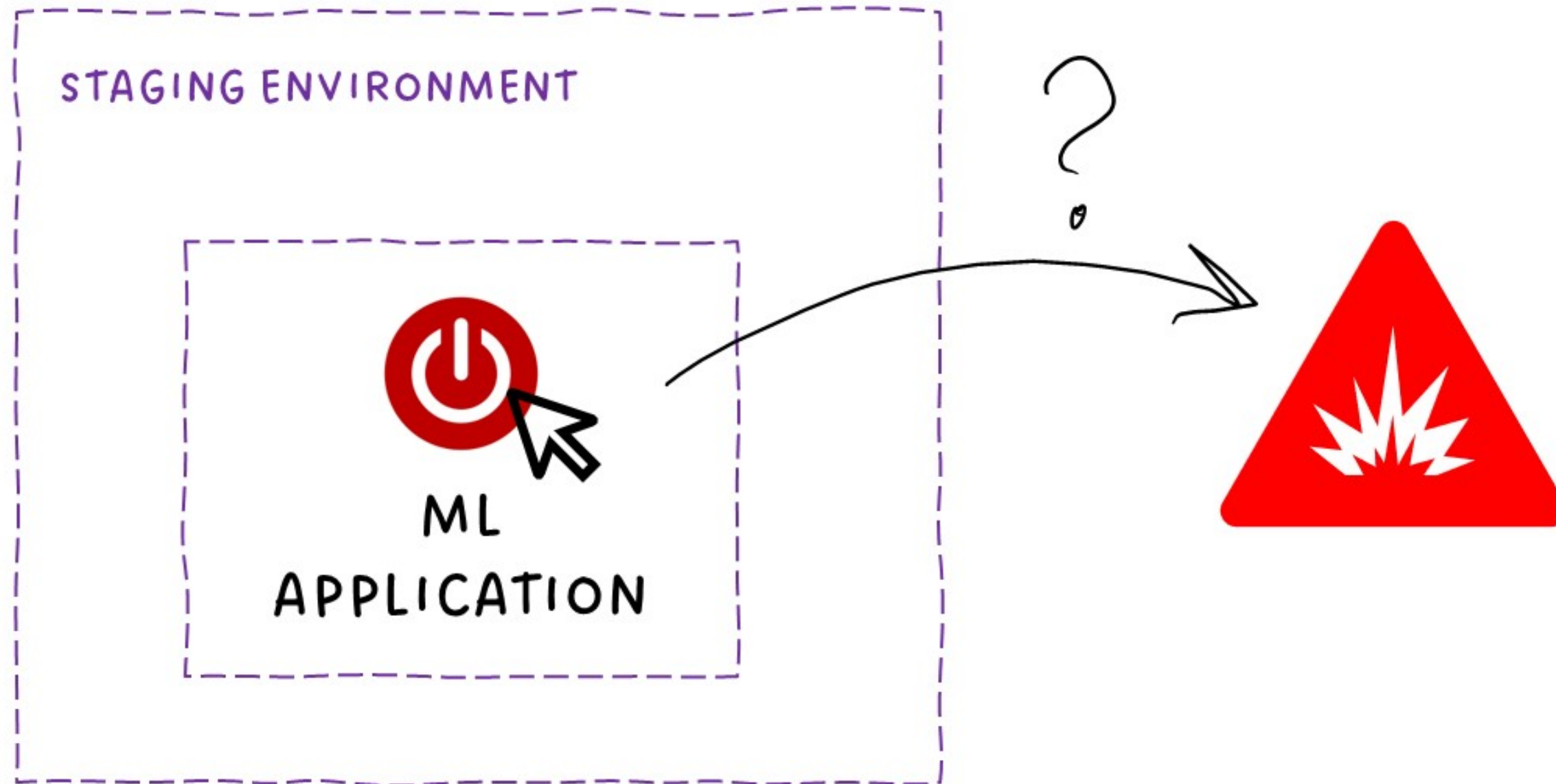




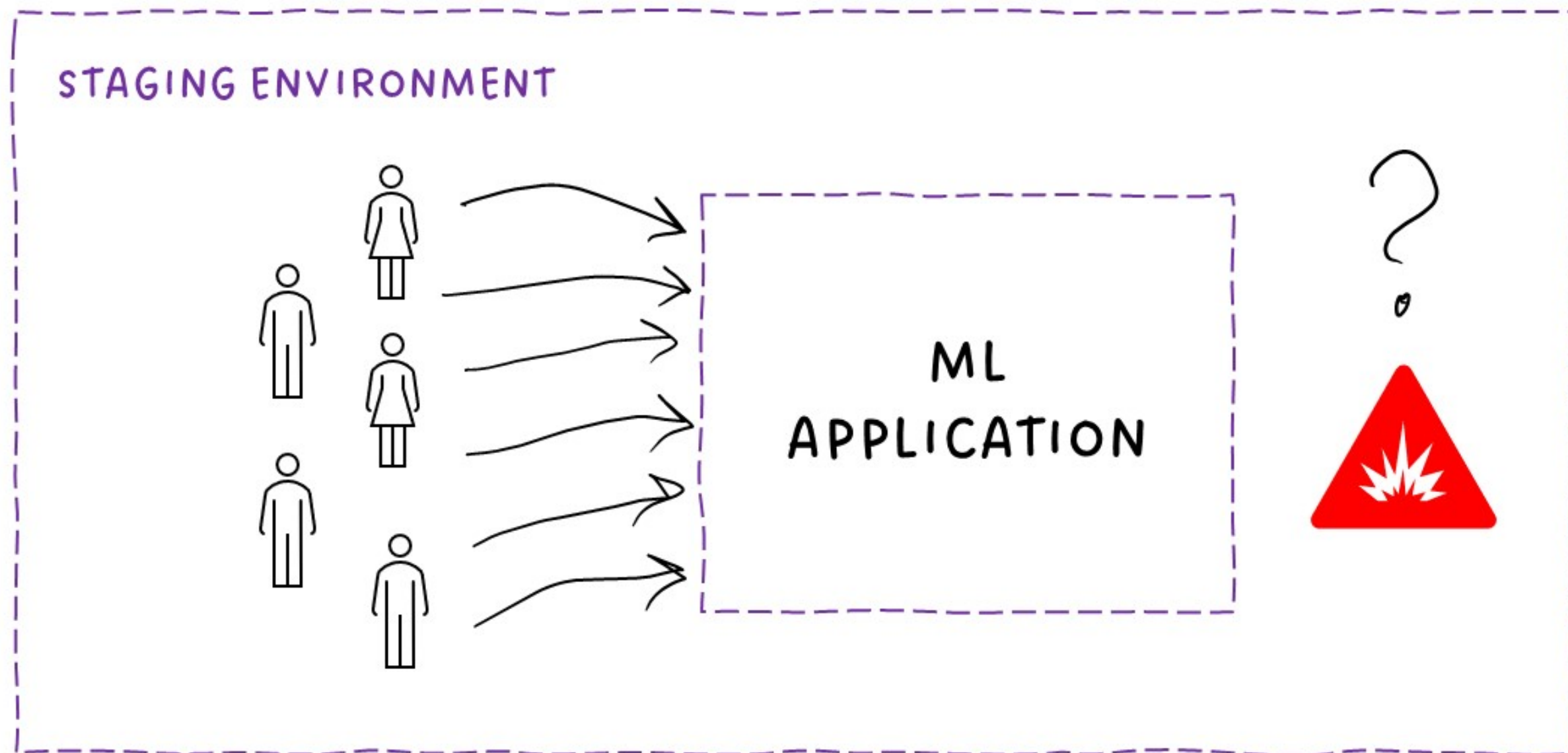


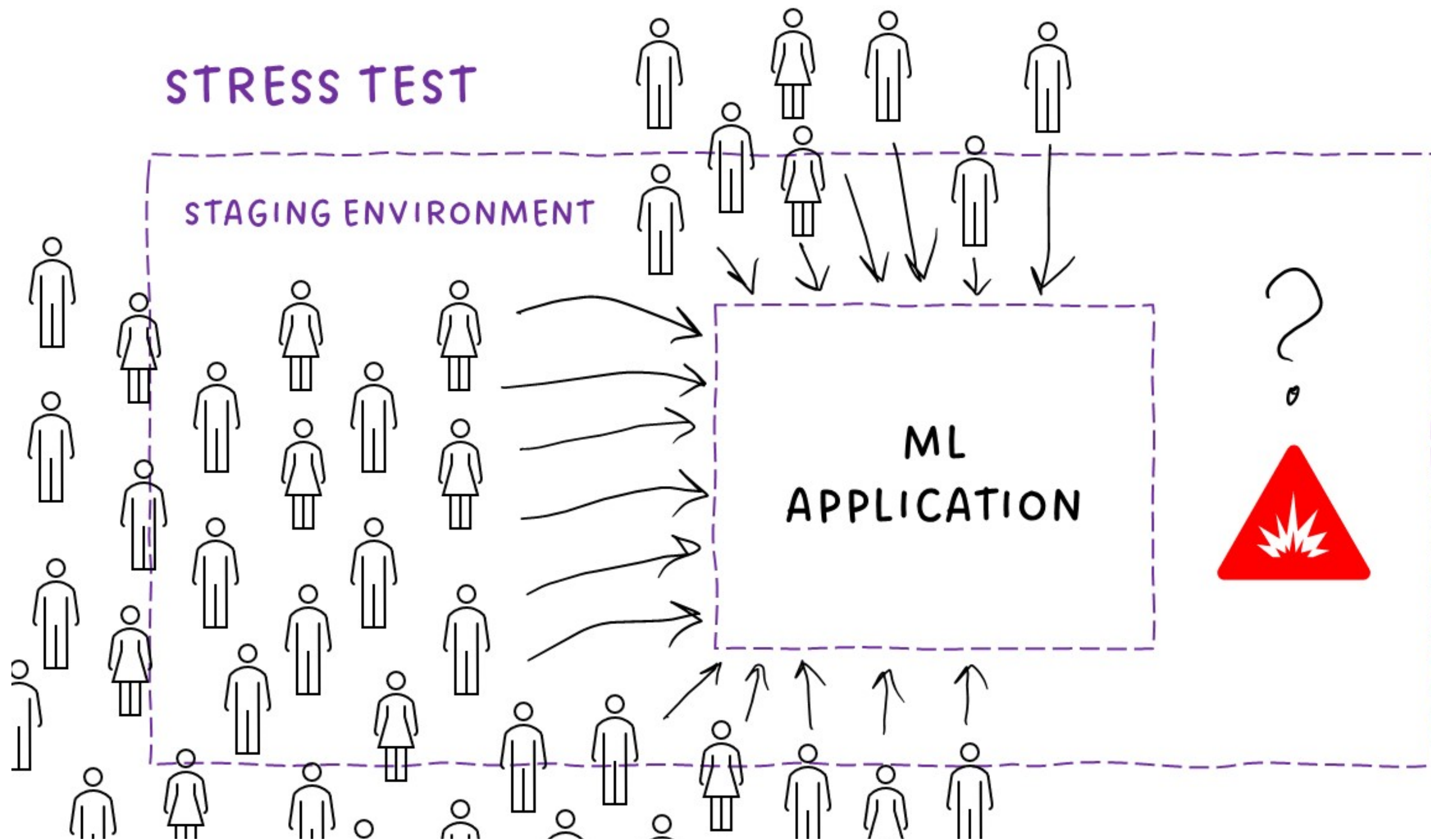


SMOKE TEST



LOAD TEST





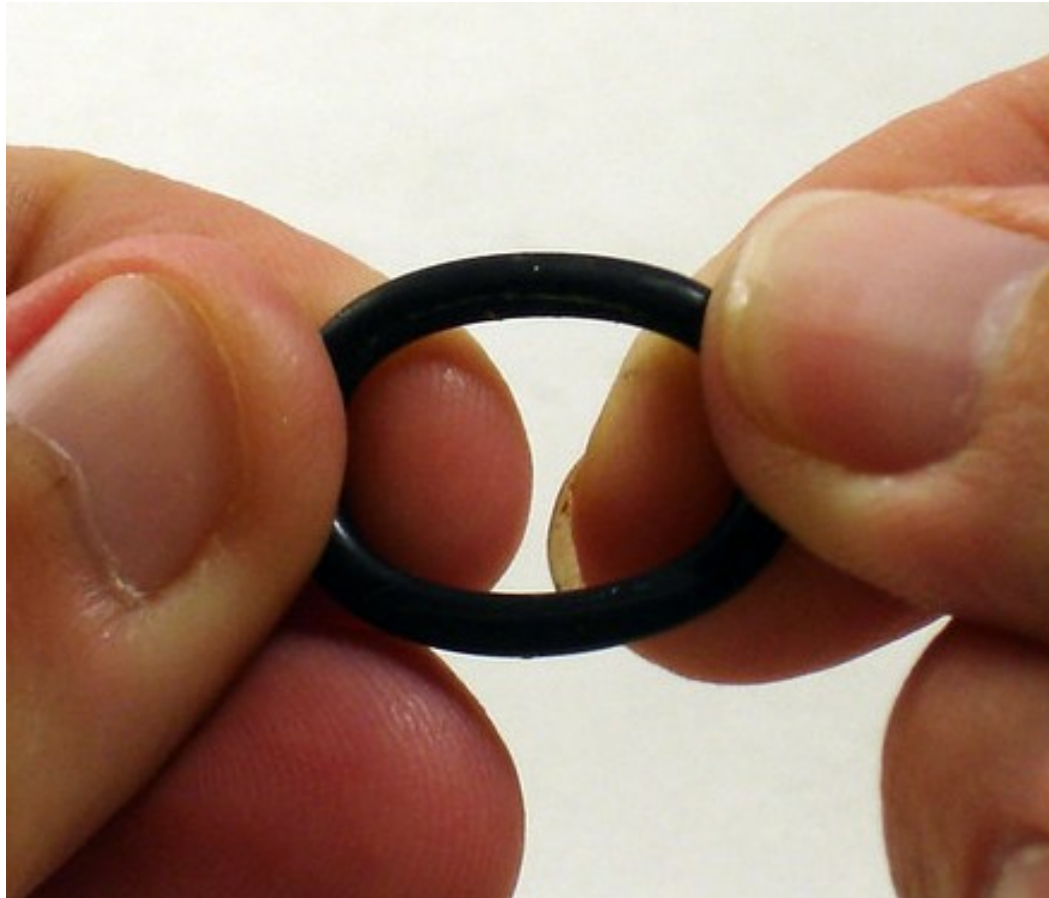
USER ACCEPTANCE TESTING (UAT)



Final "seal of approval" before deploying to production

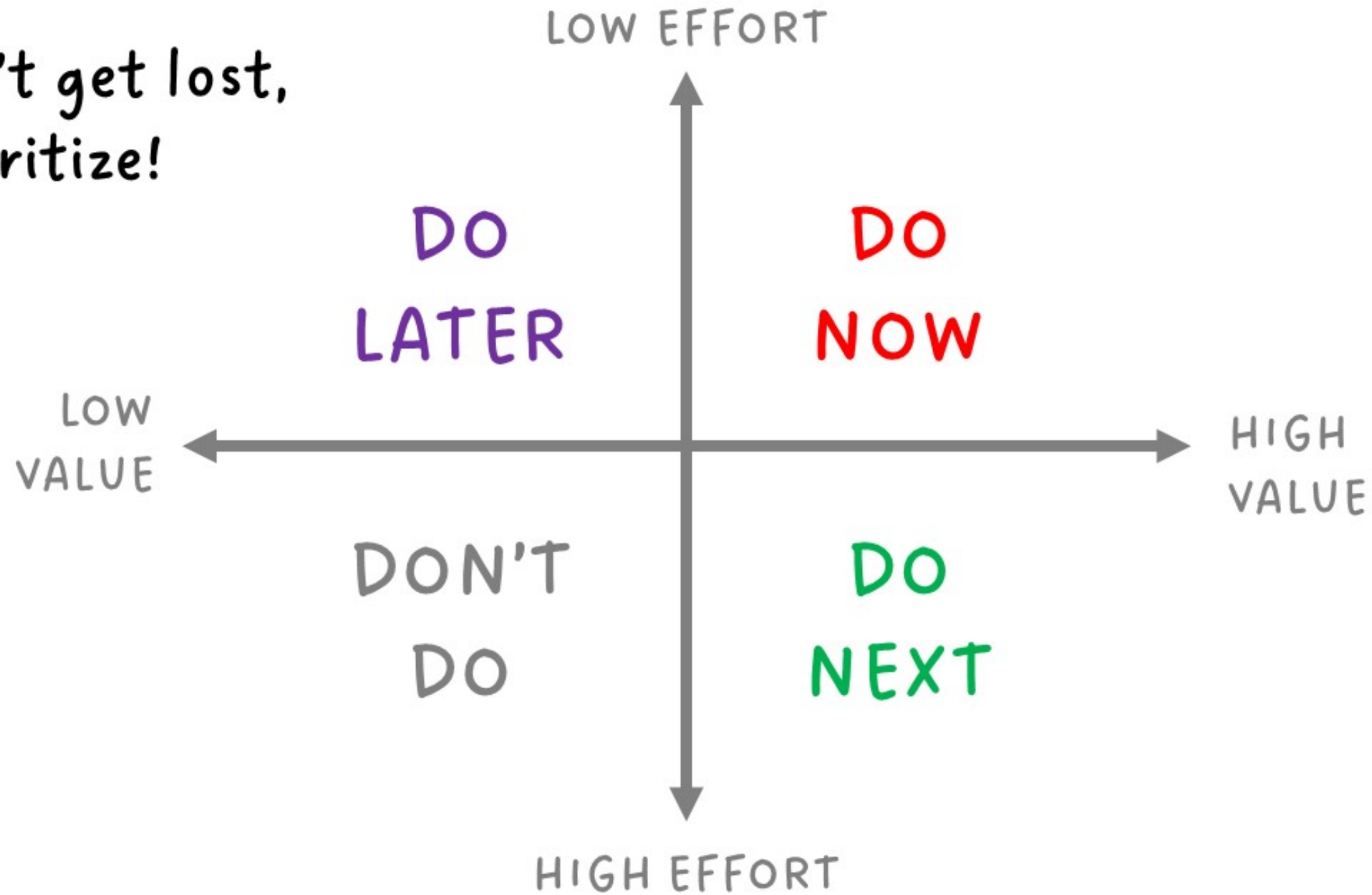
A single weak link breaks the system

We must be careful what we introduce in production.



The Challenger shuttle disaster in 1986 was caused by a failed rubber sealing ring

Don't get lost,
prioritize!



Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING

Model deployment strategies

MLOPS DEPLOYMENT AND LIFE CYCLING

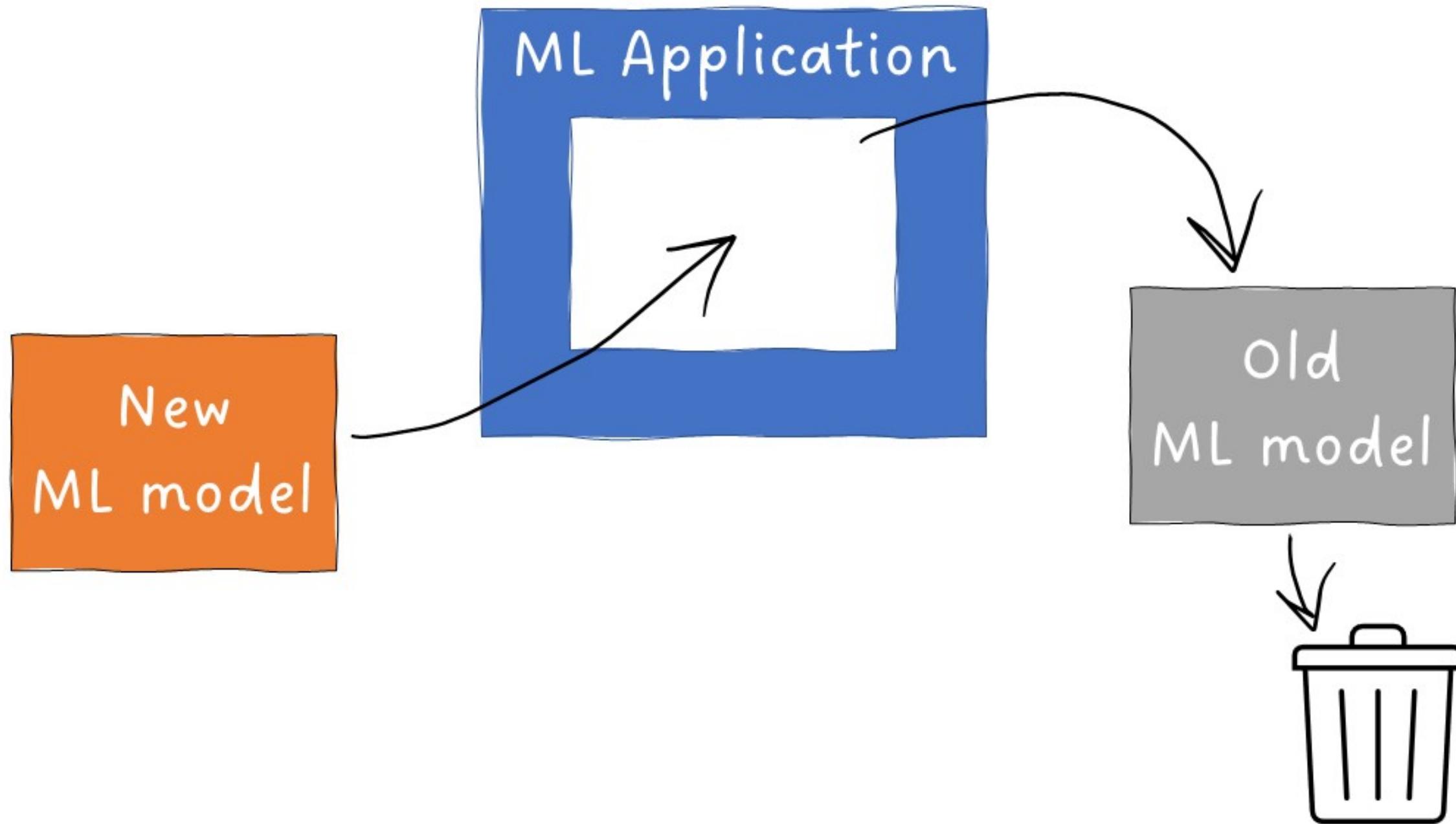


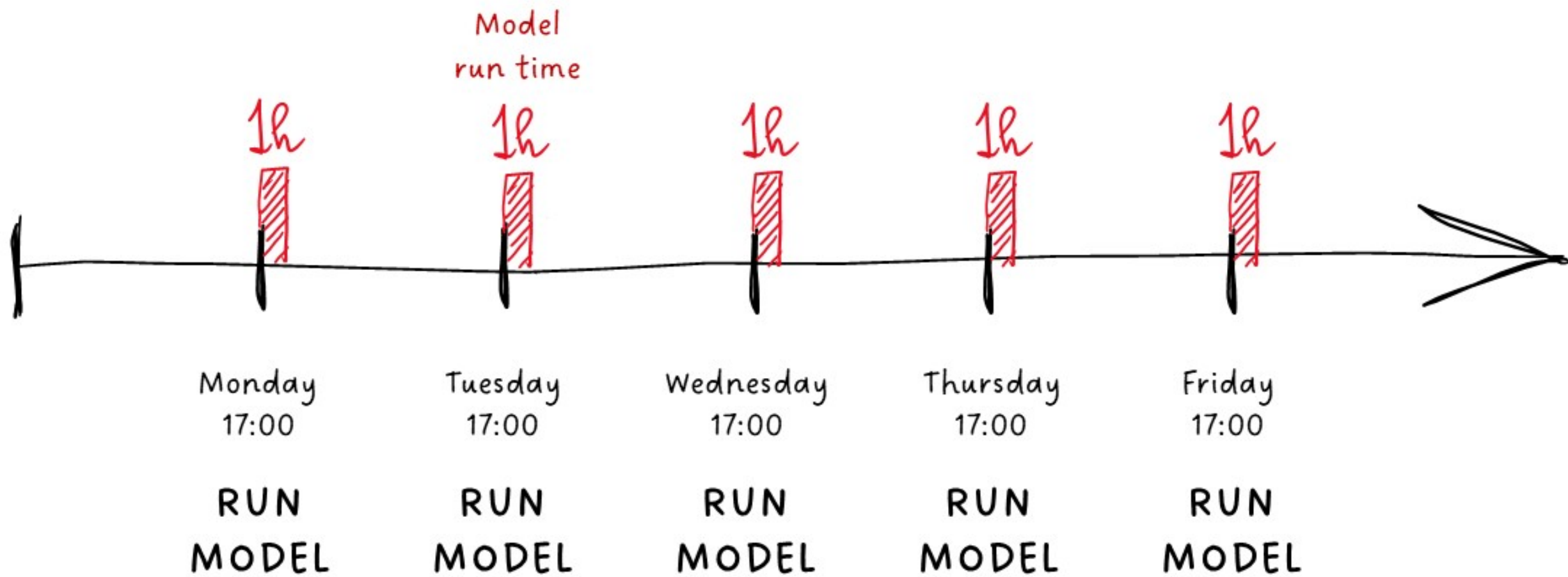
Nemanja Radojkovic

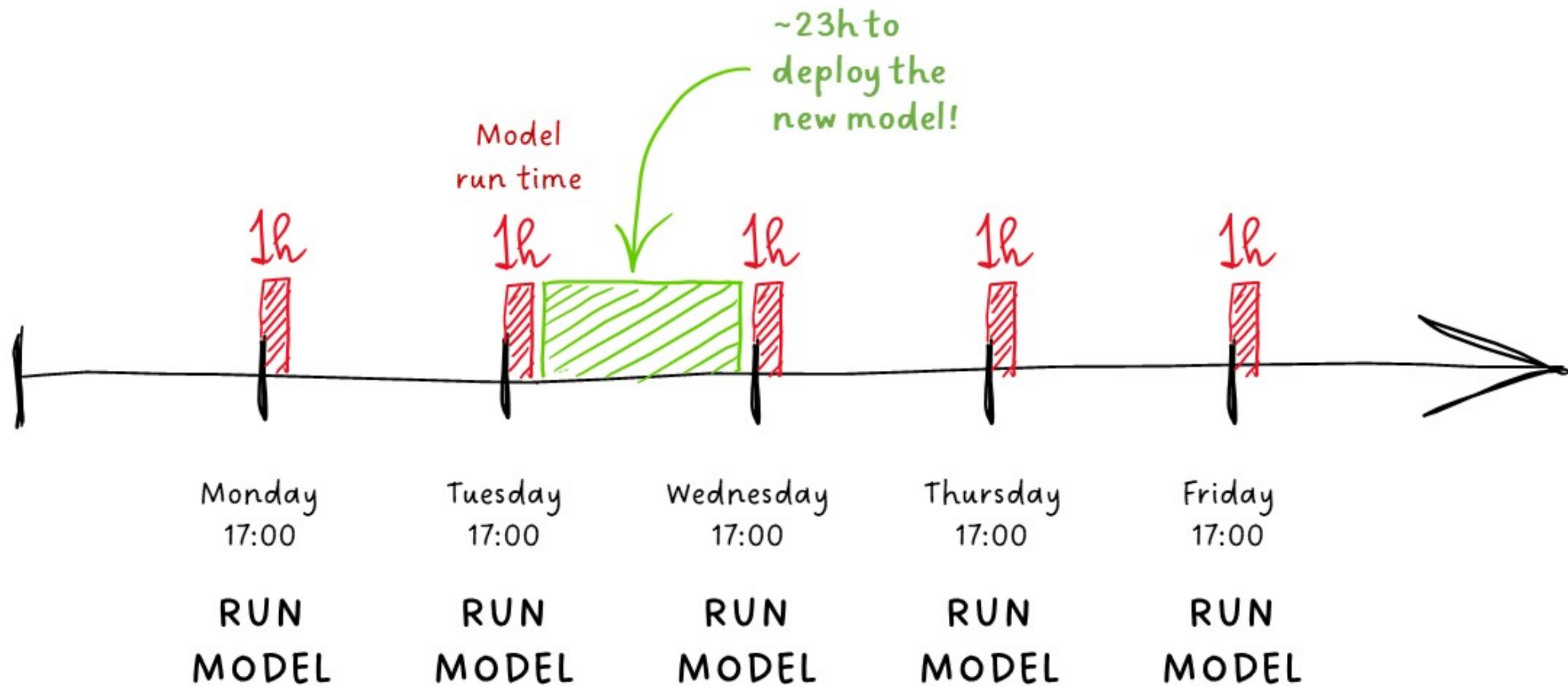
Senior Machine Learning Engineer

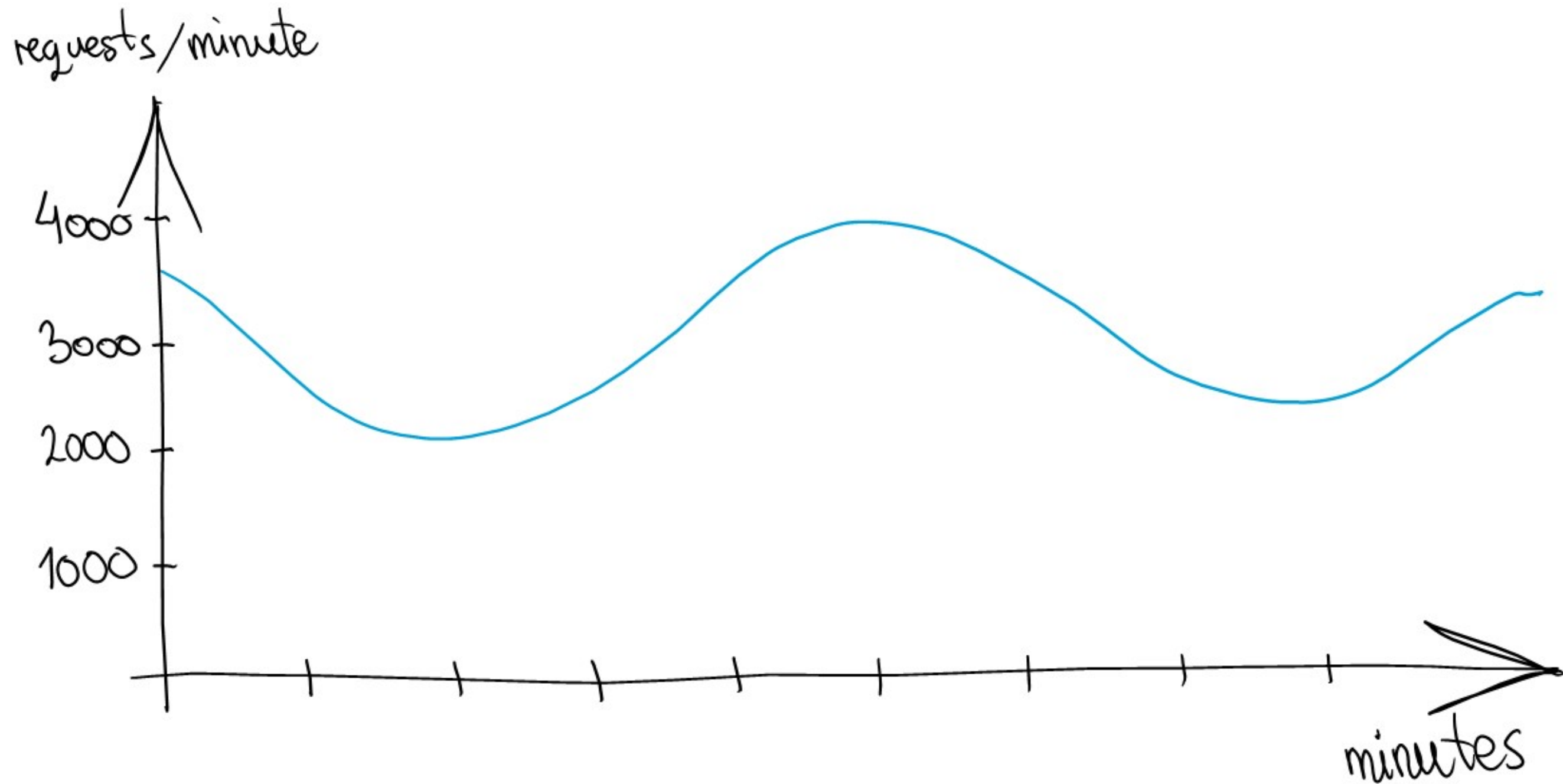
Deployment successful!

- ML app is running
- API handles 1000s of requests/hour
- One month later:
 - New, better features found
 - Collected new batch of training data
 - Ran the model build pipeline
 - New model package created

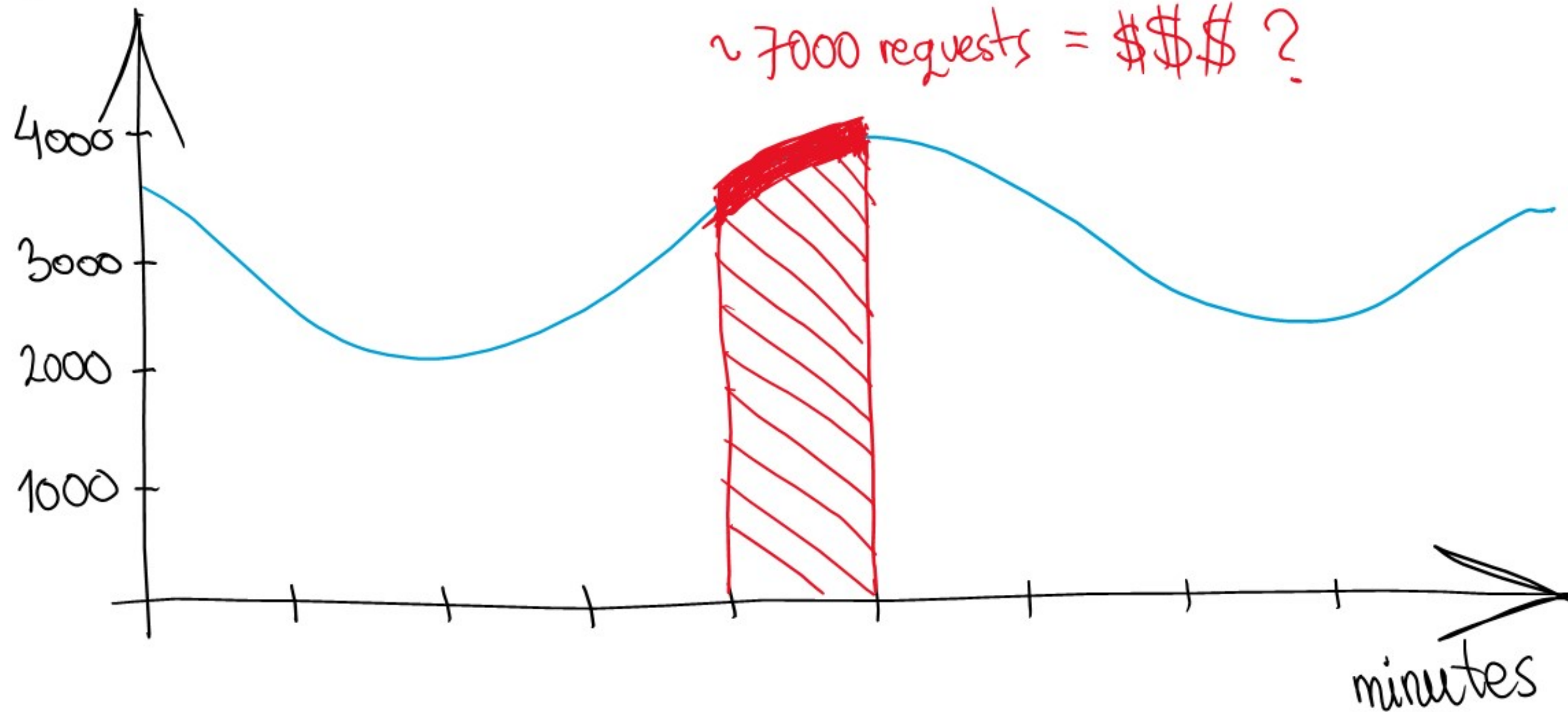


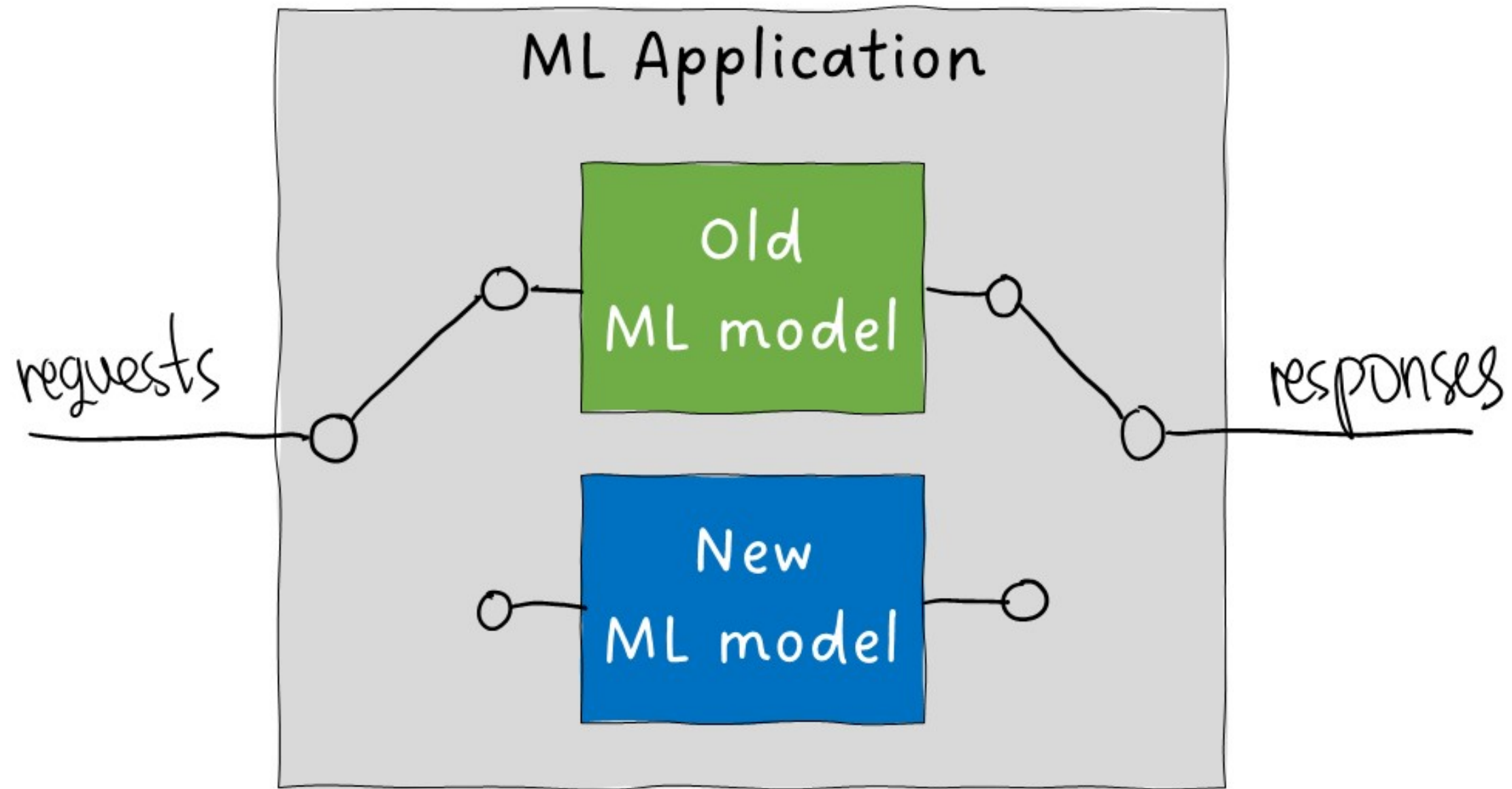


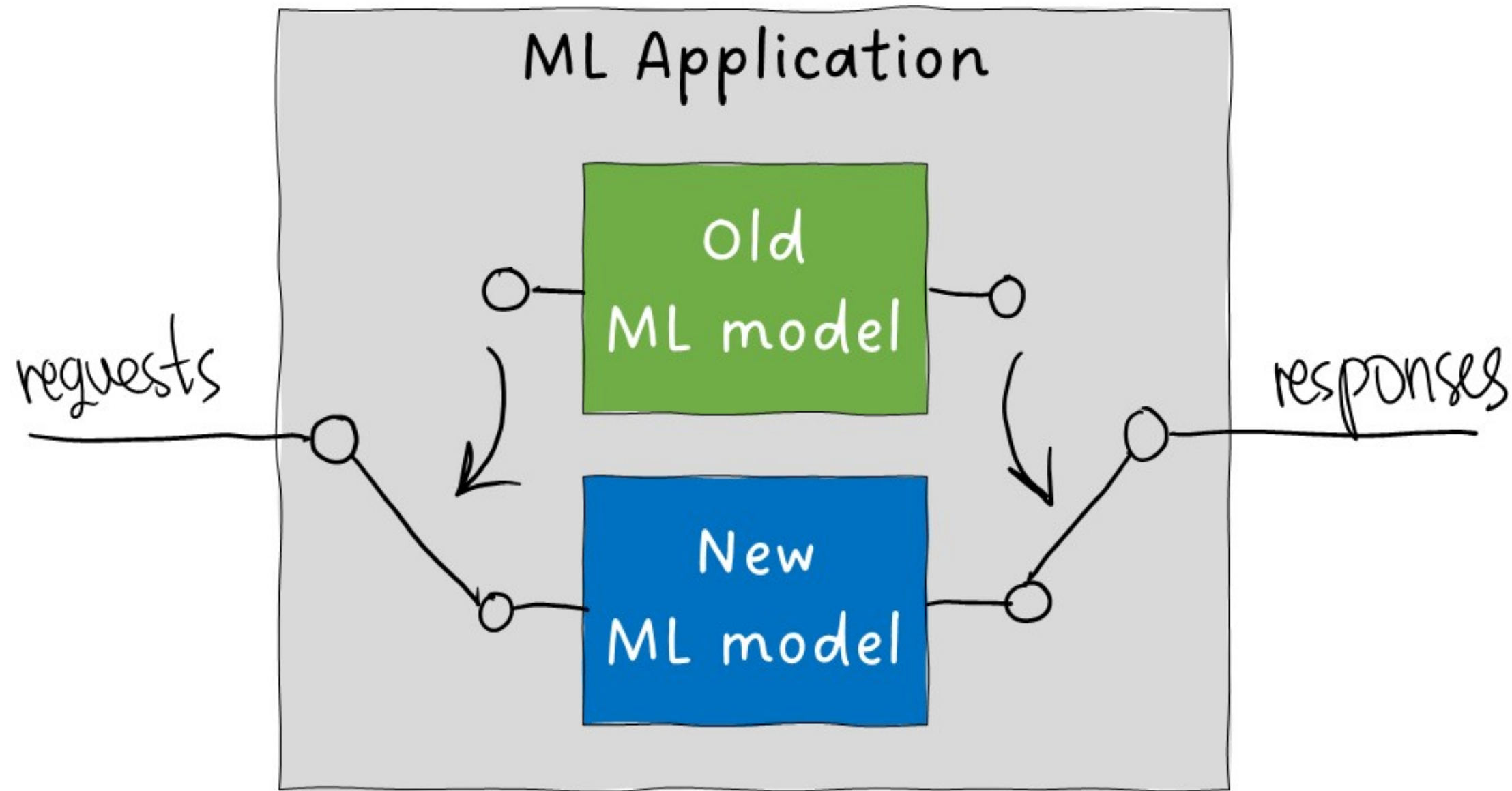




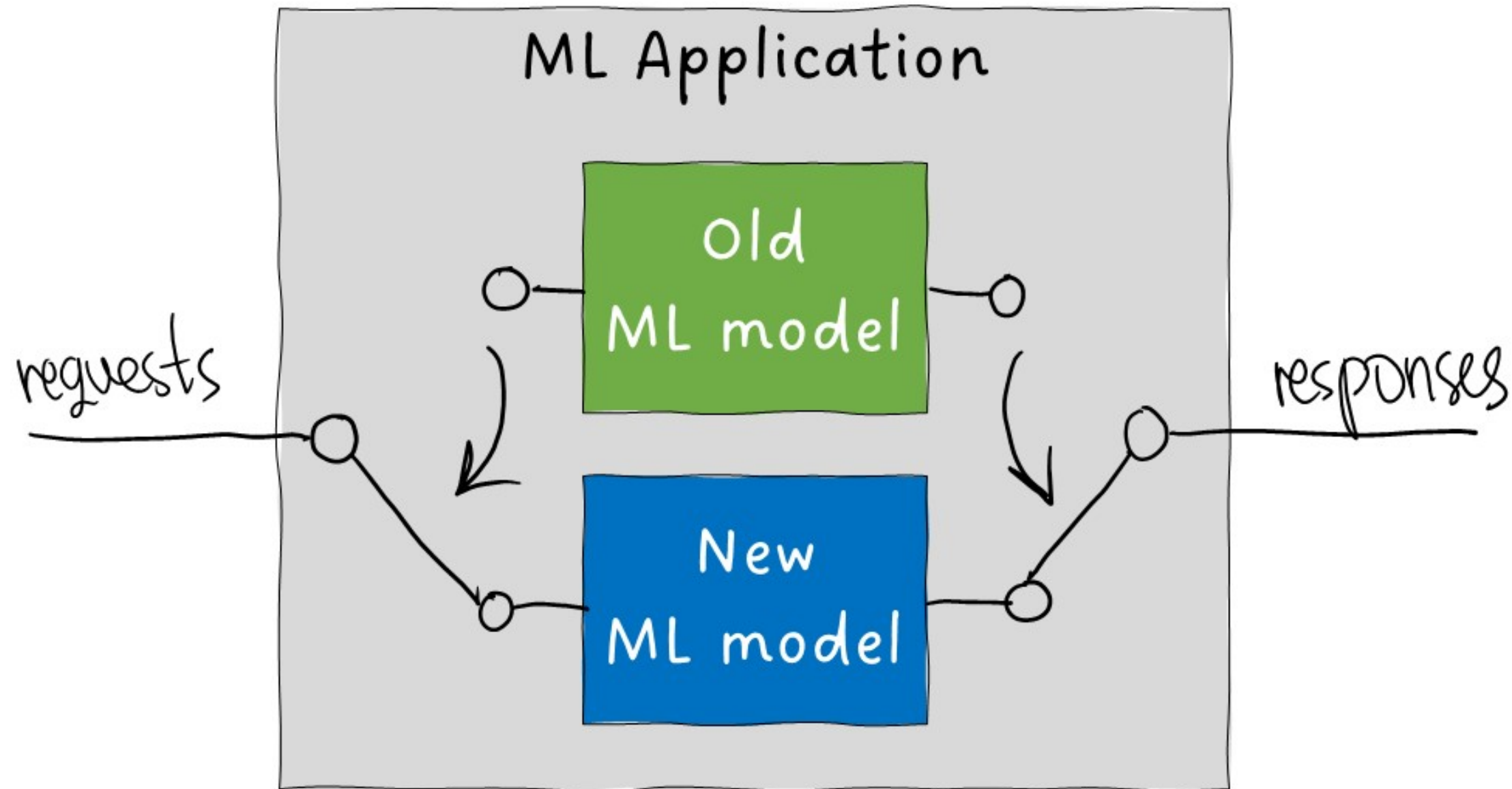
requests/minute



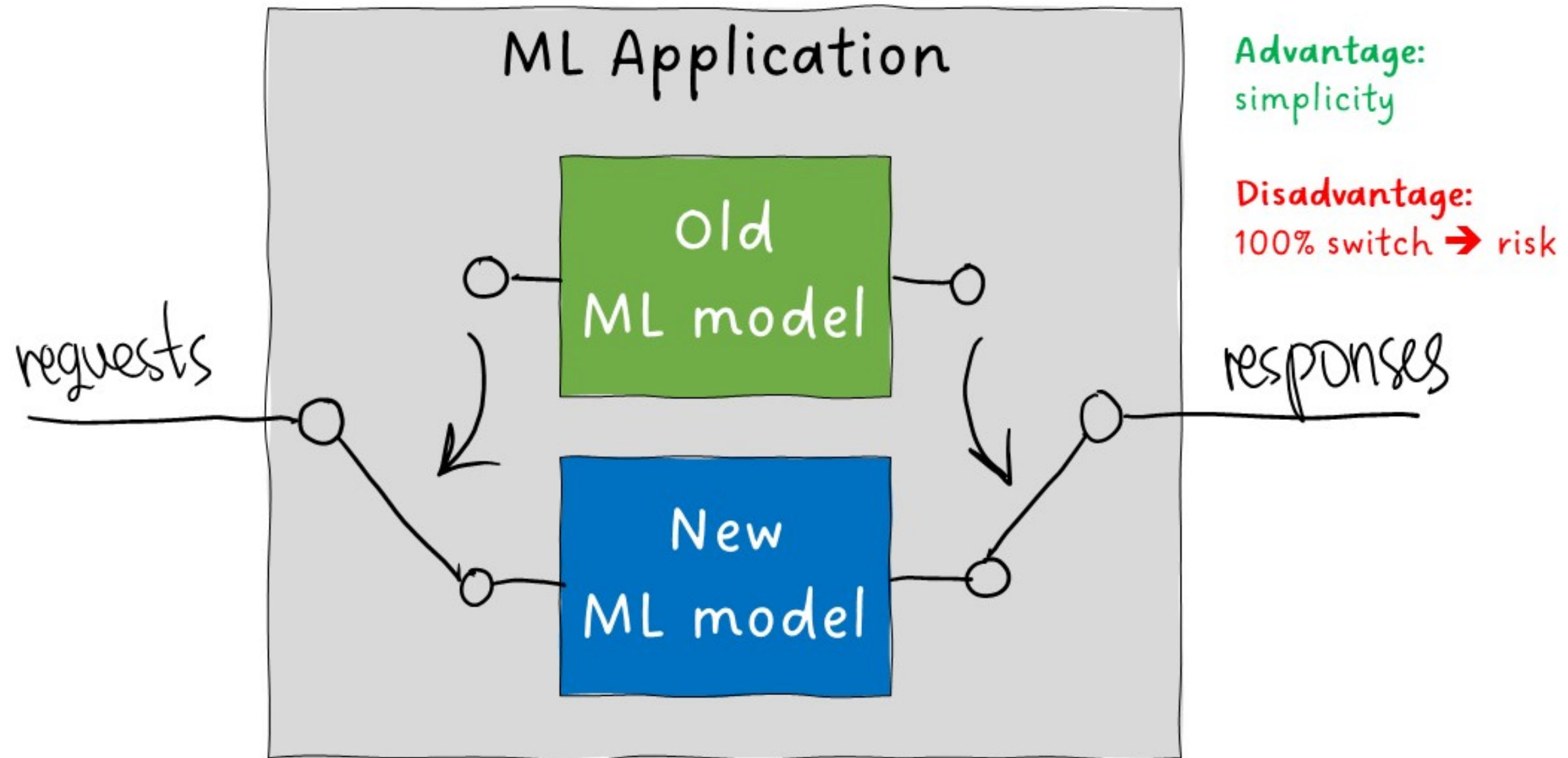




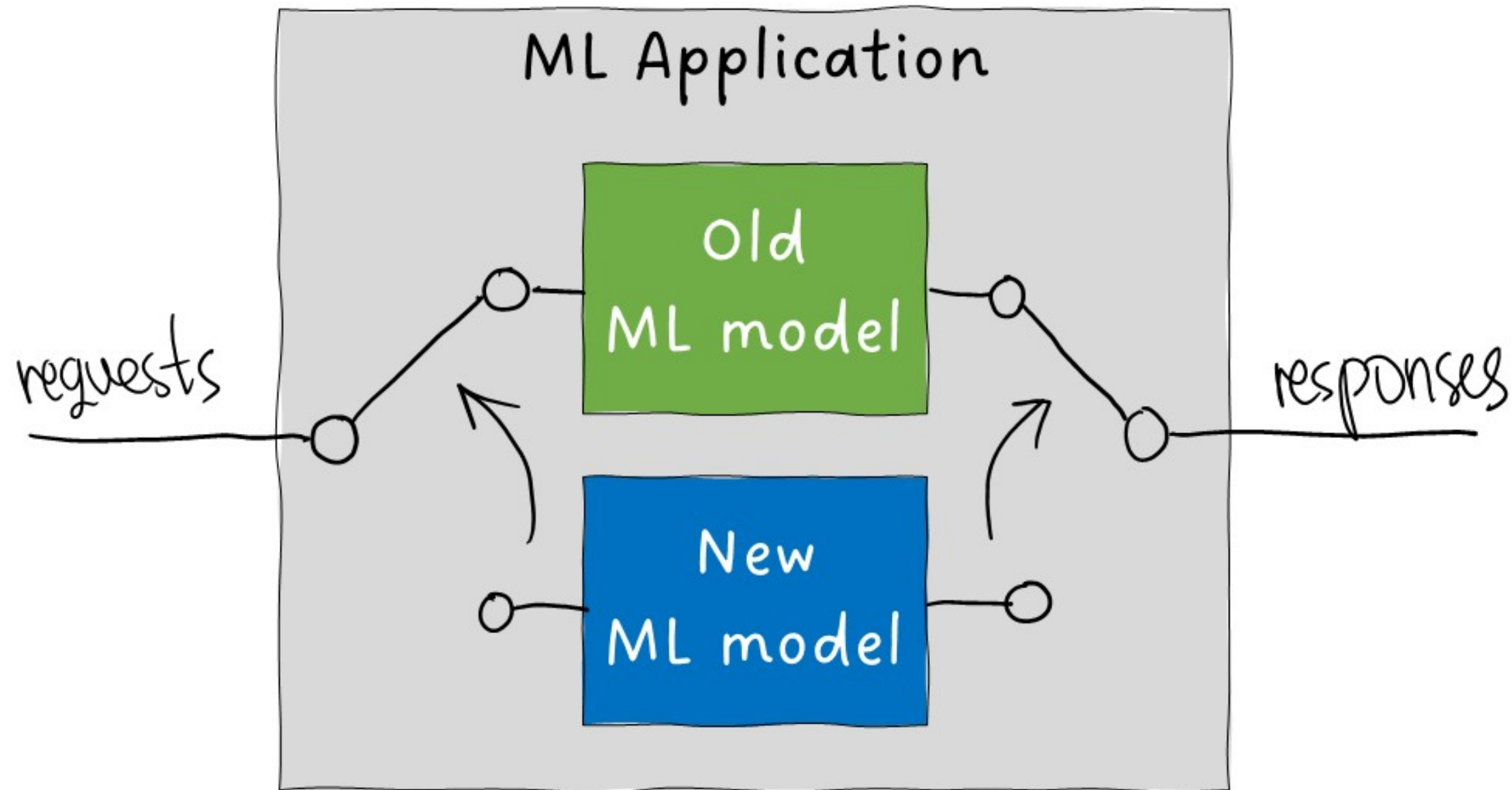
Blue/green deployment



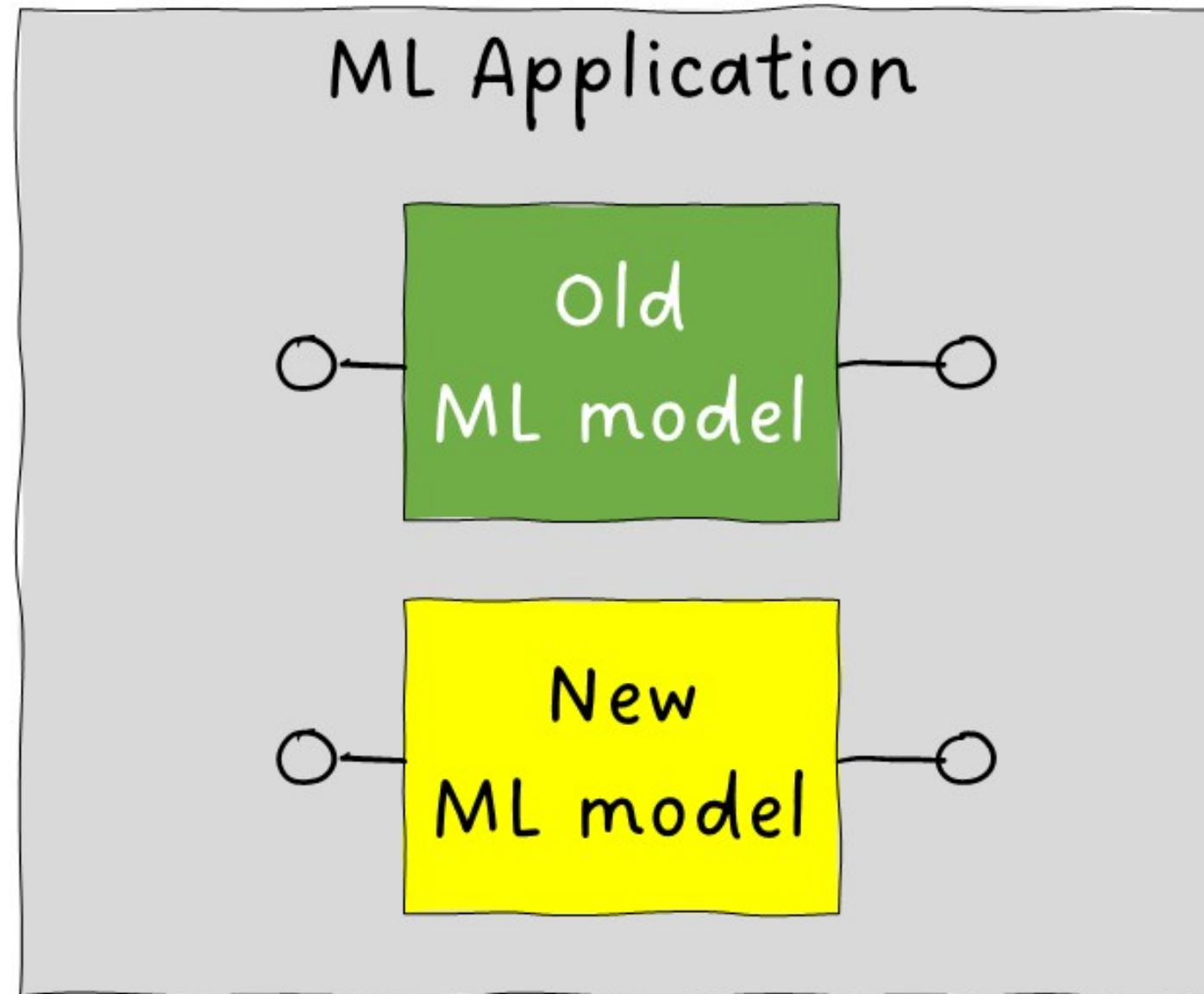
Blue/green deployment



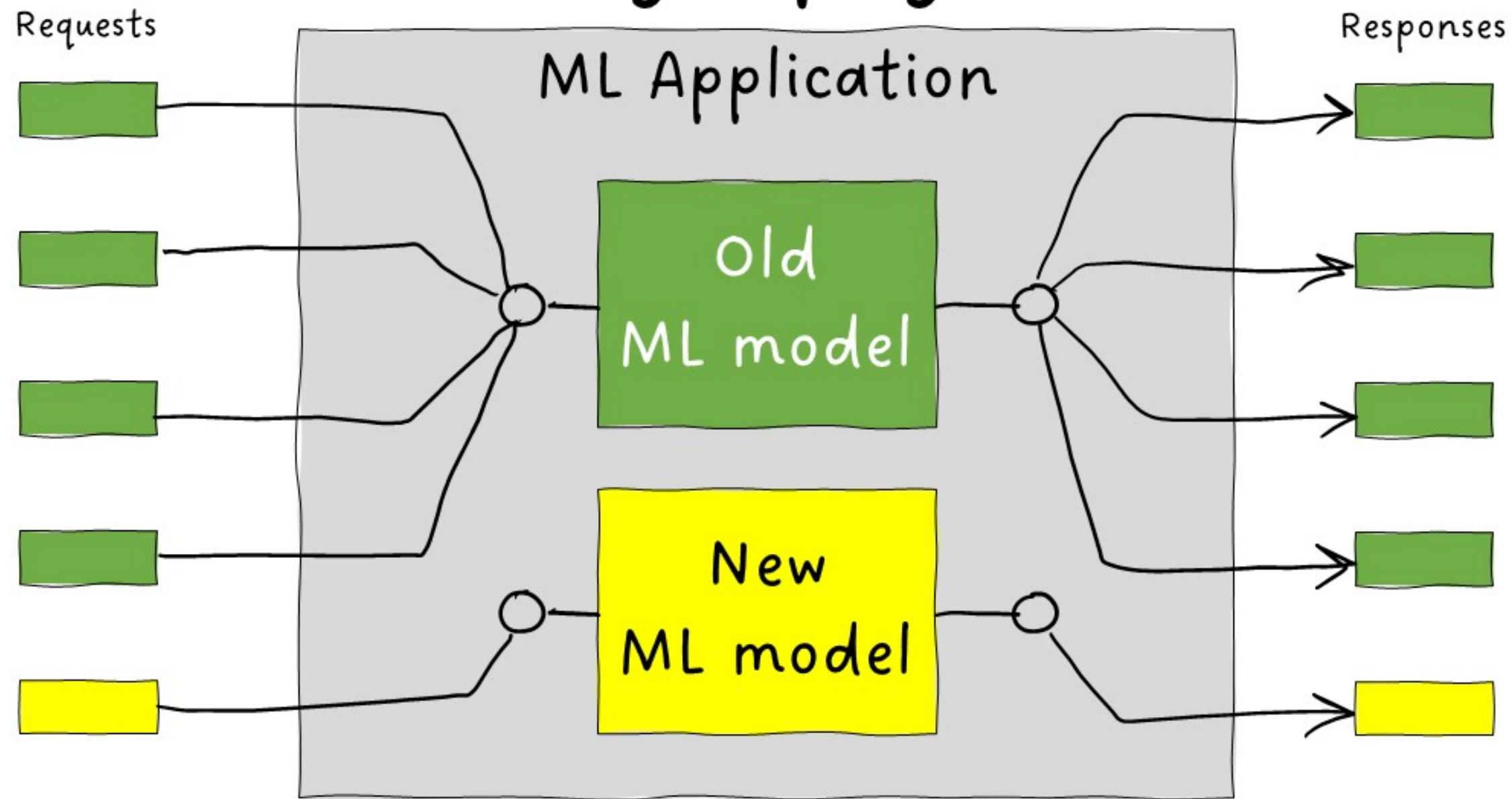
Rollback



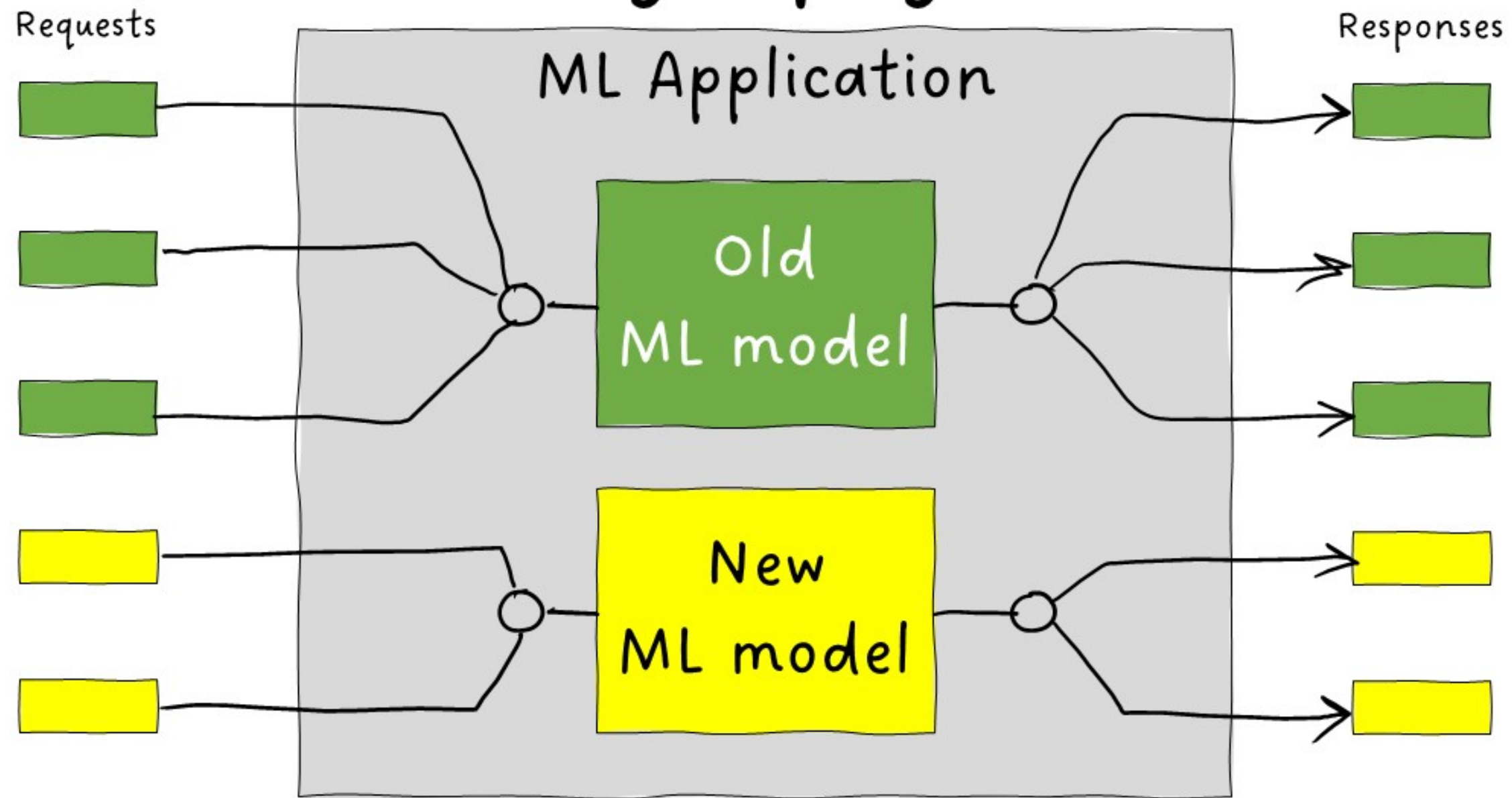
Canary deployment



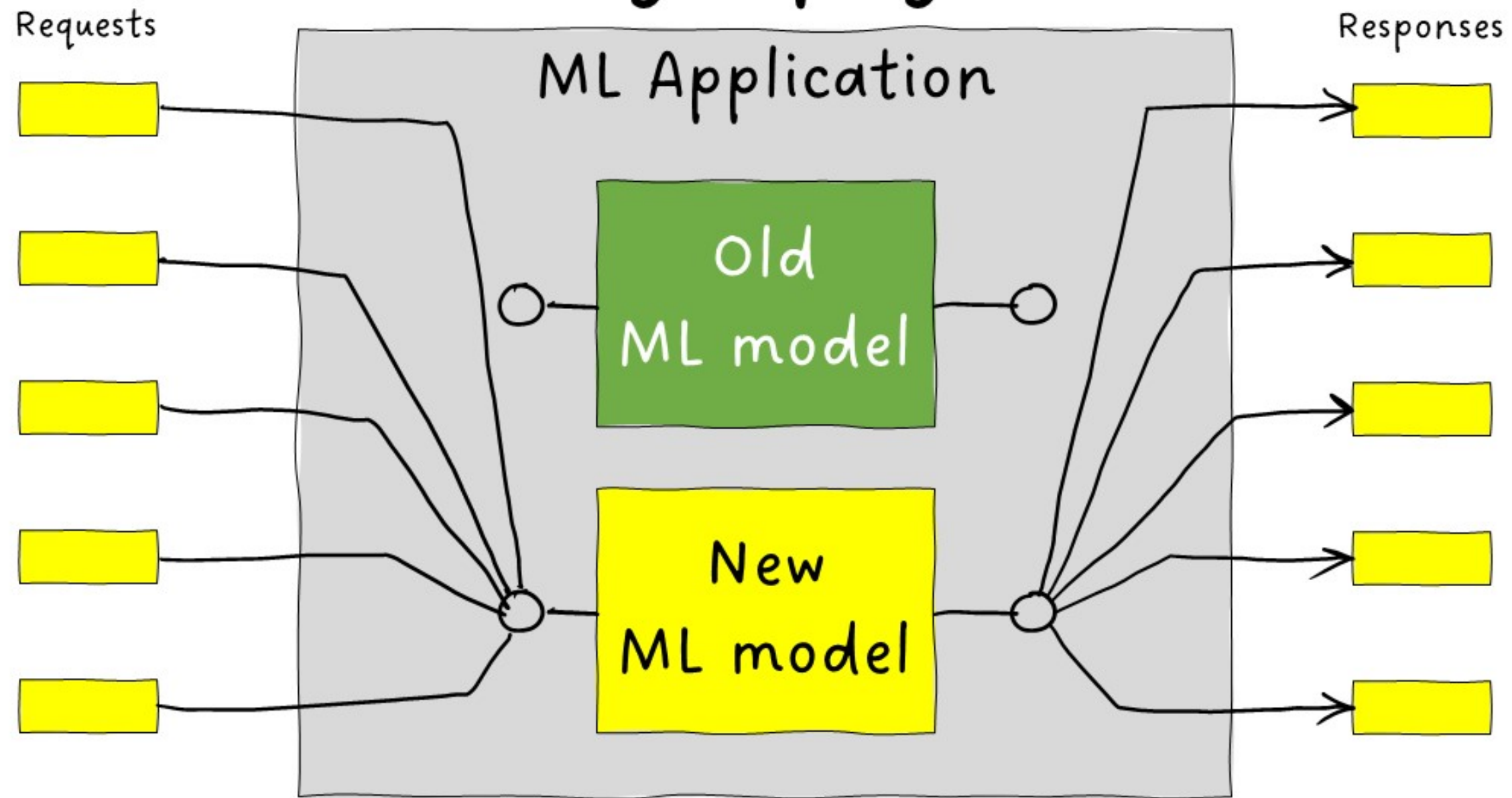
Canary deployment



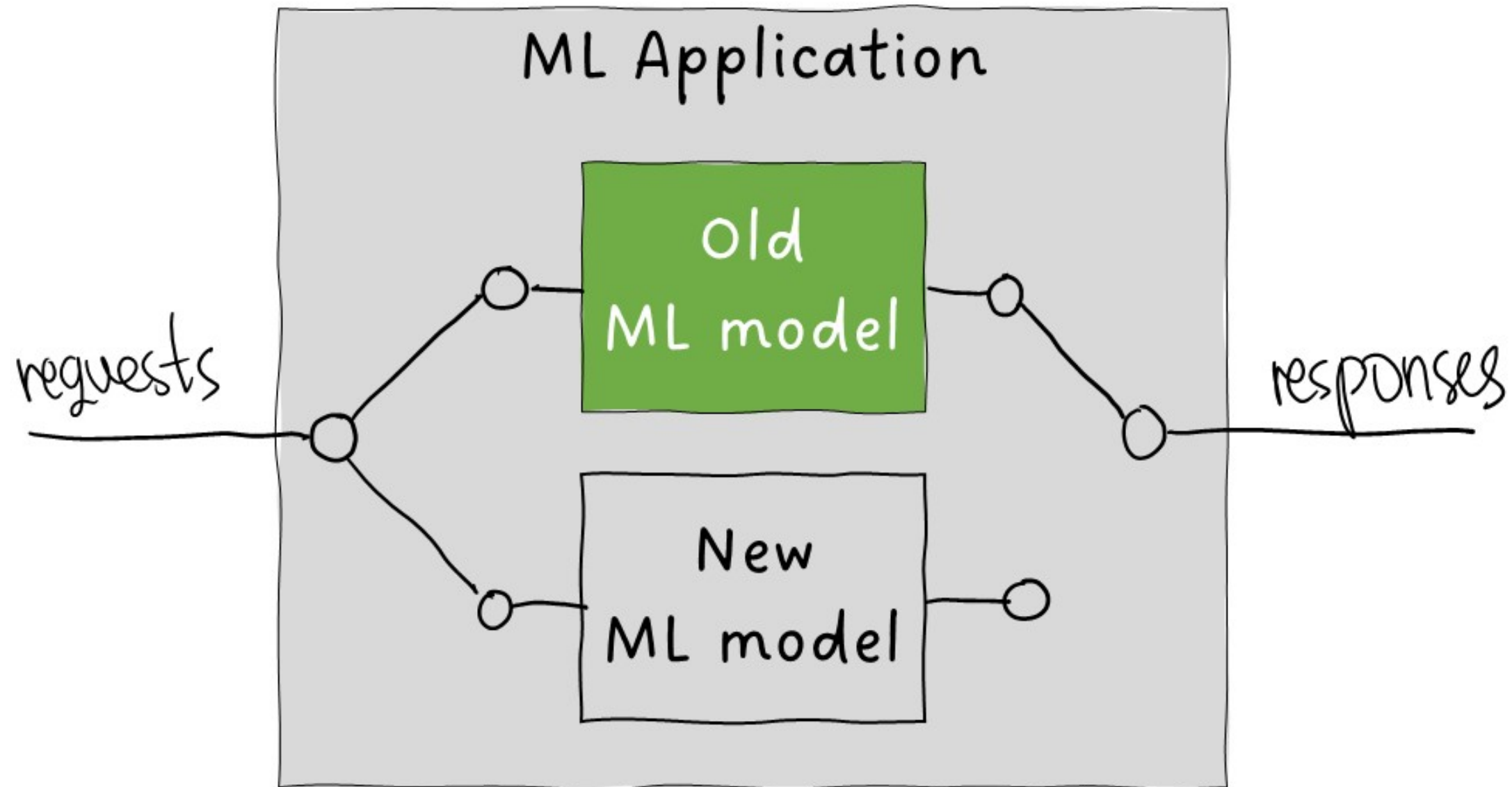
Canary deployment



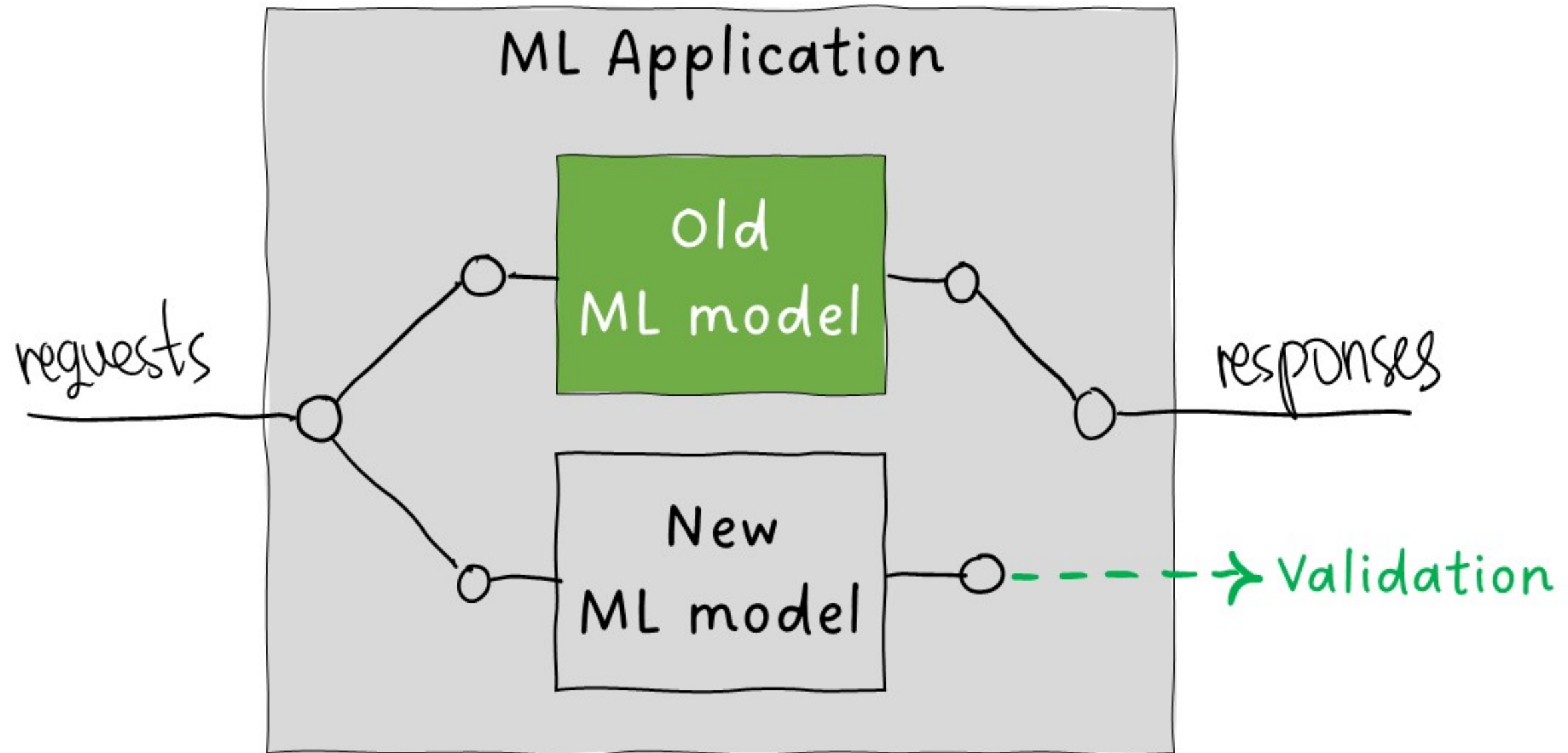
Canary deployment



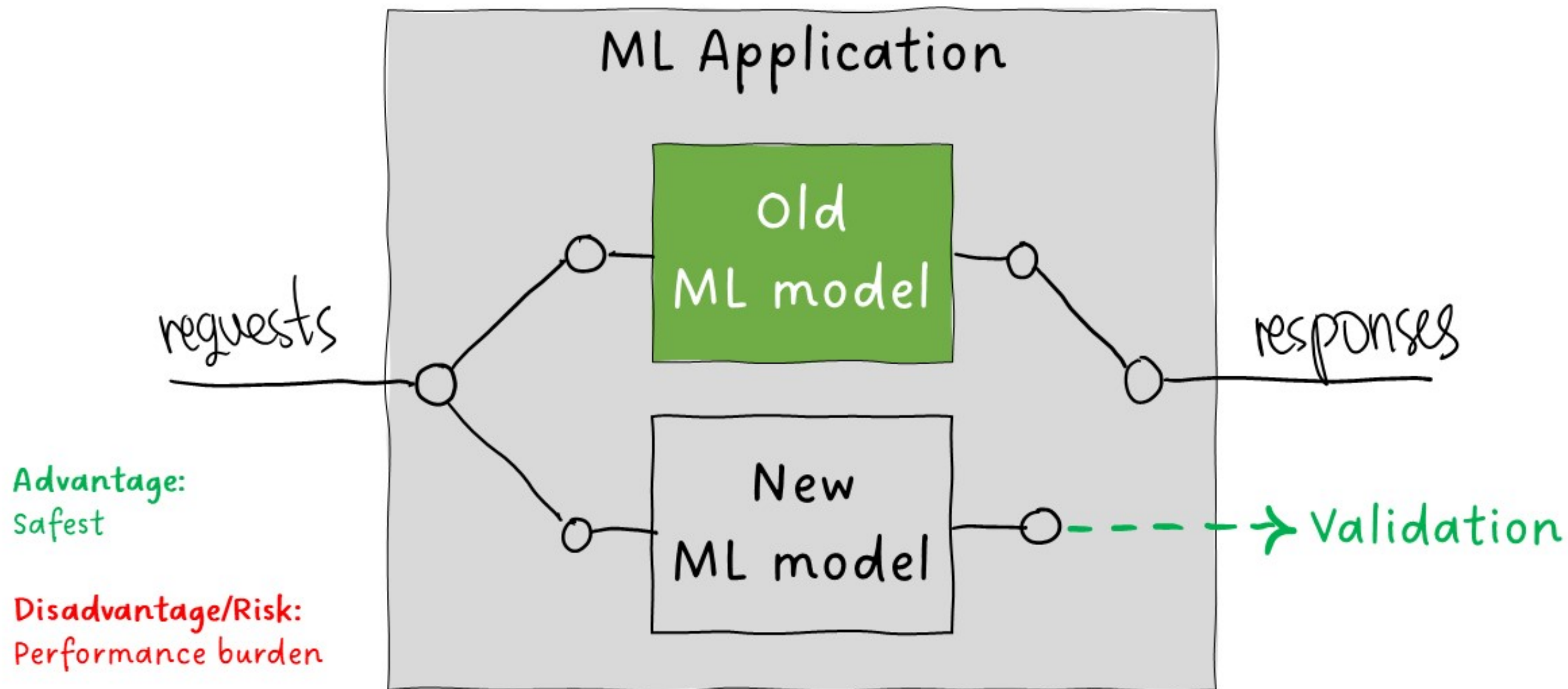
Shadow deployment



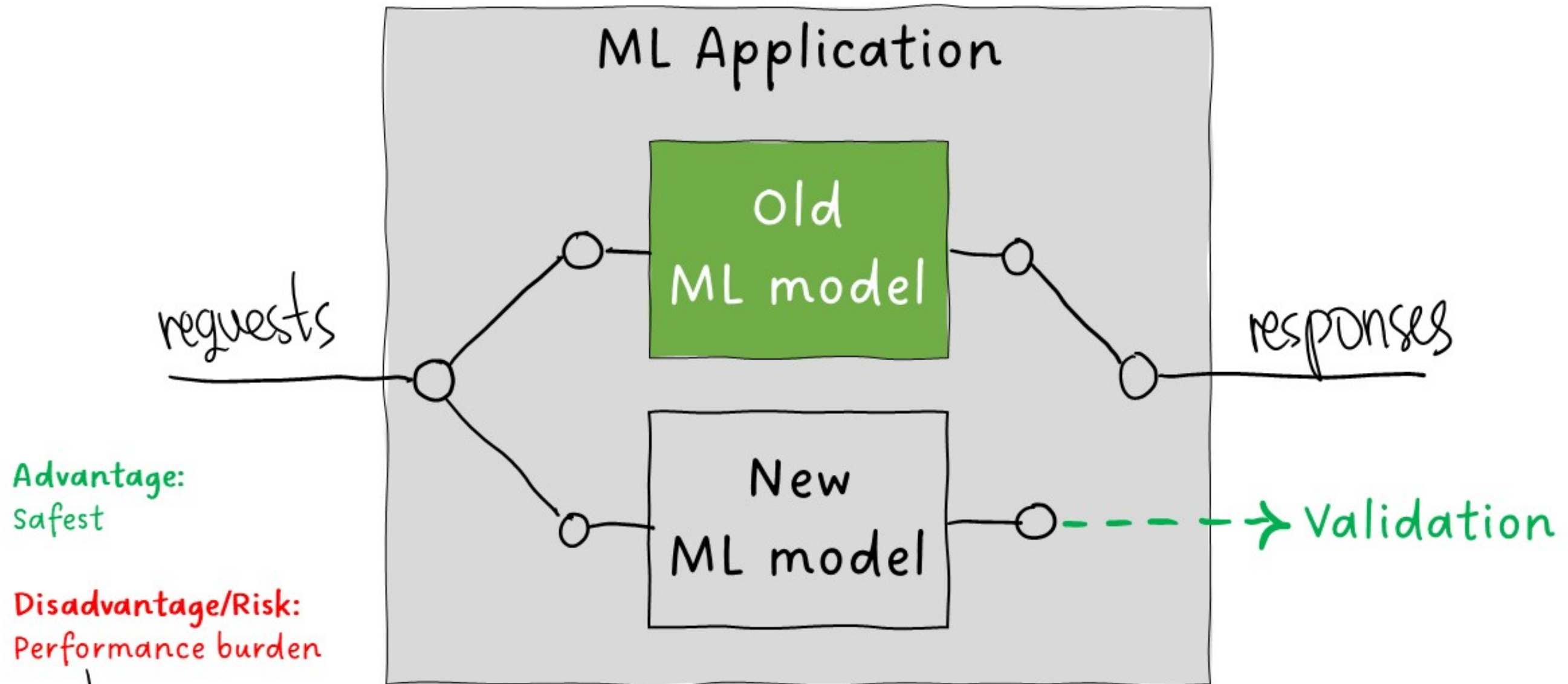
Shadow deployment



Shadow deployment



Shadow deployment



Advantage:
Safest

Disadvantage/Risk:
Performance burden

- Use shadow model on a fraction of requests ("canary shadow")
- Run shadow model offline, outside of peak hours.

Let's practice!

MLOPS DEPLOYMENT AND LIFE CYCLING